United States Patent 12409689
Kind Code B2
Date of Patent September 09, 2025
Inventor(s) Schmidt; Mark Christopher et al.

Trailer jack and transport systems employing a spherical load-bearing transport ball, and sport, recreational and utility trailer systems and methods employing the same

Abstract

A trailer jack and transport system providing improved rolling and steerable motion to sport, recreational and utility trailers, when operating on diverse kinds of ground surfaces, including pavement, dirt, sand, and mud, without use of caster-style wheels. The trailer jack and transport system employs a spherical load-bearing transport ball supported within a frame assembly and configured to allow the spherical wheel to freely rotate in any orientation, on ground surfaces of a diverse nature during system operation.

Inventors: **Schmidt; Mark Christopher (Windermere, FL), Rodriguez; Edward Joel (Kissimmee, FL)**

Applicant: **ORBIT JACK LLC** (Windermere, FL)

Family ID: **86385012**

Assignee: **ORBITJACK LLC (Greenville, SC)**

Appl. No.: **17/535561**

Filed: **November 24, 2021**

Prior Publication Data

Document Identifier	**Publication Date**
US 20230158847 A1	May. 25, 2023

Publication Classification

Int. Cl.: **B60D1/66** (20060101); **B62D63/08** (20060101)

U.S. Cl.:

CPC **B60D1/665** (20130101); **B62D63/08** (20130101);

Field of Classification Search

CPC: B60D (1/665); B62D (63/08)

USPC: 280/400

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Primary Examiner: Neacsu; Valentin

Assistant Examiner: Brittman-Alabi; Felicia L.

Attorney, Agent or Firm: THOMAS J. PERKOWSKI, ESQ., PC

Background/Summary

BACKGROUND OF INVENTION
Field of Invention
(1) The present invention is related to new and improved trailer jack and transport systems and related equipment, adapted for mounting to sport, recreational and/or utility trailers designed to carry and transport personal water crafts (PWCs), boats, mobile vehicles such as motorcycles and all-terrain vehicles (ATVs), recreational and camping shelters, and portable utility equipment such as power generators, air compressors and pumping systems, transportable over solid surfaces such as concrete and asphalt, over soft surfaces such as sandy beaches and dirt roads, and within shallow muddy waters along beaches and coastal areas.
Brief Description of the State of the Art
(2) Trailer jack and transport systems are well known in the art, and are currently provided (e.g. mounted) on the tow end or portion of sport, recreational and utility trailers designed to carry and transport personal watercrafts (PWCs), boats, mobile vehicles such as motorcycles and all-terrain vehicles (ATVs), recreational and camping shelters, and portable utility equipment such as power generators and pumping systems, capable of supporting a stationary trailer or transport over solid surfaces such as concrete and asphalt, as well as soft surfaces such as sandy beaches, dirt roads, and shallow muddy waters along beaches or coastal areas.
(3) In general, most conventional wheeled trailer jack and transport systems employ an extendable jacking post member that is rotatably mounted to the trailer's central tow beam or framework. The extendable jacking post member is terminated with one or more caster-type wheels for enabling rolling action of the supported trailer during transport along a desired direction. Trailer jacks may also be terminated with a stationary foot for supporting the trailer in a fixed position. Typically, during trailer jacking operations, the extendable jacking post member is extendable by turning a hand-crank provided on the caster-wheel trailer jack and transport system.
(4) The raising and lowering of the wheeled jack adjusts the height of a trailer's tongue, allowing for, among other purposes, the trailer coupler to be lifted above the ball of a trailer hitch, typically on a motor vehicle, and lowered onto the ball of the trailer hitch. The caster wheel offers a method of positioning the trailer coupler over the hitch through manually pushing, pulling, and steering of the trailer assembly. Trailer jacks may also include a means of rotating into a storage position on

the trailer to which the trailer jack is fastened. This non-functioning position allows for the coupled motor vehicle and trailer to operate as a physically connected system without interference by the trailer jack wheel on the rolling surface.
(5) Prior art trailer jacks, as they are often called, are available in a variety of sizes capable of carrying varying loads depending on the load capacity of the trailer to which it is affixed. For example: 500-pound, 1,000-pound or 2,000-pound capacities. Similarly, conventional wheeled trailer jacks also utilize a variety of wheel sizes, both in diameter and width, to accommodate load and offer motion over varied terrain. Common wheel diameters range from 6″ to 8″ with wheel widths ranging from 1″ to 2.5″. Some casters include two wheels side-by-side, offering better support on gravel or loose ground surfaces where trailers are often used. Trailer jacks can be welded onto trailers or bolted in place.
(6) Prior art caster-style wheels such as those employed on wheeled trailer jacks utilize two axes of rotation to accomplish steerable movement. A first axis is provided about which the wheel rotates (namely the axle, or X-axis) and a second axis is provided about which the wheel assembly consisting of forks, wheel, axle, and mechanical fasteners rotate to accomplish steering (the longitudinal or Y-axis)). The two axes are offset by design, allowing a caster wheel to “swing” into the intended direction of travel as the trailer is manually moved. A disadvantage of this type of arrangement occurs when starting from a standstill, particularly when the desired travel distance of the trailer is short. While bearing the weight of the trailer (and often its contents), it is often difficult to turn the caster wheel(s) toward an intended target (e.g. a trailer hitch) while the trailer and wheeled jack assembly are not in motion. It is improbable to have “pre-set” the direction of travel when the trailer jack wheel was previously lowered onto the ground as the desired direction of travel of the trailer assembly is not known until its target (typically a motor vehicle's trailer hitch) has returned and later positioned in close proximity to the trailer coupler.
(7) When coupling or de-coupling a trailer from a motor vehicle, it is desirable for the action to be as easy, quick, and safe as possible. Coupling or de-coupling a trailer is a critical but small preparation step of a larger goal: to transport goods (a boat, personal watercraft, snowmobile, lawn mower, furniture, camper, air compressor or any other equipment that does not contain its own terrestrial propulsion) over the road to a new destination (e.g. a boat launch, a beach, a trail, a worksite, a storage facility, a campsite, etc.). Users desire to spend as little time as needed on preparation. When a caster wheel is not rotated into the desired direction of travel, it takes time and energy to reposition the wheel through “banging” on the wheel, jockeying an entire trailer assembly, or through unwanted and unnecessary additional rolling travel. Some caster-style trailer jacks employ a larger distance between its steering axis (Y-axis) and its axle. Such a configuration offers smoother straight-line travel but requires a larger turning circle. Short X-to-Y axis caster styles exhibit shorter turning circles but can result in erratic straight-line travel as the caster is more prone to “wobbling” about the steering axis.
(8) Prior art type caster-style wheeled trailer jacks, as disclosed in U.S. Pat. No. 6,439,545 to Hansen can be easily damaged or broken when excessive force is exerted onto the caster wheel assembly by a user pushing the trailer tongue in a direction for which its two-axis system is not properly positioned. Plastic wheels can break, wheel bearings can be damaged, and a jack's typically metal-and-welded components can bend or break. Pushing a wheeled trailer-jack-enabled trailer whose caster is not properly positioned can also result in damage to the trailer hitch-equipped motor vehicle. Once in motion, a caster-style wheeled jack continues in the direction the wheel is pointed unless and until sufficient lateral force is applied to the rolling trailer to cause the caster to rotate about its second axis. The momentum of a trailer equipped with a caster-style wheeled trailer jack, particularly when carrying a load, can be difficult to overcome. When traveling toward a vehicle, this can result in a collision, causing cosmetic damage to the vehicle. Worse yet, the user can get caught between the trailer and the motor vehicle, causing bodily injury.
(9) Narrow traditional wheels have less rolling resistance and are therefore easier to get moving

and keep in motion when used on a wheeled trailer jack. The disadvantage to narrow wheels carrying weight is their tendency to sink into soft ground surfaces which are common to many trailer applications, namely non-paved surfaces like sand, gravel, grass, snow, or dirt, rendering the wheel inoperable. Wider conventional wheels and dual wheel configurations have been developed to minimize this disadvantage; but such wider configurations are more difficult to directionally steer.

(10) While improvements have been made with regard to manual trailer jacks and transport systems, there has also been an increased demand for remote controlled powered tractor-type trailer transport systems such as the Trailer Valet RVR series by Supertech S. Corp. While this system allows virtually anyone to transport trailers of many weights and sizes and offers great convenience when needing to transport trailers without a motor vehicle, it suffers from shortcomings and drawbacks such as in-place storage modes and easy-rolling manual 360-degree freewheel operation as situations may require. Consequently, prior art remote-controlled self-powered trailer transport systems fail to fulfill the diverse needs of the growing recreational vehicle (RV) industry, and the marketplace is clearly searching new and improved technologies that will increase convenient and safety.

(11) Accordingly, there is a great need in the art for new and improved trailer jack and transport systems that provide improved steerable rolling motion to sport, recreational and utility trailers, when operating on diverse kinds of ground surfaces, including pavement, dirt, sand, and mud, without the use of caster-style wheels, while avoiding the shortcomings and drawbacks of the prior art apparatus and methodologies.

SUMMARY AND OBJECTS OF INVENTION

(12) Accordingly, a primary object of the present invention is to provide a new and improved trailer jack and transport system that provides improved rolling and steerable motion to sport, recreational and utility trailers, when operating on diverse kinds of ground surfaces, including pavement, dirt, sand, and mud, without the use of caster-style wheels, while avoiding the shortcomings and drawbacks of the prior art apparatus and methodologies.

(13) Another object of the present invention is to provide a new and improved trailer jack and transport system that employs a load-bearing spherical object, such as a load-bearing transport ball, as a wheeling device that is supported within a frame assembly and configured to allow the spherical wheel to freely rotate with 360 degrees of freedom, in any orientation on bearing surfaces of a diverse nature, that may come in contact with and engage a portion of the spherical wheel during system operation.

(14) Another object of the present invention is to provide a new and improved trailer jack and transport system that employs a load-bearing spherical object, such as a load-bearing transport ball, as a wheeling device, having infinite number of virtual axes of rotation passing through a single virtual centroid located and embedded within the center of the spherical ball, about which all motion is accomplished-both directional and rotational.

(15) Another object of the present invention is to provide a new and improved trailer jack and transport system that comprises: a semispherical framework supporting a rigid load-bearing transport ball rotatably supported by bearing surfaces mounted in bearing mounting structure, and retained in the semispherical framework by a ball retaining ring structure; a set of braking frames mounted between the semispherical framework and the bearing surfaces, and having thumb slidable levers extending through the semispherical framework, and operable to be arranged in a braking-configuration or a non-braking configuration; and a jacking assembly mounted to the top center of the semispherical framework by welding or other fastening means, and having an inner jacking post that is extendible relative to an outer jacking post member so as to raise and lower the trailer and having a rotatable mounting mechanism that allows the outer post member to rotate through at least 90 degrees of movement, from a storage configuration, to a transport configuration.

(16) Another object of the present invention is to provide a new and improved trailer jack and

transport system, wherein the height-adjusting jack assembly is centered above the spherical wheel, providing 360-degree direction of travel.
(17) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein the height-adjusting jack assembly is offset from the center of the spherical wheel, reducing the complexities of rotating the wheeled jack into a storage position.
(18) Another object of the present invention is to provide a new and improved trailer equipped with a trailer jack and transport system centrally mounted within the center of the A-frame portion of the trailer frame, with ground clearance created through its mounting structure and multi-position outer jacking post configuration.
(19) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein the height-adjusting jack assembly creates the opportunity to drive the wheel through powered means for maximum convenience.
(20) Another object of the present invention is to provide new and improved trailer jack and transport system, wherein its spherical load-bearing wheel distributes an associated trailer's tongue weight over an increasingly larger area when the ground surface is soft, providing defense over gravel, sand, snow, grass, dirt, and mud.
(21) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein the load-bearing transport ball is supported by a set of bearing pads mounted within the semispherical framework and extending from its bottom aperture to allow rolling support on a ground surface, and embraced within a dual-sided ball braking system mounted between the load-bearing transport ball and the interior surfaces of the semispherical framework.
(22) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein a plurality of bearing surfaces is mounted within the interior volume of a semispherical framework, for freely supporting a load-bearing transport ball, and thereby enabling 360 degrees of freedom of movement by the transport ball while mounted within the semispherical framework of the trailer jack and transport system.
(23) Another object of the present invention is to provide a new and improved trailer jack and transport system, comprising a semispherical framework having an interior volume for rotatably mounting a load-bearing transport ball, a set of bearing pads (e.g. ball-bearing type) for supporting the load-bearing transport ball, a transport ball retaining ring member, a dual set of braking frames or pads, a transport ball surface brushing mechanism, and an hand-cranked extendable jacking member for mounting to both a trailer via an integrated trailer mount, and also to the top portion of semispherical framework.
(24) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein the extendible jacking post member is manually operated using a handle-portion projecting from a semispherical framework.
(25) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein a set of larger, spring-loaded ball bearing pads are mountable within a semispherical framework of the system, for supporting a load-bearing transport ball, thereby enabling 360 degrees of freedom of movement by the transport ball while mounted within the semispherical framework of the trailer jack and transport system.
(26) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein a set of smooth, self-lubricating bearing pads are mountable within a semispherical framework of the system, for supporting a load-bearing transport ball, thereby enabling 360 degrees of freedom of movement by the transport ball while mounted within the semispherical framework of the trailer jack and transport system.
(27) Another object of the present invention is to provide a new and improved trailer jack and transport system, employing a dual-sided ball braking system mounted between a load-bearing transport ball and the interior surfaces of a semispherical framework, for gripping the surfaces of the load-bearing transport ball during breaking operations.

(28) Another object of the present invention is to provide such a new and improved trailer jack and transport system, wherein the semispherical framework comprises at least a pair of arching support portions (i) extending towards and integrated with a central dome portion, to which the elongated jacking member is fixedly connected, and (ii) forming an elongated arcuate slot within which a ball braking slider is adapted to move up and down, to release and lock the load-bearing transport ball.
(29) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein a dual-sided side-pull type of transport ball braking system is provided with a mechanically-operated safety-lever wherein the braking shoes are spring-biased in their locking configuration by a pair of brake compression springs, and retracted in an upward manner by a pair of cables that are pulled against the brake compression springs, on the sides of the jacking post member assembly, and taken-up on a cable take-up mechanism mounted along the outer jacking post member, while mechanically-operated by a safety lever.
(30) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein a dual-sided center-pull type of transport ball braking system is provided with a mechanically-operated safety-lever shown arrangeable in a locking or braking mode of operation or a non-locking or non-braking mode of operation, wherein during the locking mode of operation, the braking shoes are spring-biased in their locking configuration by a pair of brake tension springs, and protracted in an down manner while the pair of cables are not pulled against the brake tension springs and are released from a cable take-up mechanism mounted along the outer jacking post member, while mechanically-operated by the safety lever.
(31) Another object of the present invention is to provide a new and improved trailer jack and transport system, in which a load-bearing transport ball is supported by three symmetrically spaced-apart bearing pads to evenly distribute trailer loads across these three (or more) load-bearing pads in an advantageous manner.
(32) Another object of the present invention is to provide a new and improved trailer jack and transport system provided with an auto-tilting rotational mounting mechanism which creates an acute angle (e.g., about 5 degrees) through the use of a slanted hinge mechanism, when the system is configured in its horizontal storage mode while mounted to the frame of a trailer, and when arranged in a storage configuration, disposed adjacent the trailer frame.
(33) Another object of the present invention is to provide a new and improved trailer for hitching to a motor vehicle equipped with a trailer hitch and equipped with a trailer jack and transport system having a storage configuration, wherein its jacking post is arranged parallel to the plane of the trailer frame, with about 5 degrees between the jack rotation plane and the trailer plane, to allow the semispherical framework portion of the system to be accommodated during its storage mode.
(34) Another object of the present invention is to provide a new and improved trailer jack and transport system, wherein a plurality of bearing surfaces are mounted within the interior volume of a semispherical framework, for freely supporting a load-bearing transport ball, and thereby enabling 360 degrees of movement by the transport ball mounted within the semispherical framework of the trailer jack and transport system, and wherein the length of the transverse mounting post elongates automatically as the jacking member is rotated through 90 degrees relative to the trailer mounting bracket fixed to the trailer's frame, so as to displace the jacking post member away from the trailer beam during the storage configuration and accommodate the diameter of the semispherical framework member containing the transport ball.
(35) Another object of the present invention is to provide a new and improved trailer equipped with a trailer jack and transport system having a telescoping rotational mounting mechanism so that its jacking post is arranged perpendicular to the plane of the trailer frame whereby, the telescopic rotational mounting mechanism allows a larger semispherical framework portion to be accommodated when arranged in its storage mode or configuration; wherein during the storage mode, the elongated length of the mounting mechanism maintains the semispherical framework portion and transport ball at the correct distance from the trailer's frame to enable compact and

stable storage.

(36) Another object of the present invention is to provide a new and improved trailer jack and transport system provided with a hinge mechanism installed along an extendable jacking post member mounted on a central portion of a semispherical framework, so that the semispherical framework is disposed at an angle of rotation (e.g. 5-10 degrees) about the hinge mechanism when hinged semispherical framework portion is disposed against the primary beam of the trailer during the storage configuration of the system.

(37) Another object of the present invention is to provide a new and improved trailer equipped with the trailer jack and transport system, having a rotational mounting mechanism and a hinged jacking post member, allowing its semispherical framework to adapt to the limited storage space provided in the storage configuration.

(38) Another object of the present invention is to provide a new and improved trailer jack and transport system mounted on a main beam of a trailer frame and when arranged in its active jack and transporting configuration, its hinged mechanism can be closed and locked along the extendable jacking post member mounted to the central portion of the semispherical framework.

(39) Another object of the present invention is to provide a new and improved trailer jack and transport system employing a cylindrically-shaped stationary mechanism (i.e. extendible foot) installed on the bottom portion of a semispherical framework and adapted for easy rotation about its load-bearing transport ball when applying torque to the cylindrically-shaped stationary mechanism, and a means of lifting the transport ball off the ground surface after a ½ turn, to prevent the transport ball and its mounting assembly from rolling along the ground by maintaining the semispherical framework member stationary relative to the ground surface, and functioning in way similar to a safety-brake mechanism.

(40) Another object of the present invention is to provide a cylindrical parking ring structure (i.e. extendible foot) having a first set of foldable torque generating handles adapted for arranging the cylindrical ring structure in a non-rolling configuration, and a second set of foldable torque generating handles adapted for arranging the cylindrical ring structure in a rolling configuration, wherein the first set of handles are being used to arrange the ring structure in its non-rolling configuration.

(41) Another object of the present invention is to provide a new and improved trailer jack and transport system comprising: a semispherical framework; bearing surfaces supported within a bearing surface retention ring structure mounted within the semispherical framework; a load-bearing transport ball supported by said bearing surfaces; a mounting mechanism for mounting said semispherical framework to said trailer; and cylindrical support structure operably mounted to said semispherical framework, and having an integrated torque generation system adapted for use in rotating the cylindrical support structure to selectively raise or lower its position along the longitudinal axis of said spherical framework, and thereby lifting said load-bearing transport ball off the ground surface for long term storage applications, and lowering back down when it is desired to transport the trailer, once again; wherein the torque generation system comprises: a first set of foldable handles adapted for arranging the cylindrical ring structure in a non-rolling configuration; and a second set of foldable handles adapted for arranging the cylindrical ring structure in a rolling configuration.

(42) Another object of the present invention is to provide a new and improved trailer jack and transport system provided with a bracket for supporting a removable chocking device (i.e., parking foot) that can be snapped onto the bottom portion of the bearing surface retention ring structure.

(43) Another object of the present invention is to provide a new and improved trailer jack and transport system for mounting to a trailer, and having a hinged or ball-and-socket push- and pull-type handlebar mounted on the jacking post member of the trailer jack and transport system, to offer a human operator mechanical advantage when moving the trailer in a desired direction while the trailer jack and transport system is arranged in its transport configuration (i.e., mode of

operation); and storing said handlebar in a downward storage direction.
(44) Another object of the present invention is to provide a new and improved electrically powered trailer jack and transport system for mounting on a trailer and employing a set of electric motors mounted on a semispherical framework and adapted to drive a load-bearing transport ball mounted within the semispherical framework, along 360 degrees of rotational freedom, while operated under the remote control of a user.
(45) Another object of the present invention is to provide a new and improved motor-powered trailer jack and transport system comprising a controller supporting a directional joystick, jacking controls, and controller electronics, and a jack and transport drive subsystem operably connected to the controller by way of a controller wiring harness, supporting a control and comminutions unit provided with gears, jack motor, battery module, battery charging unit, position controller, and drive motors provided with a drive motor wiring interface.
(46) Another object of the present invention is to provide a new and improved powered trailer jack and transport system comprising (i) a load-bearing transport ball rotatably and freely mounted within a semispherical framework by way of a set of five ball-bearing bearing pads, (ii) a powered jacking post mounted and extending from the top central portion of the semispherical framework, and bearing a rotatable mounting mechanism around midsection, and supporting an outboard controller, (iii) a set of three battery-powered motors mounted on the semispherical framework, operably connection to the outboard controls, and configured for driving the transport ball into controlled motion in accordance with user operator controls provided through the onboard controller, and (iv) drive motor protective housing covers, adapted for fitting over the rotating motor drive gears for user safety protection.
(47) Another object of the present invention is to provide a new and improved electrically powered trailer jack and transport system for mounting on a trailer, wherein a hand-supportable remote-control unit is removed from the top portion of its jacking post member and operated within the hand of a user operating the electrically-powered trailer jack and transport system.
(48) Another object of the present invention is to provide a new and improved motor-powered trailer jack and transport system comprising: a removable remote controller supporting a directional joystick, jacking controls, controller electronics, battery storage module, and wireless communications; a releasable electrical connector; and a jack and transport drive subsystem supporting a control and communications unit provided with gears, jack motor, battery module, a battery charging unit, position controller and drive motors provided with a drive motor wiring interface.
(49) Another object of the present invention is to provide a new and improved electrically-powered trailer jack and transport system for mounting on a trailer, wherein a mobile phone running a mobile application is operated within the hand of a user to operate and control the electrically-powered trailer jack and transport system.
(50) Another object of the present invention is to provide a new and improved system network comprising: (i) a mobile phone equipped with a mobile application (i.e. mobile app) to control and navigate a trailer equipped with a motorized trailer jack and transport system, in various applications namely; (ii) a truck-trailer hitching operation, wherein the trailer is transported towards the hitching post of a truck or vehicle and then jacked up and placed on the hitching post/ball; and (iii) trailer parking operations, wherein the trailer is transported from a starting position in a parking lot, across the parking lot and into an intended parking space, without the need to physically handle the trailer during the parking operation.
(51) Another object of the present invention is to provide a new and improved method of controlling and navigating a trailer equipped with a motorized trailer jack and transport system configured to serve and support various applications namely: (i) trailer-vehicle hitching operations, wherein the trailer is transported towards the hitching post of a truck or vehicle and then jacked up so the trailer hitch can be placed on the hitching post/ball; and (ii) trailer parking operations,

wherein the trailer equipped with the jack and transport system is transported from a starting position in a parking lot, across the parking lot and into an intended parking space, without the need to physically handle the trailer during the parking operation.
(52) Another object of the present invention is to provide a new and improved method of controlling a motorized trailer jack and transport system mounted to a trailer, from a stationary starting position in the parking lot, disconnected from its truck, across the parking lot surface and ultimately into a target parking space using the powered trailer jack and transport system remotely and wirelessly controlled by a mobile app running on a mobile phone in communication with the controller aboard the trailer jack and transport system
(53) Another object of the present invention is to provide a new and improved system adapted for automated IR-guided transport and docking of a utility trailer to an IR-guided hitch mounted on a mobile vehicle, using an electrically-powered trailer jack and transport system, wherein the system is adapted for remote control using a remote controller box button input or mobile phone remote input, as the case may be.
(54) Another object of the present invention is to provide a new and improved cloud-based system network supporting a motor vehicle provided with a trailer hitch located near a trailer to be hitched to the motor vehicle, wherein the trailer is equipped with a powered trailer jack and transport system so that this trailer jack and transport system employs an intelligent GPS-tracked and IR-ranging controller on the trailer and a GPS-tracked and IR-ranging module about the hitching post on the motor vehicle, and wherein GPS-tracked and IR-ranging controller on the trailer and a GPS-tracked and IR-ranging module about the hitching post communicate together in a wireless manner and support automated transport, docking and hitching operations without user involvement after the user inputs an automated docking request to either the outboard controller or the mobile app running on the smartphone used by the user.
(55) Another object of the present invention is to provide a new and improved cloud-based system network, enabling a person standing in a parking lot near a vehicle and boat or PWC trailer equipped with the intelligent GPS-tracking motorized trailer jack and transport system, to remotely-control the GPS-tracking motorized trailer jack and transport system using a mobile computing system, such as a smartphone running a special mobile application, and operational in a programmable mode of operation including an automated trailer parking method.
(56) Another object of the present invention is to provide a new and improved motor vehicle having a GPS-tracking and IR ranging module mounted about a trailer hitch for receiving GPS signals and IR ranging signals and communicating dual IR-range finding signals with a trailer jack and transport system during an automated GPS-guided docking and hitching process.
(57) Another object of the present invention is to provide a new and improved method of mounting the GPS-tracking and IR ranging hitch finding module (HFM) about the hitching post (i.e. ball) mounted on the rear end of the motor vehicle in the system intended to hitch to a trailer equipped with the trailer jack and transport system showing the communication of left and right IR ranging signals between the hitching module and the controller within the trailer jack and transport system using “time of flight” (TOF) calculation principles.
(58) Another object of the present invention is to provide a new and improved GPS-tracking and IR ranging hitch finding module (HFM) having a centrally positioned mounting aperture, through which a hitching post is allowed to pass and then secured by a suitable fastening mechanism.
(59) Another object of the present invention is to provide a new and improved GPS-tracking and IR ranging hitch finding module (HFM) employing left and right spaced apart IR transceiver diode spaced apart for supporting the communication of left and right channel IR signals through the IR light transmission window formed in the compact device.
(60) Another object of the present invention is to provide a new and improved GPS-tracking and IR ranging hitch finding module having an IR light transmission window, in communication with a trailer jack and transport system during automated hitch finding, tracking and docking operations,

wherein two channels of line of sight IR signal communications are maintained during system operation so that the two systems can share left and right channel distance information (L1 and L2) with the controller aboard the trailer jack and transport system and automatically generate drive control signals for the driving motors and accurately navigating the trailer towards the hitching post, using a hitch-finding algorithm running within the controller aboard the trailer jack and transport system.

(61) Another object of the present invention is to provide a new and improved automated motor-powered trailer jack and transport system interfaced with a cloud-computing environment and wireless communication infrastructure, and comprising: a controller module mounted upon the end of the jacking post member and supporting a directional joystick, jacking controls and automated finding, docking and hitching mode button, and controller electronics, battery storage module, and wireless communications; an electrical connector; a jack and transport drive subsystem supporting a control and comminutions unit provided with gears, jack motor, and battery module, and a control and communications electronics including an IR finding transceiver, a battery charging unit and a position controller, and drive motors provided with a drive motor wiring interface; and cloud computing and wireless data communication infrastructure.

(62) Another object of the present invention is to provide a new and improved method of automated docking and hitching of a trailer to a hitching post on a motor vehicle, using an automated and electrically-powered trailer jack and transport system mounted to the trailer, initiated by depressing a hard-key button on the outboard controller.

(63) Another object of the present invention is to provide a new and improved method of automated docking and hitching of a trailer to a hitching post on a motor vehicle, using an automated and electrically-powered trailer jack and transport system mounted to the trailer, initiated by depressing a button on the mobile phone app controller.

(64) Another object of the present invention is to provide a new and improved automated method of hitching a trailer to a motor vehicle using an automated trailer jack and transport system employing an intelligent GPS-tracked and IR-ranging controller on the trailer and a GPS-tracked and IR-ranging module about the hitching post on the motor vehicle, during wireless and automated transport, docking and hitching operations.

(65) Another object of the present invention is to provide a new and improved trailer equipped with a GPS-tracking motorized trailer jack and transport system having a GPS-tracking trailer module mounted on the rear of the trailer and in wireless (Bluetooth) communication with the GPS-tracking motorized trailer jack and transport system mounted on the front of the trailer for automatically transporting from a starting position in a parking lot, across the parking lot and into an intended parking space, and the GPS-coordinates of the trailer's boundaries and destination parking space are captured prior to automated parking operations so as to enable automated GPS-guided transport of the trailer from a starting position in a parking lot, across the parking lot, and into an intended destination parking space, using GPS-tracking and navigation methods without the need to physically handle the trailer during the parking operation.

(66) Another object of the present invention is to provide n new and improved automated trailer jack and transport system configured and working in cooperation with a GPS-tracking and LIDAR-mapping navigator beacon module mounted to the rear of the trailer, so as to establish and maintain a virtual trailer navigation axis, passing through the trailer, and about which the spatial boundaries of the trailer are defined and managed within local databases and cloud-based servers, for use to park the trailer into a destination parking space while avoiding collisions with neighboring with a GPS-tracking and LIDAR-mapping navigator beacon module to detect obstacles.

(67) Another object of the present invention is to provide an automated motor-powered trailer jack and transport system interfaced with a cloud-computing environment and wireless data communication infrastructure, and comprising: a wireless controller module mounted upon the end of the jacking post member and supporting a directional joystick, jacking controls and automated

finding, docking and hitching mode button, and controller electronics, battery storage module, and wireless communications; a releasable electrical connector; a jack and transport drive subsystem supporting a control and communications unit (CCU) provided with gears, jack motor, and battery module, battery recharging module, and control and communications electronics including a processor, firmware, and memory (supporting all functions of the system), a dual-channel IR hitch finding module, a battery charging unit and a position controller, GPS signal receiver with antenna for receiving and processing GPS positioning signals, WWAN module, Bluetooth module, gears, jack motor, battery, battery recharging module, and solar panel for recharging battery, and drive motors provided with drive motors and a drive motor wiring interface; a GPS navigator beacon for mounting on rear of trailer provided with a GPS receiver and antenna, a Bluetooth communication module, a processor, firmware and memory, and LIDAR ToF range finder with 180 degree FOV; and cloud computing and wireless data communication infrastructure.
(68) Another object of the present invention is to provide a new and improved GPS-tracking and LIDAR-mapping navigator beacon module mounted to the rear of the trailer to establish and maintain a virtual trailer navigation axis, passing through the central axis of the trailer, while avoiding collisions with neighboring objects.
(69) Another object of the present invention is to provide a new and improved method of automatic parking of a registered trailer that is equipped with the automated trailer jack and transport system and GPS navigator beacon.
(70) Another object of the present invention is to provide a new and improved method of registering GPS coordinate maps of a trailer equipped with the automated trailer jack and transport system, destination parking spaces, and keep out (i.e., obstacle) zones, for storage in a network database system on a cloud-based system network.
(71) Another object of the present invention is to provide a new and improved method for training the cloud-based trailer jack and transport system while configured in an automated learning mode so as to automatically capture and store the GPS coordinates of a trailer and parking locations where the trailer has been parked by a human driver during training operations.
(72) Another object of the present invention is to provide a new and improved automated method of parking a trailer equipped with an automated cloud-based trailer jack and transport system, wherein the automated trailer jack and transport system is installed on the trailer, and configured in a storage mode, uses a mobile app on a mobile phone to learn and record a GPS-specified parking pathway; then the user decouples trailer and automated trailer jack and transport system from tow vehicle; and when operating the trailer jack and transport system in its automated parking mode, stored GPS coordinates are used to automatically guide and transport the trailer safely to its intended parking location without the use of a human driver.
(73) Another object of the present invention is to provide a new and improved personal watercraft (PWC) trailer equipped with the trailer jack and transport system having storage and transport modes of operation.
(74) Another object of the present invention is to provide a new and improved trailer jack and transport system for mounting to the frame of a sport (e.g. personal watercraft or PWC) trailer, and arrangeable in (i) a jack and transport configuration, with the load-bearing transport ball contacting and supported upon a ground surface such as concrete, sand and/or muddy water, and (ii) a storage configuration where the load-bearing transport ball is stored securely against the trailer frame, without obstructions.
(75) Another object of the present invention is to provide a new and improved snowmobile trailer equipped with a trailer jack and transport system having storage and transport modes of operation.
(76) Another object of the present invention is to provide a new and improved trailer jack and transport system for mounting to the frame of a snow-mobile trailer, and arrangeable in (i) a jack and transport configuration, with the load-bearing transport ball contacting and supported upon a ground surface such as concrete, sand and/or muddy water, and (ii) a storage configuration where

the load-bearing transport ball is stored securely against the trailer frame, without obstructions.
(77) Another object of the present invention is to provide a new and improved multi-axle boat trailer equipped with the trailer jack and transport system having storage and transport modes of operation.
(78) Another object of the present invention is to provide a new and improved trailer jack and transport system for mounting to the frame of a multi-axle boat trailer and arrangeable in (i) a jack and transport configuration, with the load-bearing transport ball contacting and supported upon a ground surface such as concrete, sand and/or muddy water, and (ii) a storage configuration where the load-bearing transport ball is stored securely against the trailer frame, without obstructions.
(79) Another object of the present invention is to provide a new and improved camper or recreational vehicle (RV) trailer equipped with a trailer jack and transport system having storage and transport modes of operation.
(80) Another object of the present invention is to provide a new and improved trailer jack and transport system for mounting to a frame of a camping and/or recreational vehicle (RV) trailer, and arrangeable in (i) a jack and transport configuration, with the load-bearing transport ball contacting and supported upon a ground surface such as asphalt, dirt, and and/or snow, and (ii) a storage configuration where the load-bearing transport ball is stored securely against the trailer frame, without obstructions.
(81) Another object of the present invention is to provide a new and improved flatbed trailer equipped with a trailer jack and transport system having storage and transport modes of operation. Another object of the present invention is to provide a new and improved trailer jack and transport system for mounting to the frame of a utility trailer carrying an industrial tool such a power generator, air compressor and/or other tool for doing work on a job or construction site, and arrangeable in (i) a jack and transport configuration, with the load-bearing transport ball contacting and supported upon a ground surface such as concrete, sand and/or muddy water, and (ii) a storage configuration where the load-bearing transport ball is stored securely against the trailer frame, without obstructions
(82) Another object of the present invention is to provide a new and improved transportable system comprising a platform of rigid construction, comprising: at least three corners, each having a load-bearing transportation ball construction top-mounted through the top surface of each platform corner, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.), capable of carrying and transporting a heavy cylindrical beer keg or barrel across a lawn surface
(83) Another object of the present invention is to provide a new and improved trapezoidal transportable system comprising a platform of rigid construction, comprising: four corners, each having a load-bearing transportation ball construction edge-mounted to the side surface of each platform corner, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.) capable of carrying and transporting a heavy object, such as barbeque gas grill, across a lawn surface.
(84) Another object of the present invention is to provide a new and improved powered and remote controlled multi-wheeled transport system consisting of a trailer hitch or other means of coupling to a trailer for transport of said trailer and contents in tight spaces without the use of a traditional motor vehicle (e.g., car or truck).
(85) These and other objects will become more apparent hereinafter in view of the Detailed Description and pending Claims to Invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to understand the Objects more fully, the following Brief Description of the illustrative embodiments should be read in conjunction with the accompanying Drawings, wherein:

(2) FIG. **1**A is a first perspective view of a sport trailer carrying a personal water craft (PWC) **50** (commonly associated with brand names in the marketplace such as Jet-Ski®, Waverunner® or Sea Doo®) to which a trailer jack and transport system of the present invention is mounted and operated according to a first illustrative embodiment of the present invention, wherein the trailer jack and transport system is arranged in its jack and transport configuration, with the load-bearing transport ball contacting and supported upon a ground surface formed on a coastal region or beachfront made from sand and crushed rock;

(3) FIG. **1**B is a second perspective view of the sports trailer of FIG. **1**A shown supported on a smooth ground surface while not being hitched to any motor vehicle;

(4) FIG. **1**C is a perspective view of the sports trailer of the present invention shown in FIGS. **1**A and **1**B involved in the process of its coupler being hitched to a trailer hitch mounted to a mobile vehicle (e.g., pickup truck), while the trailer jack and transport system of the present invention is mounted on and supporting the free end of the trailer (i.e. “trailer tongue”);

(5) FIG. **1**D is a first plan view of the sport trailer in the process of being hitched to the trailer hitch of the mobile vehicle in FIG. **1**C, with the intended direction of travel to position the trailer coupler above the ball of the trailer hitch identified by a dashed arrow;

(6) FIG. **1**E is a second plan view showing the sport trailer having been transported along its path by the trailer jack and transport system to position the trailer coupler directly above the ball of the trailer hitch connected to the mobile vehicle during the process of hitching the sport trailer shown in FIGS. **1**A, **1**B, **1**C and **1**D.

(7) FIG. **1**F is a perspective view of FIG. **1**E showing the trailer coupler and trailer jack and transport system in transport mode with the trailer coupler positioned directly above the ball of the trailer hitch of the mobile vehicle, ready to be lowered onto said ball for coupling the trailer to the hitch.

(8) FIG. **1**G is a perspective view showing the trailer of FIG. **1**C being lowered onto the ball of the trailer hitch by rotating the mechanical crank of the trailer jack and transport system counterclockwise to complete the coupling (hitching) process.

(9) FIG. **1**H is a perspective view showing the trailer hitched up to the motor vehicle shown in FIGS. **1**C, **1**D, **1**E, **1**F and **1**G while the trailer jack and transport system is shown rearranged into its storage mode of operation;

(10) FIG. **2**A is a perspective view of the trailer jack and transport subsystem of the present invention arranged shown removed from its trailer, and in which a coordinate reference system is symbolically embedded for referencing the omnidirectional motion of the load-bearing transport ball freely mounted within its semispherical framework;

(11) FIG. **2**B is a perspective view of the trailer jack and transport system of the present invention shown removed from any trailer, while arranged in its jack and transporting configuration, with its load-bearing transport ball contacting the ground surface;

(12) FIG. **2**C is a perspective, partially cut away view of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B showing the spherical wheel assembly of the present invention, wherein the load-bearing transport ball is shown in greater detail (i) supported by a set of bearing pads mounted along the perimeter of the interior space of the semispherical framework and extending from its bottom aperture to allow physical engagement with and rolling support on a ground surface, and (ii) embraced within a dual-sided ball braking system mounted between the load-bearing transport ball and the interior surfaces of the semispherical framework;

(13) FIG. **2**D is a first elevated side view the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball and the interior surfaces of the semispherical framework, while the extendable jacking post member is shown extended and in a jack mode of operation;
(14) FIG. **2**E is a second elevated side view the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball and the interior surfaces of the semispherical framework, while the extendable jacking post member is shown extended and in a jack mode of operation;
(15) FIG. **2**F is an elevated partially cut-away side view the semispherical framework portion of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball and the interior surfaces of the semispherical framework;
(16) FIG. **2**G is a bottom view the semispherical framework portion of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball and the interior surfaces of the semispherical framework;
(17) FIG. **2**H is a top axial view the semispherical framework portion of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball and the interior surfaces of the semispherical framework;
(18) FIG. **2**I is an exploded perspective view of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, shown comprising a semispherical framework for supporting a set of bearing surfaces (e.g. ball bearing pads) between the interior surfaces of the semispherical framework and a load-bearing transport ball supported by bearing surfaces, and a dual-sided ball braking system mounted about the load-bearing transport ball while retained within the semispherical framework using a retaining ring structure;
(19) FIG. **2**J is an exploded side of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, shown comprising a semispherical framework for supporting a set of bearing surfaces (e.g. ball bearing pads) between the interior surfaces of the semispherical framework and a load-bearing transport ball supported by bearing surfaces, and a dual-sided ball braking system mounted about the load-bearing transport ball while retained within the semispherical framework using a retaining ring structure;
(20) FIG. **2**K is a perspective view the load-bearing transport ball supported by the bearing surfaces (e.g., ball-bearing pads) installed in the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, shown removed from its semispherical framework, dual-sided ball braking system, and retaining ring structure for purposes of exposition and illustration;
(21) FIG. **2**L is an elevated side view the load-bearing transport ball supported by the bearing surfaces (e.g., ball-bearing pads) installed in the trailer jack and transport system of the first illustrative embodiment shown in FIGS. **2**B and **2**K, shown with its semispherical framework, dual-sided ball braking system, and retaining ring structure removed for purposes of exposition and illustration;
(22) FIG. **2**M is a cross-sectional view of the load-bearing transport ball supported by the bearing surfaces (e.g., ball-bearing pads) installed in the trailer jack and transport system of the first illustrative embodiment shown in FIGS. **2**B and **2**K, taken along viewing line **2**M-**2**M in FIG. **2**K;
(23) FIG. **2**N is the spatial arrangement of bearing surfaces (e.g., ball bearing pads) shown employed in FIGS. **2**K and **2**L, with the load-bearing transport ball removed for purposes of exposition and illustration purposes;

(24) FIG. **2**O is a fragmented view of a single bearing surface engaging with a portion of the load-bearing transport ball employed in the trailer jack and transport system of the first illustrative embodiment of the present invention, shown in FIG. **2**B;
(25) FIG. **2**P is a perspective view of one of the five (5) bearing surfaces (e.g., ball bearing pads) installed in the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, adapted for rotatable support of the load-bearing transport ball in an omni-directional manner providing freedom for it to roll freely within the semispherical framework of the system;
(26) FIG. **2**Q is a perspective exploded view of one of the five (5) bearing surfaces (e.g. ball bearing pads) installed in the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, comprising (i) a set of ball bearings mounted between a slightly arcuate support base portion and an apertured arcuate surface plate, through which each ball bearing is permitted to project while mounted within the bearing surface pad, and (ii) a user-replaceable brush ring consisting of a frame supporting and brushes for expelling debris from the surface of the load-bearing transport ball;
(27) FIG. **2**R is a perspective front view of a first alternative bearing surface (e.g., ball bearing pad) adapted for installation in the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, for rotatably supporting the load-bearing transport ball in an omni-directional manner;
(28) FIG. **2**S is a perspective rear view of the first alternative bearing surface (e.g., ball bearing pad) shown in FIG. **2**R;
(29) FIG. **2**T is a perspective view of the first alternative bearing surface (e.g., ball bearing pad) in FIG. **2**R, shown engaging the load-bearing transport ball of the present invention while removed from its semispherical framework, for purposes of illustration;
(30) FIG. **2**U is a cross-sectional fragmented view of the first alternative bearing surface engaging the load-bearing transport ball of the present invention, taken along viewing line **2**U-**2**U in FIG. **2**T;
(31) FIG. **2**V is a perspective front view of a second alternative bearing surface (e.g., self-lubricating plastic bearing pad, such as Nylon plastic) adapted for installation in the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, for rotatably supporting the load-bearing transport ball in an omni-directional manner;
(32) FIG. **2**W is a perspective rear view of the second alternative bearing surface (e.g., Nylon plastic ball bearing pad) shown in FIG. **2**V;
(33) FIG. **2**X is a perspective view of the second alternative bearing surface (e.g., ball bearing pad) in FIG. **2**V, shown engaging the load-bearing transport ball of the present invention, while removed from its semispherical framework for purposes of illustration;
(34) FIG. **2**Y is a cross-sectional view of the second alternative bearing surface engaging the load-bearing transport ball of the present invention, taken along line **2**Z-**2**Z in FIG. **2**Y;
(35) FIG. **2**Z is a perspective view of a first illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a solid-core fabricated from durable material that can withstand the expected loads bearing down on the transport ball during operation;
(36) FIG. **2**AA is a cross-sectional view of the first illustrative embodiment of the load-bearing transport ball taken along viewing line **2**AA-**2**AA shown in FIG. **2**Z;
(37) FIG. **2**BB is a perspective view of a second illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a multi-layer core fabricated from different layers of durable material designed to withstand the expected loads bearing down on the transport ball during operation;
(38) FIG. **2**CC is a cross-sectional view of the second illustrative embodiment of the load-bearing transport ball taken along viewing line **2**CC-**2**CC shown in FIG. **2**BB;
(39) FIG. **2**DD is a perspective view of a third illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a spline-core fabricated from

durable material that can withstand the expected loads bearing down on the transport ball during operation;
(40) FIG. **2**EE is a cross-sectional view of the third illustrative embodiment of the load-bearing transport ball taken along viewing line **2**EE-**2**EE shown in FIG. **2**DD;
(41) FIG. **2**FF is a perspective view of a fourth illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a honeycomb-core fabricated from durable material that can withstand the expected loads bearing down on the transport ball during operation;
(42) FIG. **2**GG is a cross-sectional view of the fourth illustrative embodiment of the load-bearing transport ball taken along viewing line **2**GG-**2**GG shown in FIG. **2**FF;
(43) FIG. **2**HH is a perspective view of a fifth illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a hollow-core fabricated from durable material that can withstand the expect loads bearing down on the transport ball during operation;
(44) FIG. **2**II is a cross-sectional view of the fourth illustrative embodiment of the load-bearing transport ball taken along viewing line **2**II-**2**II shown in FIG. **2**HH;
(45) FIG. **3**A is a perspective, partially cut away view of the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B. showing the spherical wheel assembly of the present invention, wherein the dual-sided ball braking system is shown arranged in its non-locking or non-braking mode of operation, with the braking shoes retracted in an upward manner;
(46) FIG. **3**B is a perspective, partially cut away view of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its locking or braking mode of operation, with the braking shoes retracted in a downward manner;
(47) FIG. **3**C is an elevated side view of the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-locking or non-braking mode of operation, with the braking shoes retracted in an upward manner;
(48) FIG. **3**D is an elevated side view of the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its locking or braking mode of operation, with the braking shoes protracted in a downward manner;
(49) FIG. **3**E is another elevated side of view of the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-locking or non-braking mode of operation, with the braking shoes retracted in an upward manner;
(50) FIG. **3**F is another elevated side view of the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its locking or braking mode of operation, with the braking shoes protracted in a downward manner;
(51) FIG. **3**G is a first perspective side view of the trailer jack and transport system of a modified embodiment of the present invention shown in FIG. **2**B, wherein a side-pull type of transport ball braking system is provided with a mechanically-operated safety-lever shown arranged in its non-locking or non-braking mode of operation, wherein the dual-sided braking shoes are spring-biased in their locking configuration by a pair of brake compression springs, and retracted in an upward manner by a pair of cables that are pulled against the brake compression springs, on the sides of the jacking post member assembly, and taken-up on a cable take-up mechanism mounted along the outer jacking post member, while mechanically-operated by a safety lever;
(52) FIG. **3**H is second perspective view of the trailer jack and transport system of the modified embodiment shown in FIG. **3**G, wherein the side-pull type of transport ball braking system is

provided with a mechanically-operated safety-lever shown arranged in its locking or braking mode of operation, wherein the dual-sided braking shoes are spring-biased in their locking configuration by a pair of brake compression springs, and protracted in a down manner while the pair of cables are not pulled against the brake tension springs and are released from a cable take-up mechanism mounted along the outer jacking post member, while mechanically-operated by the safety lever;
(53) FIG. **3**I is a first perspective side view of the trailer jack and transport system of a modified embodiment of the present invention shown in FIG. **2**B, wherein a d center-pull type of transport ball braking system is provided with a mechanically-operated safety-lever shown arranged in its non-locking or non-braking mode of operation, wherein the dual-side braking shoes are spring-biased in their locking configuration by a pair of brake tension springs, and retracted in an upward manner by a pair of cables that are pulled against the brake tension springs, on the sides of the jacking post member assembly, and taken-up on a cable take-up mechanism mounted along the outer jacking post member, while mechanically-operated by a safety lever;
(54) FIG. **3**J is a second perspective view of the trailer jack and transport system of the first illustrative embodiment shown in FIG. **3**I, wherein the center-pull type of transport ball braking system is provided with a mechanically-operated safety-lever shown arranged in its locking or braking mode of operation, wherein the dual-sided braking shoes are spring-biased in their locking configuration by a pair of brake tension springs, and protracted in an down manner while the pair of cables are not pulled against the brake tension springs and are released from a cable take-up mechanism mounted along the outer jacking post member, while mechanically-operated by the safety lever;
(55) FIG. **4**A is a perspective view of the trailer jack and transport subsystem of a modified embodiment of the present invention shown removed from its trailer and arranged in its transport configuration mode, in which the load-bearing transport ball is supported by three symmetrically spaced apart bearing pads as illustrated in FIGS. **4**G through **4**K, to more evenly distribute trailer loads across these three (or more) load bearing pads in an advantageous manner;
(56) FIG. **4**B is a partially fragmented view of the trailer jack and transport subsystem shown in FIG. **4**A, representing a modified spherical wheel assembly of the present invention;
(57) FIG. **4**C is a first elevated side view of the trailer jack and transport system of FIG. **4**A, wherein the load-bearing transport ball is shown in greater detail supported by a set of three bearing pads mounted along and within the upper interior space of the semispherical framework;
(58) FIG. **4**D is a second elevated side view of the trailer jack and transport system of FIG. **4**A, wherein the load-bearing transport ball is shown in greater detail supported by a set of three bearing pads mounted along the upper space of the semispherical framework;
(59) FIG. **4**E is a perspective exploded view of the trailer jack and transport system shown in FIG. **4**A, wherein the load-bearing transport ball is shown supported by three symmetrically-spaced apart bearing pads mounted between the interior surfaces of the semispherical framework and the outer surface of the transport ball;
(60) FIG. **4**F is an elevated side exploded view of the trailer jack and transport system shown in FIG. **4**A, wherein the load-bearing transport ball is shown supported by three symmetrically-spaced apart bearing pads mounted between the interior surfaces of the semispherical framework and the outer surface of the transport ball, retained by the ball retainer ring structure;
(61) FIG. **4**G is a perspective view of the three symmetrically-spaced apart bearing pads mounted about the outer surface of the load-bearing transport ball, removed from the semispherical framework structure for purposes of illustration;
(62) FIG. **4**H is an elevated side view of the three symmetrically-spaced apart bearing pads mounted about the outer surface of the load-bearing transport ball shown in FIG. **4**G, removed from the semispherical framework structure for purposes of illustration;
(63) FIG. **4**I is a cross-sectional side view of the three symmetrically-spaced apart bearing pads mounted about the outer surface of the load-bearing transport ball, taken along the viewing line **4**I-

4I shown in FIG. **4**G;
(64) FIG. **4**J a first perspective view of the spatial arrangement of bearing surfaces (e.g., ball bearing pads) shown employed in FIGS. **4**A and **4**G, with the load-bearing transport ball and spherical framework both removed for purposes of exposition and illustration purposes;
(65) FIG. **4**K is a second perspective view of the three symmetrically-spaced apart bearing pads shown mounted in FIG. **4**G, with the load-bearing transport ball and spherical framework both removed for purposes of exposition and illustration purposes
(66) FIG. **5**A is a perspective view of a trailer equipped with the trailer jack and transport system of the first illustrative embodiment of the present invention, shown being hitched to a mobile vehicle having a trailer hitch and ball for connection with the end of the trailer, and the trailer jack and transport system mounted to the trailer frame and having an auto-tilting rotational mounting mechanism according to the present invention;
(67) FIG. **5**B is a plan view of the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **5**A, taken along line **5**B-**5**B in FIG. **5**A, showing the set of rotational blocks comprising rotational mechanism being disposed at an angle of tilt of about 5 degrees to accommodate the semispherical framework of the system when rotated into its storage/transport configuration;
(68) FIG. **5**C is a perspective view of the trailer shown in FIG. **5**A, equipped with the trailer jack and transport system arranged in its storage configuration and hitched to a motor vehicle demonstrating the clearance created for the semispherical framework of the system by the angled rotational mechanism, creating an acute angle between the plane of the trailer and the plane of the jacking plane;
(69) FIG. **5**D is a plan view of the trailer, trailer jack and transport system and tow hitch-equipped motor vehicle shown in FIG. **5**C;
(70) FIG. **5**E is a plan view of the trailer shown in FIG. **5**D, with the trailer jack and transport system arranged in its storage configuration depicting the angled orientation of the trailer jack and transport system relative to the trailer as created by the set of rotational blocks comprising the rotational mechanism being disposed at an angle of tilt;
(71) FIG. **5**F is the plan view of the trailer jack and transport system shown in FIG. **5**E, with the trailer jack and transport system arranged in its storage configuration highlighting the plane of the trailer and the trailer jack and transport system rotational plane, an about 5-degree acute angle between the jack rotation plane and the trailer plane, as shown;
(72) FIG. **6**A is a perspective view of a portion of a trailer equipped with the trailer jack and transport system of the present invention, having a telescoping rotational mounting mechanism, and shown arranged in its transport configuration so that the jacking post is arranged perpendicular to the plane of the trailer frame;
(73) FIG. **6**B is a perspective view of a portion of the trailer equipped with the trailer jack and transport system of the present invention shown in FIG. **6**A, having a telescoping rotational mounting mechanism, and shown being rotatably arranged in its storage configuration so that the jacking post is arranged parallel to the plane of the trailer frame;
(74) FIG. **6**C is a plan view of the trailer shown in FIG. **6**A, equipped with the trailer jack and transport system of the present invention, having a telescoping rotational mounting mechanism, and shown arranged in its transport configuration so that the jacking post is arranged perpendicular to the plane of the trailer frame and its telescoping rotational mounting mechanism in the retracted position;
(75) FIG. **6**D a plan view of a portion of the trailer shown in FIG. **6**B, equipped with the trailer jack and transport system of the present invention, having a telescoping rotational mounting mechanism, and shown being rotatably arranged in its storage configuration so that the jacking post is arranged parallel to the plane of the trailer frame and its telescoping rotational mounting mechanism in the extended position, creating clearance for the semispherical structure of the trailer

jack and transport system;

(76) FIG. **6**E is a perspective view of the telescoping rotational mounting mechanism employed in the trailer jack and transport system of FIG. **6**A, arranged in its retracted configuration;

(77) FIG. **6**F is a cross-sectional view of the telescoping rotational mounting mechanism employed in the trailer jack and transport system of FIG. **6**A, taken along line **6**F-**6**F in FIG. **6**E;

(78) FIG. **6**G is a perspective view of the telescoping rotational mounting mechanism employed in the trailer jack and transport system of FIG. **6**A, arranged in its protracted or extended configuration;

(79) FIG. **6**H is a cross-sectional view of the telescoping rotational mounting mechanism employed in the trailer jack and transport system of FIG. **6**A, taken along line **6**H-**6**H in FIG. **6**G;

(80) FIG. **7**A a perspective view of a portion of the trailer equipped with the trailer jack and transport system of the present invention as shown in FIG. **2**B, having a rotational mounting mechanism and modified with a hinged jacking post member, being shown rotatably arranged in its transport configuration so that the jacking post is arranged perpendicular to the plane of the trailer frame;

(81) FIG. **7**B is a perspective view of a portion of the trailer equipped with the trailer jack and transport system of the present invention as shown in FIG. **7**A, having a rotational mounting mechanism and modified with a hinged jacking post member, and being shown rotatably arranged in its storage configuration so that the jacking post is arranged in the plane of the trailer frame and arranged so that its hinge allows the semispherical framework to adapt to the storage space provided in this configuration;

(82) FIG. **7**C is a plan view of a portion of the trailer equipped with the trailer jack and transport system of the present invention as shown in FIG. **7**B, having a rotational mounting mechanism and modified with a hinged jacking post member, and being shown rotatably arranged in its storage configuration so that the jacking post is arranged parallel to the plane of the trailer frame and arranged so that its hinge allows the semispherical framework to adapt to the storage space provided in this configuration;

(83) FIG. **7**D is an enlarged view of the trailer jack and transport system of the present invention as shown in FIG. **7**C, wherein the jacking post is arranged parallel to the plane of the trailer frame and arranged so that its hinge allows the semispherical framework to adapt to the storage space provided in this configuration;

(84) FIG. **7**E is a plan view of the trailer jack and transport system of the present invention as shown in FIG. **7**A, having a rotational mounting mechanism and modified with a hinged jacking post member, and being shown rotatably arranged in its transport configuration, but removed from its trailer for purposes of illustration;

(85) FIG. **7**F is an enlarged view of the trailer jack and transport system of the present invention as shown in FIG. **7**E, wherein the hinged jacking post is arranged perpendicular to the plane of the semispherical framework with its jacking post hinge in a closed state;

(86) FIG. **8**A is a perspective view of the trailer jack and transport subsystem of the present invention arranged shown removed from a trailer, and having a cylindrical ring structure with a torque generation system removably attached to the end portion of the semispherical framework for use in rotating the cylindrical portion and extending its position along the semi-spherical framework, so as to lift the load-bearing transport ball from the ground surface and prevent rolling and/or movement;

(87) FIG. **8**B is a side exploded view of the trailer jack and transport subsystem of the present invention in FIG. **8**A, showing all of its components including semispherical framework, bearing surfaces, bearing surface retention ring structure, jacking post member and rotational mounting mechanism, a set of braking shoes for gripping the load-bearing transport ball, and a cylindrical ring structure with a torque generation system for lifting the load-bearing ball off the ground for long term storage applications;

(88) FIG. **8**C is a perspective view of the cylindrical parking ring structure (i.e. extendible foot) having a first set of foldable torque generating handles adapted for arranging the cylindrical ring structure in a non-rolling configuration, and a second set of foldable torque generating handles adapted for arranging the cylindrical ring structure in a rolling configuration, wherein the first set of handles are being used to arrange the ring structure in its non-rolling configuration;
(89) FIG. **8**D is a perspective view of the cylindrical parking ring structure (i.e. extendible foot) having a first set of foldable torque generating handles adapted for arranging the cylindrical ring structure in a non-rolling configuration, and a second set of foldable torque generating handles adapted for arranging the cylindrical ring structure in a rolling configuration, wherein the second set of handles are being used to arrange the ring structure in its rolling configuration;
(90) FIG. **8**E is an elevated side view of the trailer jack and transport system of the present invention as shown in FIG. **8**A, wherein the first set of handles are being used to arrange the cylindrical ring structure in its non-rolling configuration;
(91) FIG. **8**F is an elevated side view of the trailer jack and transport system of the present invention as shown in FIG. **8**A, wherein the second set of handles are being used to arrange the cylindrical ring structure in its rolling configuration;
(92) FIG. **9**A is a perspective view of the trailer jack and transport system of the present invention as shown mounted on a boat trailer with an accessory cylindrical ring structure (i.e., “parking foot”) snapped onto the semispherical structure to create a stationary system;
(93) FIG. **9**B is an elevated side view of the trailer jack and transport system of the present invention as shown in FIG. **9**A mounted on a boat trailer, wherein the parking foot accessory is snapped onto the bottom portion of the bearing surface retention ring structure to create a stationary system;
(94) FIG. **9**C is a perspective view of the trailer jack and transport system of the present invention as shown in FIG. **9**A, removed from any trailer for purposes of illustration, and provided with a bracket supporting the parking foot accessory that has been removed from the semispherical structure;
(95) FIG. **9**D is an exploded view of the trailer jack and transport system of the present invention as shown in FIG. **9**C, comprising the primary components employed in prior described embodiments of the present invention, including an accessory cylindrical parking foot adapted for snap-on attachment to the bottom portion of the bearing surface retention ring structure;
(96) FIG. **9**E is a front elevated side view of the trailer jack and transport system of the present invention shown in FIG. **9**C, showing its cylindrical parking foot device stored on the jacking post portion of the system;
(97) FIG. **9**F is a side elevated side view of the trailer jack and transport system of the present invention shown in FIG. **9**C, showing its cylindrical parking foot device stored on the bracket fixedly attached to the jacking post portion of the system;
(98) FIG. **9**G is a first perspective view of the trailer jack and transport system of the present invention shown in FIG. **9**, showing its cylindrical parking foot device removed from the bracket of the jacking post portion of the system;
(99) FIG. **9**H is an elevated front view of the trailer jack and transport system of the present invention shown in FIG. **9**C, showing its cylindrical parking foot device removed from the jacking post portion of the system;
(100) FIG. **9**I is a top perspective view of the removable cylindrical parking foot device used with the trailer jack and transport system of the present invention shown in FIG. **9**C;
(101) FIG. **9**J is a bottom perspective view of the removable parking foot device used with the trailer jack and transport system of the present invention shown in FIG. **9**C;
(102) FIG. **9**K is a first elevated side view showing the removable cylindrical parking foot device being connected to the bottom portion of the trailer jack and transport system of the present invention shown in FIG. **9**C;

(103) FIG. **9**L is a second elevated side view showing the removable cylindrical parking foot device being snap-connected to the bottom portion of the trailer jack and transport system of the present invention shown in FIG. **9**C;
(104) FIG. **10**A is a perspective view of the trailer jack and transport system of the present invention in FIG. **2**A shown mounted on a boat trailer, wherein an anti-rolling disc (i.e., chock) is placed beneath its load-bearing transport ball to prevent rolling or other movement;
(105) FIG. **10**B is an elevated side view of the trailer jack and transport system of the present invention in FIG. **2**A shown mounted on a boat trailer, wherein an anti-rolling disc is placed beneath its load-bearing transport ball to prevent rolling or other movement;
(106) FIG. **10**C is a perspective view of the trailer jack and transport system of the present invention in FIG. **2**A shown removed from a trailer, its spherical wheel chock disc is shown stored on the jacking post member of the system;
(107) FIG. **10**D is an elevated front view of the trailer jack and transport system of the present invention in FIG. **2**A shown removed from a trailer, with its spherical wheel chock disc shown stored on the bracket of the jacking post member of the system;
(108) FIG. **10**E is an elevated side view of the trailer jack and transport system of the present invention in FIG. **2**A shown removed from a trailer, with its spherical wheel chocking disc shown stored on the bracket of the jacking post member of the system;
(109) FIG. **10**F is a front perspective view of the spherical wheel chocking disc removed from storage on the jacking post member of the system, and applied to a ground surface, for preventing movement of the load-bearing transport ball supported within the trailer jack and transport system of the present invention shown in FIG. **10**A;
(110) FIG. **10**G is a rear perspective view of the spherical wheel chock shown in FIG. **10**F;
(111) FIG. **10**H is a perspective partially cut away view of the trailer jack and transport system of the present invention as shown in FIG. **10** shown with its load-bearing transport ball supported within the spherical wheel chocking disc, preventing movement of the system relative to the ground surface;
(112) FIG. **11**A is a perspective view of the personal watercraft (PWC) and trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, wherein a hinged or ball-and-socket push-bar type handle is mounted on the jacking post member of the system, to enable a human operator to simply push and move the trailer in a desired direction while the trailer jack and transport system is arranged in its transport configuration (i.e. mode of operation);
(113) FIG. **11**B is a perspective view of the modified trailer jack and transport system of FIG. **11**A, shown arranged in its transport mode, removed from any trailer system or assembly, and having a push-type handlebar accessory mounted on the jacking post assembly;
(114) FIG. **11**C is a partially fragmented perspective view of the push-type handlebar accessory mounted on the jacking post assembly shown in FIG. **11**B;
(115) FIG. **11**D is partially fragmented perspective view of a trailer to which the trailer jack and transport system of FIG. **11**A is mounted, showing the push-type handlebar assembly configured for and being manually pushed in a forward direction (e.g., towards trailer hitch);
(116) FIG. **11**E is partially fragmented perspective view of a trailer to which the trailer jack and transport system of FIG. **11**A is mounted, showing the push-type handlebar assembly configured for and being manually pushed in a reversed direction (e.g., away from trailer hitch);
(117) FIG. **11**F is an elevated side view of the trailer jack and transport system shown mounted in FIG. **11**A, showing its push-type handlebar assembly being configured for storage;
(118) FIG. **11**G is a perspective partially-fragmented view of the trailer jack and transport system shown mounted in FIG. **11**A, showing its push-type handlebar assembly arranged in its storage configuration;
(119) FIG. **11**H is a perspective partially fragmented view of the trailer jack and transport system

shown mounted in FIG. **11**A, showing its push-type ball assembly configured for and being manually pushed in a forward direction (e.g., toward the trailer hitch);
(120) FIG. **11**I is a perspective view of the trailer jack and transport system of FIG. **11**H showing the push-type ball assembly for manually pushing, pulling and/or steering the trailer equipped with the trailer jack and transport system;
(121) FIG. **11**J is an elevated side view of the trailer jack and transport system shown in FIG. **11**I:
(122) FIG. **12**A is a perspective view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its transport mode and being used to transport and jack a trailer during the hitching of the trailer to a trailer hitch and ball mounted on a mobile vehicle such as a pickup truck;
(123) FIG. **12**B is a perspective view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its storage mode, after the trailer has been completely hitched to of the trailer hitch of the mobile vehicle;
(124) FIG. **12**C is a plan view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its transport mode and being used to transport and jack a trailer during the hitching of the trailer to a trailer hitch mounted on a mobile vehicle such as a truck;
(125) FIG. **12**D is a plan view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its storage mode, after the trailer has been completely hitched to the trailer hitch of the mobile vehicle;
(126) FIG. **12**E is an elevated front plan view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its transport mode, showing that the side-mounted trailer jack and transport system is physically offset from the trailer frame enabling the use of a simple rotational mounting mechanism, without telescopic or tilting mechanisms, and providing a trailer tongue clearance zone as shown;
(127) FIG. **13**A is a perspective view of the trailer jack and transport system of the alternative embodiment of the present invention whereby the jacking post member is tangent to the semispherical structure of the system (i.e., “offset”), shown arranged in its transport mode, while removed from any trailer, with its load-bearing transport ball contacting the ground surface;
(128) FIG. **13**B is a first elevated side view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its transport mode, while removed from a trailer;
(129) FIG. **13**C is a second elevated side view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, shown arranged in its transport mode, while removed from a trailer;
(130) FIG. **13**D is an exploded perspective view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, comprising a semispherical framework integrated to an elongated jacking post member assembly having a rotational mounting and jacking mechanism, a set of five ball-bearing bearing surface pads mounted within a bearing pad mounting ring structure, a load-bearing transport ball, a set of braking shoes, and a transport ball retention ring structure;
(131) FIG. **13**E is an elevated side exploded view of the trailer jack and transport system of the alternative embodiment of the present invention shown in FIG. **13**A, comprising a semispherical framework integrated to an elongated jacking post member assembly having a rotational mounting mechanism, a set of five ball-bearing bearing surface pads mounted within a bearing pad mounting ring structure, a load-bearing transport ball, a set of braking shoes, and a transport ball retention ring structure;
(132) FIG. **14**A is a first perspective view of a trailer equipped with a trailer jack and transport system of the present invention centrally mounted within the center of the A-frame portion of the trailer frame, and arranged in its transport configuration, by lowering the load-bearing transport

ball within its semispherical framework to contact the ground surface and showing the direction of travel necessary to position the trailer coupler above the trailer hitch ball in preparation for coupling the trailer to the mobile vehicle's trailer hitch;
(133) FIG. **14**B is a second perspective view of the trailer equipped with the trailer jack and transport system of the present invention shown in FIG. **14**A, with the trailer coupler positioned above the trailer hitch ball and being lowered onto the same during the hitching process by rotating the hand-crank mechanism to raise the jacking post member and correspondingly lower the trailer hitch;
(134) FIG. **14**C is a side view of the A-frame trailer equipped with the trailer jack and transport system of the present invention shown in FIGS. **14**A and **14**B, showing the semispherical framework and transport ball stored within the A-frame portion of the trailer frame, after the trailer and motor vehicle are hitched together and the external jacking post member is raised using its spring-loaded coarse adjusting pin and socket feature mounted on the A-frame mount;
(135) FIG. **14**D is a perspective view of FIG. **14**C, namely the trailer equipped with the trailer jack and transport system of the present invention shown in FIGS. **14**A and **14**B with the semispherical framework and transport ball stored within the A-frame portion of the trailer frame and its own A-frame support structure, after the trailer and motor vehicle are hitched together;
(136) FIG. **14**E is a perspective view of the dual-axle A-frame trailer equipped with the trailer jack and transport system of the present invention shown in FIGS. **14**A and **14**B, with the semispherical framework and load-bearing transport ball stored within the A-frame jack and transport system support structure which creates ground clearance when in its storage mode, shown without any motor vehicle for illustrative purposes;
(137) FIG. **14**F is a perspective view of the dual-axle A-frame utility trailer equipped with the trailer jack and transport system of the present invention shown in FIG. **14**E, with its semispherical framework and load-bearing transport ball lowered from the A-frame portion of the trailer frame and engaging the ground surface, so that the trailer can be manually transported and/or jacked up for hitching up with the trailer hitch mounted on the rear of a motor vehicle;
(138) FIG. **14**G is an elevated side view of the dual-axle A-frame utility trailer equipped with the trailer jack and transport system of the present invention shown in FIG. **14**F, with its semispherical framework and load-bearing transport ball lowered from the A-frame portion of the trailer frame and engaging the ground surface, so that the trailer can be manually transported or jacked up for hitching to the trailer hitch mounted on the rear of a motor vehicle;
(139) FIG. **14**H is a plan view of the dual-axle A-frame utility trailer equipped with the trailer jack and transport system of the present invention shown in FIGS. **14**F and **14**G;
(140) FIG. **15**A is a perspective view of a trailer equipped with a motorized (i.e., motor-powered) trailer jack and transport system of the present invention, shown arranged in its transport configuration and mode of operation, with a human operator manually controlling the direction of powered motion of the trailer, via the trailer jack and transport system, towards the trailer hitch of a nearby motor vehicle;
(141) FIG. **15**B is a perspective view the powered trailer jack and transport system shown in FIG. **15**A illustrating the use of the onboard mounted controllers on the top of the jacking post member, for moving the trailer in the direction of desired travel for transport and motor-driven jacking function for hitching operations;
(142) FIG. **15**C is a partially cut away perspective view of the semispherical framework portion of the motor-powered transport jack and semispherical framework portion shown in FIGS. **15**A and **15**B, showing in greater detail the three electric battery-powered motors, and rotocaster drive wheel rotors for driving omnidirectional rotation of the load-bearing transport ball supported within its semispherical framework;
(143) FIG. **16** is a schematic block diagram of the motor-powered trailer jack and transport system of FIGS. **15**A, **15**B and **15**C, shown comprising (i) a controller supporting a directional joystick,

jacking controls, and controller electronics, (ii) a controller wiring harness, and (iii) a jack and transport drive subsystem supporting a control and comminutions unit provided with gears, jack motor, battery module, solar panel and a control and communications electronics including a processor, firmware, memory, battery charging unit, a position controller, and drive motors provided with a drive motor wiring interface;

(144) FIG. **17** is a flow chart describing the primary steps involved in the method of operating the powered trailer jack and transport system with onboard controls mounted to a trailer according to the principles of the present invention, as shown in FIGS. **15**A, **15**B, and **15**C;

(145) FIG. **18**A is a perspective view of the motorized (i.e. motor-powered) trailer jack and transport system shown in FIG. **15**A, removed from the trailer and arranged its transport configuration, revealing the onboard controller mounted atop of its extendable jacking post member, and its load-bearing transport ball supported within its semispherical framework, and driven into controlled motion by its three battery-powered drive motors mounted external to the semispherical framework and controlled by the onboard controller;

(146) FIG. **18**B is a partially fragmented view of the semispherical framework portion of the powered trailer jack and transport system shown in FIG. **18**A;

(147) FIG. **18**C is a top axial view of the powered trailer jack and transport system shown in FIG. **18**A, demonstrating the spatial arrangement of the equally-spaced drive motors;

(148) FIG. **18**D is a bottom axial view of the powered trailer jack and transport system shown in FIG. **18**A;

(149) FIG. **18**E is an elevated front view of the powered trailer jack and transport system shown in FIG. **18**A;

(150) FIG. **18**F is a cross-sectional view of the powered trailer jack and transport system shown in FIG. **19**E, taken along line **18**F-**18**F;

(151) FIG. **18**G is an elevated side view of the powered trailer jack and transport system shown in FIG. **18**A;

(152) FIG. **18**H is a cross-sectional view of the powered trailer jack and transport system shown in FIG. **18**G, taken along line **18**H-**18**H;

(153) FIG. **18**I is an exploded perspective view of the powered trailer jack and transport system shown in FIG. **18**A, comprising (i) a load-bearing transport ball rotatably and freely mounted within a semispherical framework by way of a set of four ball-bearing bearing pads, (ii) a powered jacking post mounted and extending from the top central portion of the semispherical framework, and bearing a rotatable mounting mechanism around the midsection, and supporting an onboard controller, and (iii) a set of three battery-powered motors mounted on the semispherical framework, operably connection to the onboard controls, and configured for driving the transport ball into controlled motion in accordance with user operator controls provided through the onboard controller;

(154) FIG. **18**J is an exploded side view of the powered trailer jack and transport system shown in FIGS. **18**A and **18**I;

(155) FIG. **18**K shows a perspective view of the powered trailer jack and transport system shown in FIG. **18**A, provided with drive motor protective housing covers, adapted for fitting over the rotating motor drive gears for user safety protection;

(156) FIG. **18**L shows a perspective view of the powered trailer jack and transport system shown in FIG. **18**K, wherein a drive motor protective housing cover is shown being snap-fitted over one of the rotating motor drive gears for user safety protection;

(157) FIG. **18**M shows an enlarged partially-fragmented view of the powered trailer jack and transport system shown in FIG. **18**L, wherein a drive motor protective housing cover is shown being snap-fitted over one of the rotating motor drive gears for user safety protection;

(158) FIG. **18**N shows a first elevated front view of the powered trailer jack and transport system shown in FIG. **18**K;

(159) FIG. **18**O shows a second elevated side view of the powered trailer jack and transport system shown in FIG. **18**K;
(160) FIG. **19**A is a schematic illustration of a system network of the present invention, showing a person using a mobile computing device (e.g. a mobile phone) equipped with a mobile application (i.e. mobile app) of the present invention to remotely control and navigate a trailer equipped with the motorized trailer jack and transport system of the present invention, in various applications namely, (i) a truck-trailer hitching operation illustrated in FIG. **19**B, wherein the trailer is transported towards the hitching post of a truck or vehicle and then jacked up and placed on the trailer hitch/ball, and (ii) trailer parking operations illustrated in FIG. **19**C, wherein the trailer is transported from a starting position in a parking lot, across the parking lot and into an intended parking space, without the need to physically handle the trailer during the parking operation;
(161) FIG. **19**B is a schematic illustration of a motor vehicle provided with a trailer hitch located near a trailer to be hitched to the motor vehicle, wherein the trailer is equipped with the powered trailer jack and transport system of the present invention as shown in FIG. **19**A, and suitably modified so that the trailer jack and transport system employs a wireless outboard controller that is removable and hand-supportable within an operator's hand, and also remotely controllable by a mobile computing device (e.g. a mobile phone) running a suitable mobile application (APP) to enable wireless remote control operations within the trailer jack and transport system;
(162) FIG. **19**C is a plan view illustration showing the user in FIG. **19**A transporting the trailer from its stationary starting position in the parking lot, disconnected from its tow vehicle, into its target parking space using the powered trailer jack and transport system remotely controlled by either the removeable wireless outboard controller or a mobile computing device equipped with the mobile application and in wireless communication with the control and communications module aboard the motor-powered trailer jack and transport system;
(163) FIG. **20** is a schematic block diagram of the motor-powered trailer jack and transport system of FIG. **19**A, shown comprising (i) a removable remote controller supporting a directional joystick, jacking controls, controller electronics, battery module, and wireless communications, (ii) a releasable electrical connector, and (iii) a jack and transport drive subsystem supporting a control and communications unit provided with gears, jack motor, and battery module, and a control and communications electronics a processor, firmware, memory, local wireless communications, wireless wide area communication (i.e. cellular communication), drive motor position controller, a battery charging unit, a solar panel and drive motors provided with a drive motor wiring interface; and (iv) a mobile computing device (e.g. mobile phone) equipped with a mobile app
(164) FIG. **21**A shows a mobile phone type client computer system deployed on the system network of the present invention, comprising a touch-screen GUI screen;
(165) FIG. **21**B shows a tablet-type client computer system deployed on system network of the present invention, comprising a touch-screen GUI screen;
(166) FIG. **21**C shows a laptop-type client computer system deployed on the system network of the present invention, comprising a keyboard interface and GUI display screen;
(167) FIG. **21**D shows a schematic representation of the general system architecture of a mobile client system deployed on the system network of the present invention, as shown in FIGS. **21**A, **21**B and **21**C, and comprising the system components: a Processor(s); a Memory Interface; Memory for storing Operating System Instructions, Electronic Messaging Instructions, Communication Instructions, GUI Instructions, Sensor Processing Instructions, Phone Instructions, Web Browsing Instructions, Media Processing Instructions, GPS/Navigation Instructions, Camera Instructions, Other Software Instructions, and GUI Adjustment Instructions; Peripherals Interface; Touch-Screen Controller; Other Input Controller(s); Touch Screen; Other Input/Control Devices; I/O Subsystem; Other Sensor(s); Motion Sensor; Light Sensor; Proximity Sensor; Camera Subsystem; Wireless Communication Subsystem(s); and Audio Subsystem;
(168) FIG. **22** is a schematic illustration showing the use of the wireless hand-supportable remote

controller removed from the powered trailer jack and transport system of the present invention mounted to a trailer, and being used to remotely control the jack and transport movement operations of the system, when transporting and/or hitching the trailer to a motor vehicle or the like, as the use case may be;
(169) FIG. **23** is a flow chart describing the primary steps involved in the method of remotely controlling the powered trailer jack and transport system of FIGS. **19**A, **19**B and **22** mounted to a trailer, using a removable wireless controller according to the principles of the present invention illustrated in FIG. **22**;
(170) FIG. **24** is a schematic illustration showing the use of a mobile computing device (e.g., a mobile phone) running a suitable mobile application to remotely control the jack and transport movement operations of the system when transporting and/or hitching a trailer to a motor vehicle as the use case may be;
(171) FIG. **25** is a flow chart describing the primary steps involved in the method of remotely controlling the powered trailer jack and transport system of FIGS. **19**A, **19**B, and **19**C mounted to a trailer, using a mobile computing device running a suitable mobile application according to the principles of the present invention as illustrated in FIG. **24**;
(172) FIG. **26**A is a schematic illustration of a cloud-based system network of the present invention, showing a motor vehicle provided with a trailer hitch located near a trailer to be hitched to the motor vehicle, wherein the trailer is equipped with a powered trailer jack and transport system of the present invention as shown in FIGS. **19**A through **19**B, suitably modified so that this trailer jack and transport system employs an intelligent GPS-tracked and IR-ranging controller on the trailer and a GPS-tracked and IR-ranging module about the trailer hitch on the motor vehicle, and wherein GPS-tracked and IR-ranging controller on the trailer and a GPS-tracked and IR-ranging module about the trailer hitch communicate together in a wireless manner and support automated transport, docking and hitching operations without user involvement after the user inputs an automated docking request to either the outboard controller or the mobile app running on the mobile computing device used by the user;
(173) FIG. **26**B is an elevated side view of the motor vehicle, trailer and wireless powered trailer jack and transport system shown in FIG. **26**A, showing the GPS-tracking and IR ranging module mounted about the trailer hitch and performing the automated GPS-guided docking and hitching process of the present invention by receiving GPS signals and communicating dual IR range finding signals with the trailer jack and transport system using “time of flight” (TOF) calculations principles well known in the IR distance ranging art;
(174) FIG. **26**C is an elevated side view illustrating a method of mounting the GPS-tracking and IR ranging hitch finding module (HFM) of the system of the present invention about the trailer hitch ball mounted on the rear end of the motor vehicle in the system of FIG. **26**A, intended to hitch to a trailer equipped with the trailer jack and transport system of the present invention;
(175) FIG. **26**D is a perspective view of the GPS-tracking and IR ranging hitch finding module (HFM) of the system of the present invention, showing its centrally positioned mounting aperture, through which the trailer ball mounting thread is allowed to pass and then secured by a suitable nut;
(176) FIG. **26**E is an elevated side view of the GPS-tracking and IR ranging hitch finding module (HFM) of the present invention, showing its IR light transmission window;
(177) FIG. **26**F is a cross-sectional view of the GPS-tracking and IR ranging hitch finding module (HFM) of the system taken along line **26**F-**26**F in FIG. **26**E, showing its left and right IR transceiver diodes spaced apart and allowed to transmit and receive left and right channel IR signals through the IR light transmission window formed in the compact device;
(178) FIG. **26**G is a plan view of the trailer jack and transport system and hitch finding module of FIGS. **26**A and **26**B in communication with each other during automated hitch finding, tracking and docking operations, wherein two channels of line of sight IR signal communications are shown unobstructed during system operation so that the two systems can share left and right channel

distance information (L1 and L2) with the controller aboard the trailer jack and transport system and automatically generate drive control signals for the driving motors and accurately navigating the trailer towards the trailer hitch, using a hitch-finding algorithm running within the controller aboard the trailer jack and transport system;

(179) FIG. **27** is a schematic block diagram of the automated motor-powered trailer jack and transport system of FIG. **26**, interfaced with a cloud-computing environment and wireless communication infrastructure, and comprising (i) a controller module mounted upon the end of the jacking post member and supporting a directional joystick, jacking controls and automated finding, docking and hitching mode button, and controller electronics, battery storage module, and wireless communications, (ii) an electrical connector, (iii) a jack and transport drive subsystem supporting a control and comminutions unit provided with gears, jack motor, battery module, solar panel and control and communications electronics including a dual-channel IR finding transceiver, a GPS receiver, wireless wide area communications, local wireless communication, a battery charging unit, a position controller, and drive motors provided with a drive motor wiring interface, (iv) a mobile computing device equipped with a suitable mobile application (v) cloud computing and wireless data communication infrastructure and (vi) a wireless hitch finder supporting a GPS receiver, local wireless communication, dual-channel IR finding transceiver, docking switch, processor, firmware, memory and a battery module;

(180) FIG. **28** is a flow chart describing the primary steps involved in the method of automated docking and hitching of a trailer to a hitching post on a motor vehicle, using an automated and electrically-powered trailer jack and transport system of FIGS. **26**A and **27** mounted to the trailer, initiated by depressing a hard-key button on the outboard controller, and operated according to the principles of the present invention, as illustrated in FIG. **27**;

(181) FIG. **29** is a flow chart describing the primary steps involved in the method of automated docking and hitching of a trailer to a trailer hitch on a motor vehicle, using an automated and electrically-powered trailer jack and transport system of FIGS. **26** and **27** mounted to the trailer, initiated by depressing a button on the mobile computing device app controller, and operated according to the principles of the present invention, as illustrated in FIG. **27**;

(182) FIG. **30**A is a schematic illustration of a cloud-based system network, showing a person standing in a marina parking lot near a parked vehicle and boat trailer equipped with the intelligent GPS-tracking motorized trailer jack and transport system of the present invention, that is remotely-controllable using a mobile computing system, such as a smartphone running a special mobile application, and operational in a programmable mode of operation including an automated trailer parking method illustrated in FIG. **32**, wherein the trailer is equipped with a GPS-tracking motorized trailer jack and transport system of the present invention having a GPS-tracking trailer module mounted on the rear of the trailer and in wireless (Bluetooth) communication with the GPS-tracking motorized trailer jack and transport system mounted on the front of the trailer, and automatically transported from a starting position in a parking lot, across the parking lot and into an intended parking space, using the GPS coordinates of the trailer's boundaries and destination parking space captured prior to automated parking operations so as to enable automated GPS-guided transport of the trailer from a starting position in a parking lot, across the parking lot, and into an intended destination parking space, employing GPS-tracking and navigation methods without the need to physically handle the trailer during the parking operation;

(183) FIG. **30**B is a perspective illustration of the automated trailer jack and transport system shown in FIG. **24**, configured and working in cooperation with a GPS-tracking and LIDAR-mapping navigator beacon module of the present invention mounted to the rear of the trailer, so as to establish and maintain a virtual trailer navigation axis, passing through the trailer, and about which the spatial boundaries of the trailer are defined and managed within local databases and cloud-based servers, for use to park the trailer into a destination parking space while avoiding collisions with neighboring objects with a GPS-tracking and LIDAR-mapping navigator beacon

module of the present invention;
(184) FIG. **30**C is a perspective of the GPS-tracking and LIDAR-mapping (i.e., collision avoidance) navigator beacon module of the present invention shown in FIGS. **30**A and **30**B;
(185) FIG. **31** is a schematic block diagram of the automated motor-powered trailer jack and transport system of FIGS. **30**A through **30**C, interfaced with a cloud-computing environment and wireless data communication infrastructure, and comprising (i) a wireless controller module mounted upon the end of the jacking post member and supporting a directional joystick, jacking controls and automated finding, docking and hitching mode button, and controller electronics, battery storage module, and wireless communications, (ii) a releasable electrical connector, (iii) a jack and transport drive subsystem supporting a control and communications unit (CCU) provided with gears, jack motor, and battery module, battery recharging module, solar panel and control and communications electronics including a processor, firmware, and memory (supporting all functions of the system), a dual-channel IR hitch finding module, a battery charging unit and a position controller, GPS signal receiver with antenna for receiving and processing GPS positioning signals, WWAN module, Bluetooth module, gears, jack motor, battery, battery recharging module, and solar panel for recharging battery, and drive motors provided with drive motors and a drive motor wiring interface, (iv) a GPS navigator beacon for mounting on rear of trailer provided with a GPS receiver and antenna, a Bluetooth communication module, a processor, firmware and memory, and LIDAR ToF range finder with 180 degree FOV, and (v) cloud computing and wireless data communication infrastructure;
(186) FIG. **32** is a plan schematic illustration illustrating major steps carried out during the method of automatic parking of a registered trailer that is equipped with the automated trailer jack and transport system and GPG navigator beacon, showing the trailer's navigation vector at different stages along a projected pathway of travel towards the destination parking space;
(187) FIG. **33** is a flow chart describing the primary steps carried out during the practice of the method of automatic parking of a registered trailer equipped with the automated trailer jack and transport system of the present invention;
(188) FIG. **34**A is a plan schematic illustration illustrating major steps carried out during the registration of GPS coordinate maps of a trailer equipped with the automated trailer jack and transport system and destination parking spaces;
(189) FIG. **34**B is a plan schematic illustration illustrating major steps carried out during the registration of GPS coordinate maps of keep out zones associated with automated parking of a trailer equipped with an automated trailer jack and transport system;
(190) FIG. **35** is a flow chart describing the primary steps carried out during the method of registering of GPS coordinate maps of (i) a trailer equipped with the automated trailer jack and transport system of the present invention, (ii) destination parking spaces, and (iii) keep out (i.e., obstacle) zones, for storage in a network database system on a cloud-based system network;
(191) FIG. **36** is a plan schematic illustration illustrating major steps carried out during the storing of a home position and associated parking path traversed by a trailer equipped with the automated trailer jack and transport system during a learning mode supported by the system for subsequent use in automated trailer parking operations;
(192) FIG. **37** is a flow chart describing the primary steps carried out during the method of storing a home position and associated entrance path of a trailer equipped with the automated trailer jack and transport system of the present invention, illustrated in FIG. **36**;
(193) FIG. **38** is a plan schematic illustration illustrating major steps carried out during the method of automatic parking of a trailer equipped with the automated trailer jack and transport system of the present invention illustrated in FIGS. **30**A, **30**B and **31** and for which a home position has been learned by the system;
(194) FIG. **39** is a flow chart describing the primary steps carried out during the method of automatic parking of trailer equipped with the automated trailer jack and transport system of the

present invention illustrated in FIG. **38**;

(195) FIG. **40**A is a first perspective view of a personal watercraft (PWC) and trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q shown being used on a sandy beach surface;

(196) FIG. **40**B is a second perspective view of personal watercraft (PWC) and trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q shown on a generally smooth surface such as concrete or asphalt;

(197) FIG. **40**C is a perspective view of a personal watercraft (PWC) and trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, arranged in its transport configuration, and located near a motor vehicle which is preparing to be hitched to the trailer via its hitch ball;

(198) FIG. **41**A is a perspective view of a snowmobile trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, arranged in its transport mode on a snow-covered surface;

(199) FIG. **41**B is a perspective enlarged view of a section of the snowmobile trailer shown in FIG. **41**A, equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, while arranged in its transport configuration (i.e., mode of operation);

(200) FIG. **42**A is a perspective view of a dual-axle boat trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, arranged in its transport mode, and not hitched to the motor vehicle;

(201) FIG. **42**B is a perspective view of the dual-axle boat trailer equipped with the trailer jack and transport system of FIG. **2**B through **2**Q, shown arranged in its transport mode, with the trailer hitched to the motor vehicle;

(202) FIG. **42**C is a perspective view of the dual-axle boat trailer equipped with the trailer jack and transport system of FIG. **2**B through **2**Q, shown arranged in its storage mode, with the trailer hitched to the motor vehicle;

(203) FIG. **43**A is a perspective view of a camper trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, arranged in its transport mode, and shown on a grassy and gravel surface;

(204) FIG. **43**B is a perspective enlarged view of a section of the camper trailer shown in FIG. **43**A, equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **2**B through **2**Q, while arranged in its transport configuration (i.e., mode of operation);

(205) FIG. **44**A is a perspective view of an A-frame flatbed utility trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **14**A through **14**H, arranged in its storage mode, and shown not hitched to any motor vehicle;

(206) FIG. **44**B is a perspective view of the flatbed trailer of FIG. **44**A equipped with the trailer jack and transport system shown and described in FIGS. **14**A through **14**H, arranged in its transport mode, and not hitched to any motor vehicle;

(207) FIG. **45**A is a perspective view of an A-frame industrial tool trailer equipped with the trailer jack and transport system of the present invention shown and described in FIGS. **14**A through **14**H, arranged in its transport mode and shown on a gravel surface

(208) FIG. **45**B is a perspective view of the flatbed trailer of FIG. **44**A equipped with the trailer jack and transport system shown and described in FIGS. **14**A through **14**H, arranged in its storage mode, but not hitched to any motor vehicle;

(209) FIG. **46**A is a perspective view of a first transportable system having trapezoidal platform of rigid construction, with three corners, each having a load-bearing transportation ball construction side-mounted from the side surface of each platform corner, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be

transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.);

(210) FIG. **46**B is a perspective view of the first transportable system of FIG. **45**A, shown carrying and transporting a heavy cylindrical beer keg or barrel across a lawn surface;

(211) FIG. **47**A is perspective view of a second transportable system having a trapezoidal platform of rigid construction, with three corners, each having a load-bearing transportation ball construction of the present invention top-mounted through the top surface of each platform corner, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.);

(212) FIG. **47**B is perspective view of the second transportable system of FIG. **47**B, shown carrying and transporting a heavy cylindrical beer keg or barrel across a lawn surface;

(213) FIG. **48**A is perspective view of a third transportable system having trapezoidal platform of rigid construction, with four corners, each having a load-bearing transportation ball construction of the present invention edge-mounted to the side surface of each platform corner, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.);

(214) FIG. **48**B is perspective view of the third trapezoidal transportable platform of FIG. **48**A, shown carrying and transporting a heavy object, such as barbeque gas grill, across a lawn surface;

(215) FIG. **49**A is perspective view of a fourth transportable system having trapezoidal platform of rigid construction, with four corners, each having a load-bearing transportation ball construction of the present invention top-mounted through the top surface of each platform corner, defining a plane of load-bearing support for carrying a load (e.g. such as an ice chest, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.);

(216) FIG. **49**B is a perspective view of the third transportable system of FIG. **49**A, shown carrying and transporting a heavy object, such as an ice chest, across a lawn surface;

(217) FIG. **50**A is a perspective illustration of a remote-controlled (RC) powered transport system utilizing a configuration of multiple load-bearing spherical wheel assemblies of the present invention, shown arranged and hitched to a trailer, or other recreational or work vehicle, and being remotely controlled by a user using a mobile phone running associated mobile app configured for supporting navigation, (optionally jacking) and transport functions in accordance with the principles of the present invention;

(218) FIG. **50**B is an elevated side view of the remotely controllable (RC) powered spherical wheel transport system of the present invention, operatively coupled to the boat trailer as shown in FIG. **50**A; and

(219) FIG. **50**C is a perspective view of the remotely controllable powered spherical wheel transport system of the present invention shown in FIGS. **50**A and **50**B, but not coupled to a trailer, for purposes of illustration.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS OF THE PRESENT INVENTION

(220) Referring to the figures in the accompanying Drawings, the illustrative embodiments of the system will be described in great detail, wherein like elements will be indicated using like reference numerals.

(221) As a primary object of the present invention, each of the illustrative embodiments of the

trailer jack and transport system of the present invention **1** is equipped with a load-bearing spherical object **101** such as a load-bearing spherical transport ball that provides and supports improved rolling and steerable motion to sport, recreational and utility trailers **5**, when operating on diverse kinds of ground surfaces **90**, including pavement, dirt, sand and mud, without the use of caster-style wheels.

(222) Functioning as a wheeling device, the load-bearing spherical transport ball **101** is supported within a framework assembly **103**, preferably having a spherical or other suitable geometry, and configured to allow the spherical wheel to freely rotate with 360 degrees of freedom, in any orientation on bearing surfaces **102** of a diverse nature, that may come in contact with and engage a portion of the spherical wheel during system operation.

(223) As illustrated in FIG. **2**A, the load-bearing spherical transport ball **101** has infinite number of virtual axes of rotation passing through a single virtual centroid **141** located and embedded within the center of the spherical ball **101** about which all motion is accomplished-both directional and rotational.

(224) In one preferred embodiment, the height-adjusting jack assembly **12** is centered above the spherical wheel **101**, providing 360-degree direction of travel. In another illustrative embodiment, the height-adjusting jack assembly **12** is offset from the center of the spherical wheel **101**, reducing the complexities of rotating the wheeled jack into a storage position.

(225) By virtue of its design, the spherical wheel assembly **10** creates the opportunity to drive the wheel through battery-powered means for maximum convenience.

(226) By virtue of its design, the spherical load-bearing transport ball (e.g., spherical wheel) **101** distributes an associated tongue weight of the trailer **5** over an increasingly larger area when the ground surface **90** is soft, providing defense over gravel, sand, snow, grass, dirt and mud.

(227) These and other features and benefits of the present invention will become more apparent hereinafter and in the Claims.

(228) Before specifying the technical details of each illustrative embodiment of the trailer jack and transport system of the present invention, it will be helpful to provide an overview of a primary use case or end-user application for the system. Additional end-user applications for the trailer jack and transport system of the present invention, shown across different fields of human activity and recreational, commercial, and industrial (including military) activity over varied terrain can be found in FIGS. **40** through **45**.

(229) Specification of a Sport Trailer Carrying a Personal Watercraft (PWC), to which a Trailer Jack and Transport System of the Present Invention is Mounted and Operated According to a First Illustrative Embodiment of the Present Invention

(230) FIG. **1**A shows a sport trailer **5** carrying a personal watercraft (PWC) **50**, to which a trailer jack and transport system of the present invention **1** is mounted and operated according to a first illustrative embodiment of the present invention. As shown, the trailer jack and transport system **1** is arranged in its jack and transport configuration, with the load-bearing transport ball **101** contacting and supported upon a ground surface **90** formed on a coastal region or beachfront made from sand and crushed rock.

(231) FIG. **1**B shows the sports trailer **5** of FIG. **1**A supporting a personal watercraft **50** and resting on a smooth ground surface while not hitched to any motor vehicle with its free end **6** (i.e., “trailer tongue”) supported.

(232) FIG. **1**C shows a sports trailer of the present invention **5** shown in FIGS. **1**A and **1**B being hitched to a mobile vehicle (e.g., pickup truck) **9**. As shown, the trailer hitch **8** mounted on the bumper of the motor vehicle **9** is located in close proximity to the trailer coupler (ball socket) **7** mounted on the tongue of the trailer frame **6**, while the trailer **5** is jacked and transported by the trailer jack and transport system **1** of the present invention.

(233) FIG. **1**D shows the trailer **5** and the path along which the trailer jack and transport system **1** needs to be transported to position the coupler **7** above the ball of the trailer hitch **8** during the

process of hitching the sport trailer of the present invention to the mobile vehicle **9** as shown.

(234) FIG. **1**E shows the trailer coupler **7** positioned directly above the trailer hitch **8**, having been transported along the direct path identified in FIG. **1**D.

(235) FIG. **1**F shows the relationship of the trailer coupler **7** to the trailer hitch **8** as described in of FIG. **1**E, whereby the trailer jack and transport system **1** is arranged in its transport/jacking mode of operation and is now ready to be lowered onto the ball of the trailer hitch **8**.

(236) FIG. **1**G shows the trailer tongue **6** and trailer coupler **7** of FIG. **1**F lowered onto the trailer hitch **8** through counter-clockwise motion of the jacking crank **124** and corresponding retraction of the jacking post member **121** and load-bearing spherical wheel assembly **10** to which it is fixedly attached.

(237) FIG. **1**H shows the trailer **5** of FIG. **1**C hitched up to the motor vehicle **9**, while the trailer jack and transport system **1** is shown rearranged into its storage mode of operation, closely arranged against the frame of the trailer.

(238) Specification of the Trailer Jack and Transport Subsystem of the Present Invention

(239) FIG. **2**A shows the trailer jack and transport system of the present invention **1** arranged shown removed from a trailer, and in which a coordinate reference system **140** is symbolically embedded for referencing the omni-directional motion of the load-bearing transport ball **101** within its spherical wheel assembly **10**. As shown, the coordinate reference system **140** is shown characterized by polar rotational coordinate directions **142** and **143**, defined about a virtual center located at point **141** in the center of the spherical wheel assembly **10**, mounted to a rotatable extendable hand-operated jacking post member assembly **12**.

(240) FIG. **2**B shows the trailer jack and transport system of the present invention **1** shown removed from any trailer, while arranged in its jack and transport configuration, with its load-bearing transport ball contacting the ground surface. As shown, the trailer jack and transport system **1** comprises: (i) a spherical wheel assembly **10** having a semispherical framework **103** supporting a rigid load-bearing transport ball **101** rotatably supported by bearing surfaces **102-1** through **102-5** mounted in the semispherical framework **103** and ring structure **104**, and retained in the framework by a ball retainer ring structure **105**; (ii) a set of braking frames **106-1** and **106-2** mounted between the semispherical framework **103** and the load-bearing ball **101** and each having an aperture for the bearing surfaces **102-2** and **102-4**, and having thumb or foot-slidable levers **106**A-**1** and **106**A-**2** extending through the semispherical framework **103**, and operable to be arranged in a braking-configuration or a non-braking configuration; and (iii) a hand-operated and rotatably jacking post assembly **12** rigidly mounted to the top center of the semispherical framework **103** by welding or other fastening means, and having a rotatable mounting mechanism **123** that allows the outer post member **122** to rotate through at least 90 degrees of movement, from a storage configuration shown in FIGS. **7**B, **7**C and **7**D, to a transport configuration shown in FIG. **7**A, and a hand-crank **124** that extends or retracts the inner jacking post **121** relative to the outer jacking post member **122** in response to rotation of the hand-crank **124**.

(241) FIG. **2**C shows the trailer jack and transport system of the first illustrative embodiment 1 in FIG. **2**B. As shown, the load-bearing transport ball **101** is shown in greater detail (i) supported by a set of bearing pads **102-5** mounted directly beneath the top center of the semispherical framework **103** and bearing pads **102-1** through **102-4** mounted along the perimeter of the interior space of the semispherical framework **103** and ring structure **104** and extending from its bottom aperture to allow physical engagement with and rolling support on a ground surface, and (ii) embraced within a dual-sided ball braking system **106-1** and **106-2** mounted between the load-bearing transport ball **101** and the interior surfaces of the semispherical framework **103**.

(242) FIG. **2**D shows the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the dual-sided ball braking system **106** is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball and the interior surfaces of the semispherical framework, while the extendable jacking post member **121** is shown

extended and in a jack mode of operation.
(243) FIG. **2**E shows the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the dual-sided ball braking system **106** is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball **101** and the interior surfaces of the semispherical framework **103**, while the extendable jacking post member **121** is shown extended and in a jack mode of operation.
(244) FIG. **2**F shows the semispherical framework portion of the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the dual-sided ball braking system **106** is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball **101** and the interior surfaces of the semispherical framework **103**.
(245) FIG. **2**G shows the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the dual-sided ball braking system **106** is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball **101** and the interior surfaces of the semispherical framework **103**.
(246) FIG. **2**H shows the semispherical framework portion **103** of the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the dual-sided ball braking system **106** is shown arranged in its non-braking (i.e., unlocked) mode of operation, between the load-bearing transport ball **101** and the interior surfaces of the semispherical framework **103**.
(247) FIG. **2**I shows the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the system comprises: a semispherical framework **103** and ring structure **104** for supporting a set of bearing surfaces (e.g. ball bearing pads) **102-1** through **102-5** between the interior surfaces of the semispherical framework **103** and a load-bearing transport ball **101** supported by bearing surfaces **102**, and a dual-sided ball braking system **106-1** and **106-2** mounted about the load-bearing transport ball **101**, all of which is captured by the ball retainer ring structure **105**, retaining the transport ball **101** in manner that allows it to rotate freely.
(248) FIG. **2**J shows the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, the system comprises: a semispherical framework **103** for supporting a set of bearing surfaces (e.g. ball bearing pads) **102-1** through **102-5** between the interior surfaces of the semispherical framework **103** and a load-bearing transport ball **101** supported by bearing surfaces **102-1** through **102-5** using support ring structure **104**, and retaining ring structure **105**, also within said assembly is a dual-sided ball braking system **106-1**, **106-2** mounted about the load-bearing transport ball **101** while the transport ball **101** is retained within the semispherical framework **103** and the bearing pads **102** arranged about its perimeter;
(249) FIG. **2**K shows the load-bearing transport ball supported by the bearing surfaces (e.g. ball-bearing pads) installed in the trailer jack and transport system of the first illustrative embodiment **101** shown in FIG. **2**B, shown without its semispherical framework **103**, dual-sided ball braking system **106**, pad mounting ring structure **104**, and ball retaining ring structure **105**, each being removed for purposes of exposition and illustration of how the bearing pads **102-1** through **102-5** engage the surface of the load-bearing transport ball **101** to enable and allow free low-frictional rotational movement within the spherical wheel assembly **10** of the system **1**.
(250) FIG. **2**L shows the load-bearing transport ball **101** supported by the bearing surfaces (e.g., ball-bearing pads) **102-1** through **102-5** installed in the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B, shown without its semispherical framework **103**, dual-sided ball braking system **106-1**, **106-2**, pad mounting ring structure **104**, and retaining ring structure **105** each removed for purposes of exposition and illustration.
(251) FIG. **2**M is a cross-section of the FIG. **2**L showing the concentric bearing surfaces **102** supporting the load-bearing transport ball **101** taken from a few along **2**M-**2**M shown in FIG. **2**K
(252) FIG. **2**N shows the set of bearing surfaces **102-1** through **102-5** shown employed in FIGS. **2**K and **2**L, with the load-bearing transport ball **101** removed for purposes of exposition and

illustration purposes.
(253) FIG. **2**O shows a single bearing surface engaging with a portion of the load-bearing transport ball **101** employed in the trailer jack and transport system of the first illustrative embodiment **101** shown in FIG. **2**B.
(254) FIG. **2**P shows one of the five (5) bearing surfaces (e.g., ball bearing pads) **102** installed in the trailer jack and transport system of the first illustrative embodiment **101** shown in FIG. **2**B, to rotatably support the load-bearing transport ball **101** in an omni-directional manner.
(255) FIG. **2**Q shows one of the five (5) bearing surfaces (e.g., ball bearing pads) **102-1** through **102-5** installed in the trailer jack and transport system of the first illustrative embodiment 1 shown in FIG. **2**B. As shown, each bearing pad **102** comprises: a set of ball bearings **102**B mounted between a slightly arcuate support base portion **102**A and an apertured arcuate surface plate **102**C, through which each ball bearing **102**B is permitted to project while mounted within the bearing surface pad **102**, by screws **102**F passing through support base **102**A and threaded into bearing aperture plate **102**C. Snapped thereto is a brush frame **102**D that secures bristles **102**E about the perimeter of the apertured arcuate surface plate **102**C and ball bearings **102**B.
(256) As shown, replaceable flexible bristles **102**E are mounted about the perimeter region of each pad on a user-replaceable snap-fit brush ring **102**D to scrape, flick and remove debris such as sand, dirt, and grit from entering between the bearing surfaces **102** and the exterior surface of the transport ball **101**. By removing sand and debris from entering the engaging bearing surfaces of the transport ball **101**, during normal operation of the trailer jack and transport system **1**, the bearing pads **102** are less likely to wear away and more likely to minimize frictional abrasion and extend the operating lifetime and need for servicing and/or replacement of the bearing surface pads.
(257) Specification of First Alternative Bearing Surface (e.g. Ball Bearing Pad) Adapted for Installation in the Trailer Jack and Transport System of the First Illustrative Embodiment of the Present Invention
(258) FIG. **2**R shows a first alternative bearing surface (e.g., ball bearing pad) **102′** adapted for installation in the trailer jack and transport system **1** shown in FIG. **2**B, for rotatably supporting the load-bearing transport ball **101** in an omni-directional manner.
(259) FIG. **2**S shows the rear surface of the first alternative bearing surface (e.g., ball bearing pad) **102′** shown in FIG. **2**R.
(260) FIG. **2**T shows the first alternative bearing surface (e.g., ball bearing pad) **102′** in FIG. **2**S engaging the load-bearing transport ball **101**, while removed from its semispherical framework **103** and other system components for purposes of illustration.
(261) FIG. **2**U shows the first alternative bearing surface **102** engaging the load-bearing transport ball of the present invention, from a view taken along line **2**U-**2**U in FIG. **2**T. As shown, each bearing pad **102′** comprises a base portion **102**A′ through which are formed apertures that permit steel ball bearings **102**B′ to project slightly while retained in a spring-biased **102**F′ chamber integrated with the base portion **102**A′, thereby enabling the transport ball **101** to roll about on ball bearing surfaces **102**B′ projecting from the base portion **102**A of the bearing pad arranged about the transport ball **101** within the semispherical framework **103**.
(262) As shown, replaceable flexible bristles **102**E are mounted about the perimeter region of each pad on a user-replaceable snap-fit brush ring **102**D to scrape, flick and remove debris such as sand, dirt, and grit from entering between the bearing surfaces **102** and the exterior surface of the transport ball **101**. By removing sand and debris from entering the engaging bearing surfaces of the transport ball **101**, during normal operation of the trailer jack and transport system **1**, the bearing pads **102′** are less likely to wear away and more likely to minimize frictional abrasion and extend the operating lifetime and need for servicing and/or replacement of the bearing surface pads.
(263) Specification of Second Alternative Bearing Surface (e.g. Self-Lubricating Bearing Pad) Adapted for Installation in the Trailer Jack and Transport System of the First Illustrative Embodiment of the Present Invention

(264) FIG. **2**V is a perspective front view of a second alternative bearing surface (e.g., self-lubricating plastic bearing pad, such as Nylon plastic) adapted for installation in the trailer jack and transport system of the first illustrative embodiment shown in FIG. **2**B, for rotatably supporting the load-bearing transport ball in an omni-directional manner.
(265) FIG. **2**W is a perspective rear view of the second alternative bearing surface (e.g., Nylon plastic ball bearing pad) shown in FIG. **2**V.
(266) FIG. **2**X is a perspective view of the second alternative bearing surface (e.g., ball bearing pad) in FIG. **2**V shown engaging the load-bearing transport ball of the present invention, while removed from its semispherical framework for purposes of illustration.
(267) FIG. **2**Z is a cross-sectional view of the first alternative bearing surface engaging the load-bearing transport ball of the present invention, taken along line **2**Z-**2**Z in FIG. **2**Y.
(268) As shown, a replaceable dense bristle pad **102**D″ is mounted about the perimeter region of each pad on a user-replaceable snap frame **102**C″ to scrape, flick and remove debris such as sand, dirt, and grit from entering between the self-lubricating bearing surfaces **102** and the exterior surface of the transport ball **101**. By removing sand and debris from entering the engaging bearing surfaces of the transport ball **101**, during normal operation of the trailer jack and transport system **1**, the self-lubricating bearing pads **102**″ are less likely to wear away and more likely to minimize frictional abrasion and extend the operating lifetime and need for servicing and/or replacement of the bearing surface pads.
(269) Specification of Internal Construction of Load-Bearing Transport Balls of the Illustrated Embodiments of the Present Invention
(270) FIGS. **2**Z and **2**AA shows the first illustrative embodiment of the load-bearing transport ball, characterized by having a solid core fabricated from one or more durable materials designed to withstand the expected loads bearing down on the transport ball during operation.
(271) FIGS. **2**BB and **2**CC show a second illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a multi-layer core fabricated from different layers of durable material designed to withstand the expected loads bearing down on the transport ball during operation.
(272) FIGS. **2**DD and **2**EE shows a third illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a spline-core fabricated from durable material that can withstand the expected loads bearing down on the transport ball during operation.
(273) FIGS. **2**FF and **2**GG show a fourth illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a honeycomb-core fabricated from durable material that can withstand the expected loads bearing down on the transport ball during operation.
(274) FIGS. **2**HH and **2**II show a fifth illustrative embodiment of the load-bearing transport ball of the present invention, characterized by having a hollow-core fabricated from durable material that can withstand the expected loads bearing down on the transport ball during operation.
(275) Specification of External Surface of Load-Bearing Transport Ball of the Illustrated Embodiments of the Present Invention
(276) As shown in FIGS. **2**Z through **2**II, the load-bearing transport ball **101** employs a textured or patterned surface for, among other things, channeling water to prevent slippage and sliding while rolling on the ground surface. It is also understood that said surface texture or pattern can be crafted to offer traction for engagement with the drive rotors of the powered (i.e. motorized) alternative embodiment of the present invention as taught in FIG. **15**A and shown in detail in FIG. **15**C.
(277) Specification of the Trailer Jack and Transport System, Wherein the Dual-Sided Ball Braking System is Arranged for Locking and Non-Braking Modes of Operation
(278) FIGS. **3**A, **3**C and **3**E show the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its non-locking or non-braking mode of operation, with the braking shoes retracted in an upward manner.

(279) FIGS. **3**B, **3**D and **3**F show the trailer jack and transport system of the first illustrative embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is shown arranged in its locking or braking mode of operation, with the braking shoes protracted in a downward manner.
(280) As shown, these braking shoes can be manually deployed into their respective positions using the user's hand or fingers, or foot-operations, as the case may be. Below, alternative methods of braking operation are taught with great advantage and benefit.
(281) Specification of the Trailer Jack and Transport System of the Present Invention, Wherein the Dual-Sided Side-Pull Type of Transport Ball Braking System is Provided with a Mechanically-Operated Safety-Lever for Locking and Non-Braking Modes of Operation
(282) FIG. **3**G shows the trailer jack and transport system of a modified embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is provided with a mechanically-operated side-pull cabling system with safety-lever **107**B shown arranged in its non-locking or non-braking mode of operation. As shown, the braking shoes **106**B-**1** and **106**B-**2** are spring-biased in their locking configuration by a pair of brake compression springs **107**E-**1** and **107**E-**2**, and retracted in an upward manner by a pair of cables **107**C-**1** and **107**C-**2** that are pulled against the brake compression springs **107**E-**1** and **107**E-**2**, on the sides of the jacking post member assembly **12**, and taken-up on a cable take-up mechanism **107**G mounted along the outer jacking post member **122**, while mechanically-operated by a safety lever **107**B.
(283) FIG. **3**H shows the trailer jack and transport system of modified embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is provided with a mechanically-operated side-pull cabling system with safety-lever **107**B shown arranged in its locking or braking mode of operation. As shown, the braking shoes **106**B-**1** and **106**B-**2** are spring-biased in their locking configuration by a pair of brake compression springs **107**E-**1** and **107**E-**2**, and are protracted in a downward manner when the pair of cables **107**C-**1** and **107**C-**2** are not pulled against the brake compression springs **107**E-**1** and **107**E-**2**, and the cables are released from the cable take-up mechanism **107**G mounted along the outer jacking post member **122**, while mechanically-operated by a safety lever **107**B.
(284) Specification of the Trailer Jack and Transport System of the Present Invention, Wherein the Dual-Sided Center-Pull Type of Transport Ball Braking System is Provided with a Mechanically-Operated Safety-Lever for Locking and Non-Braking Modes of Operation
(285) FIG. **3**I shows the trailer jack and transport system of a modified embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is provided with a mechanically-operated center-pull cabling system **107**D′ with safety-lever **107**B′ shown arranged in its non-locking or non-braking mode of operation. As shown, the braking shoes **106**B-**1** and **106**B-**2** are spring-biased in their locking configuration by a pair of brake tension springs **107**F-**1** and **107**F-**2**, and retracted in an upward manner by a pair of cables **107**C-**1**′ and **107**C-**2**′ that are pulled against the brake tension springs **107**F-**1** and **107**F-**2**, on the sides of the jacking post member assembly **12**, and taken-up on a cable take-up mechanism **107**G′ mounted along the outer jacking post member **122**, while mechanically-operated by a safety lever **107**B′ _.
(286) FIG. **3**J shows the trailer jack and transport system of a modified embodiment of the present invention shown in FIG. **2**B, wherein the dual-sided ball braking system is provided with a mechanically-operated center-pull cabling system with safety-lever **107**B′ shown arranged in its locking or braking mode of operation. As shown, the braking shoes **106**B-**1** and **106**B-**2** are spring-biased in their locking configuration by a pair of brake tension springs **107**F-**1** and **107**F-**2**, and are retracted in a downward manner by the tension springs **107**F-**1** and **107**F-**2** when the safety lever **107**B′ is turned to allow the cable to be released by the cable take-up mechanism **107**G′ mounted along the outer jacking post member **122**.
(287) Specification of Alternative Trailer Jack and Transport System of the Present Invention Having a Tripodal Bearing Surface System Installed about the Load-Bearing Transport Ball

(288) FIGS. **4**A through **4**F show the trailer jack and transport system of a modified embodiment of the present invention **1**′ shown removed from its trailer and arranged in its transport configuration mode. As shown, the load-bearing transport ball **101** is supported by three symmetrically spaced apart bearing pads. **102-1**, **102-2**, and **102-3** as illustrated in FIGS. **4**G through **4**K, to evenly distribute trailer tongue weight loads across these three bearing pads **102-1**, **102-2**, and **102-3** in an advantageous manner. As shown, the load-bearing transport ball **101** is supported by a set of three bearing pads **102-1**, **102-2**, and **102-3** mounted along the upper space of the semispherical framework. FIGS. **4**G through **4**K illustrate how the symmetrically-spaced apart bearing pads, mounted about the outer surface of the load-bearing transport ball **101**, allows for even distribution of loads imposed on the transport ball
(289) Specification of an Angled Rotatable Mounting Mechanism for the Trailer Jack and Transport System of the First Illustrative Embodiment of the Present Invention
(290) FIG. **5**A shows a trailer equipped with the trailer jack and transport system of the first illustrative embodiment 1, shown decoupled from a mobile vehicle **9** having a trailer hitch adapted for connection with the end of the trailer. As shown, the trailer jack and transport system **1** mounted to the trailer frame and having an auto-tilting rotational mounting mechanism **123**, and being arranged in its transport configuration, with its jacking post **122** disposed away from the frame of the trailer **5**.
(291) FIG. **5**B shows how the post rotational mounting mechanism **123** has angled plate components **123**A and **123**B stacked and rotatably mounted to bracket **123**C by way of a mounting post. As shown, the angled rotational mechanism comprises: a set of rotational blocks **123**A and **123**B mounted along a common axis and disposed at an angle of tilt of about 5 degrees to accommodate the semispherical framework of the system. The mounting bracket **123**C is welded to mounting plate **123**D that is fixed to the trailer frame **5** using bolts, welding and/or other fastening methods well known in the art.
(292) FIG. **5**C shows a trailer equipped with the trailer jack and transport system of the first illustrative embodiment 1, shown hitched to a mobile vehicle **9** equipped with a trailer hitch. As shown, the trailer jack and transport system **1** is mounted to the frame of trailer **5** and has an auto-tilting rotational mounting mechanism **123**, shown arranged in its storage configuration, with its jacking post **122** closely aligned and creating an acute angle (e.g. about 5 degrees) with the trailer frame **5** so as to offer clearance for the spherical wheel assembly **10** of the trailer jack and transport system **1**.
(293) FIG. **5**D shows the trailer **5** shown in FIG. **5**C, hitched to a motor vehicle **9** with the trailer jack and transport system **1** arranged in its storage configuration, rotated about the rotational mounting mechanism **123** so that the jacking post **122** is within the plane of the trailer frame **5** and being disposed at an angle so as to create clearance for the spherical wheel assembly **10** against the frame of trailer **5**
(294) FIG. **5**E shows the trailer **5** in FIG. **5**A, with the trailer jack and transport system **1** arranged in its storage configuration whereby the rotational mounting mechanism **123** and its angled plate components **123**A and **123**B work in concert to create clearance for the spherical wheel assembly **10** of the trailer jack and transport system **1** when rotated 90 degrees from its transport orientation. FIG. **5**F shows the trailer **5** in FIG. **5**D, with the trailer jack and transport system **1** arranged in its storage configuration, rotated 90 degrees from its transport orientation as shown in FIG. **5**A about the rotational mounting mechanism **123** with a jack rotation plane approximately 5 degrees as measured to the plane of the trailer frame **5**. The 5 degrees of tilting enabled by the rotational mounting mechanism **123** allows the semispherical framework portion **103** of the system **1** be accommodated during its storage mode.
(295) Specification of the Trailer Jack and Transport System of the Present Invention Having a Telescoping Rotational Mounting Mechanism
(296) FIG. **6**A shows a trailer **5** equipped with the trailer jack and transport system of the present

invention **1**, having a telescoping rotational mounting mechanism **123**′, and shown arranged in its transport configuration so that the jacking post **122** is arranged substantially perpendicular to the ground surface **90**.

(297) FIGS. **6**B, and **6**D show the trailer equipped with the trailer jack and transport system **1** shown in FIG. **6**A, having a telescoping rotational mounting mechanism **123**′, and shown being rotatably arranged in its storage configuration so that the jacking post **122** is arranged parallel to the plane of the trailer frame **5**. By virtue of the telescopic rotational mounting mechanism **123**′, the extended mechanism **123**′ allows the larger semispherical framework portion **103** to be accommodated when arranged in its storage mode or configuration, as so clearly shown and illustrated in FIGS. **6**B, and **6**D

(298) FIG. **6**C show the trailer equipped with the trailer jack and transport system **1** shown in FIG. **6**A in its transport mode with the telescoping rotational mounting mechanism **123**′ in its retracted position with the load-bearing spherical wheel **101** and semispherical framework portion **103** partially beneath the trailer frame **5** for minimized cantilever torque on the rotational mechanism mount **123**C′

(299) FIGS. **6**E and **6**F show the telescoping rotational mounting mechanism **123**′ employed in the trailer jack and transport system **1** of FIG. **7**A, shown without select additional parts for illustrative purposes, arranged in its retracted configuration. In the transport mode, the shortened length of the mounting mechanism **123**′ brings the semispherical framework portion **103** and transport ball **101** of the system closer to the trailer's principal plane to provide improved balance and load balancing.

(300) FIGS. **6**G and **6**H shows the telescoping rotational mounting mechanism **123**′ employed in the trailer jack and transport system of FIG. **7**A, shown without select additional parts for illustrative purposes, arranged in its protracted or elongated configuration. In the storage mode, the elongated length of the mounting mechanism **123**′ maintains the semispherical framework portion **103** and transport ball **101** of the system at the correct distance from the trailer's frame **5** to enable compact and stable storage.

(301) Specification of a Trailer Equipped with the Trailer Jack and Transport System of the Present Invention Having a Rotational Mounting Mechanism Modified with a Hinged Jacking Post Member

(302) FIG. **7**A shows the trailer **5** equipped with the trailer jack and transport system **1** as shown in FIG. **2**B, having a rotational mounting mechanism **123** and modified with a hinged jacking post member **121**′, and being shown rotatably arranged in its transport configuration so that the inner and outer jacking post members **121**′ and **122**′ is arranged perpendicular to the plane of the trailer frame **5**.

(303) FIG. **7**B shows the trailer **5** equipped with the trailer jack and transport system as shown in FIG. **7**A. The trailer jack and transport system **1** has a rotational mounting mechanism **123** that is modified with a hinged inner jacking post member **121**′, and is shown rotatably arranged in its storage configuration, so that the inner jacking post **121**′ is arranged parallel to the plane of the trailer frame **5** and arranged so that its hinge members **121**A′ and **121**B′ allow the spherical wheel assembly **10** to adapt to the storage space provided in this configuration.

(304) FIGS. **7**C and **7**D show the trailer **5** equipped with the trailer jack and transport system **1** as shown in FIG. **7**B. The trailer jack and transport system **1** has a rotational mounting mechanism **123** and modified with a hinged jacking post member **121**′, and is shown rotatably arranged in its storage configuration so that the jacking post **121**′ is arranged parallel to the plane of the trailer frame **5** and arranged so that its hinge **121**A′, **121**B′ allows the spherical wheel assembly **10** to adapt to the storage space provided in this configuration.

(305) FIG. **7**E shows the trailer jack and transport system **1** in FIG. **7**A, with its hinged jacking post member **121**′, arranged in its transport configuration. FIG. **7**F shows the trailer jack and transport system **1** of FIG. **7**E, in greater detail, with its hinged inner post member **121**′ mounted to the top central portion of the hemispherical framework **103**.

(306) Specification of the Trailer Jack and Transport System of the Present Invention in which a Torque Generation System is Provided at the End Portion of the Semispherical Framework for Rotating the Cylindrical Portion, Extending its Position Along the Framework, and Lifting the Load-Bearing Transport Ball from the Ground Surface to Prevent Rolling and/or Movement
(307) FIG. **8**A shows the trailer jack and transport system **1**″ shown removed from a trailer, and in which a torque generation system **161** is attached to the ring structure **104** for rotating a cylindrical portion and extending its position along the central axis of the spherical wheel assembly **16**. By virtue of this extension operation, the load-bearing transport ball **101** is lifted off of and away from the ground surface **90** so as to prevent rolling and/or movement along the ground surface. This device effectively prevents the trailer jack and transport system from rolling without the use of blocks or chocks.
(308) FIG. **8**B shows the trailer jack and transport system **1**″ shown removed from a trailer, showing all of its components comprising: a hemi-semispherical framework **103**; bearing surfaces **102-1** through **102-5**; a load-bearing transport ball **103**; bearing surface retention ring structure **104**; inner and out jacking post members **121** and **122**; a rotational mounting mechanism **123**; a set of brakes **106-1** and **106-2** for gripping the load-bearing transport ball **101** in the braking mode of operation; and a cylindrical parking ring structure (i.e. cylindrical support structure) **161** having an integrated torque generation system **161**A and **161**B, for use in rotating the cylindrical structure **161** to raise or lower its position along the longitudinal axis of the spherical wheel assembly **16**, and thereby lifting the load-bearing ball **101** off the ground surface for long term storage applications, and lowering back down thereunto when it is desired to transport the trailer, once again.
(309) FIG. **8**C shows the cylindrical parking ring structure (i.e., an extendible foot) comprising: a first set of foldable torque generating handles **161**A-**2** and **161**-A**4** adapted for arranging the cylindrical ring structure in a non-rolling configuration; and a second set of foldable torque generating handles **161**A-**1** and **161**-A**3** adapted for arranging the cylindrical ring structure **161** in a rolling configuration. As shown in FIG. **8**D, the second set of handles **161**A-**1** and **161**-A**3** are used to arrange the ring structure in its rolling configuration. By simply grabbing these pairs of handles, the user can simply rotate the cylindrical ring structure **161** and raise and lower the transport ball **101** off the ground surface as desired.
(310) FIG. **8**E shows the trailer jack and transport system **1**″ in FIG. **8**A, wherein the first set of handles **161**A-**1** and **161**-A**3** are used to rotate the cylindrical ring structure **161**, lower the transport ball **101** onto the ground surface **90**, and arrange the ring structure in its rolling-enabled configuration.
(311) FIG. **8**F shows the trailer jack and transport system **1**″ in FIG. **8**A, wherein the second set of handles **161**A-**2** and **161**-A**4** are used to lift the transport ball **101** off the ground surface **90** and arrange the ring structure in its rolling-disabled (i.e., non-rolling) configuration.
(312) Specification of the Trailer Jack and Transport System of the Present Invention Mounted on a Boat Trailer, Wherein a Snappable Accessory Cylindrical Ring (i.e. Stationary Foot) is Arranged to Create a Non-Rolling Load-Bearing Trailer Jack with Large Surface Area
(313) FIG. **9**A shows the trailer jack and transport system **1** in FIG. **2**B mounted on a boat trailer **5**. As shown, a snap-on stationary foot accessory **161**′ is attached to the spherical wheel assembly **10** to transform the transport system into a stationary jack, preventing rolling or other movement on the ground surface **90** and creating a circular landing area approximately the diameter of the transport ball to avoid sinking into the ground surface **90** under the weight of the trailer **5**.
(314) FIG. **9**B shows the trailer jack and transport system in FIG. **9**A mounted on the boat trailer **5** in its stationary jacking configuration whereby its load-bearing transport ball **101** is prevented from reaching the ground surface, instead the weight of trailer tongue **6** is distributed on the ground surface by the stationary foot **161**′.
(315) FIG. **9**C shows the trailer jack and transport system **1** in FIG. **9**A, removed from any trailer

for purposes of illustration, and provided with a bracket for storing the removable stationary foot **161′** when disconnected from the bottom portion of the ball retention ring structure **105**, suitably adapted for snap-fit connection and removal.

(316) FIG. **9**D shows the trailer jack and transport system **1** in FIG. **9**C, comprising: the primary components employed in prior embodiments, described above, in addition to a removable stationary foot **161′** adapted for snap-fit attachment onto the bottom portion of the ball retention ring structure **105** and the addition of a storage mounting feature **175** fixedly mounted to the outer jacking post member **122**

(317) FIGS. **9**E and **9**F show the trailer jack and transport system in FIG. **9**C, with its parking foot device **161′** stored on the storage mounting feature **175** fixed attached to the jacking post portion **122** of the system.

(318) FIGS. **9**G and **9**H show the trailer jack and transport system in FIG. **9**C, with its parking foot device removed from the jacking post portion **122** of the system.

(319) FIGS. **91** and **9**J show the removable parking foot device **161′** for use with the trailer jack and transport system **1** shown in FIG. **9**C.

(320) FIG. **9**K shows the removable parking foot device **161′** in the process of being connected to the bottom portion of the trailer jack and transport system **1** shown in FIG. **9**C.

(321) FIG. **9**L shows the removable parking foot device being snap-connected to the bottom of the spherical wheel assembly **10** of the trailer jack and transport system **1** shown in FIG. **9**C.

(322) Specification of the Trailer Jack and Transport System of the Present Invention Mounted on a Boat Trailer, Wherein an Anti-Rolling Disc is Placed Beneath its Load-Bearing Transport Ball to Prohibit Rolling or Other Movement

(323) FIGS. **10**A and **10**B show the trailer jack and transport system **1** in FIG. **2**B mounted on a boat trailer **5**. As shown, an anti-rolling disc (i.e., spherical wheel chock) **171** is placed beneath its load-bearing transport ball **101** to prohibit rolling or other movement.

(324) FIGS. **10**C, **10**D and **10**E show the trailer jack and transport system in FIGS. **10**A and **10**B removed from the trailer **5**, with its spherical wheel chock **171** shown stored on the storage mounting feature **175** fixed attached to the jacking post member **122** of the system.

(325) FIGS. **10**F and **10**G show the spherical wheel chock **171** removed from physical connection to the trailer jack and transport system, and applied to a ground surface, for preventing movement of the load-bearing transport ball **101** supported within the trailer jack and transport system **1** in FIG. **10**A when placed thereupon.

(326) FIG. **10**H shows the spherical wheel assembly **10** of the trailer jack and transport system **1** in FIG. **10**A, with its load-bearing transport ball **101** supported within the disc-like anti-rolling wheel chock **171**, preventing movement of the system relative to the ground surface.

(327) Specification of the Trailer Jack and Transport System of the Present Invention Having a Manually-Extendable Push-Type Hand Bar Assembly, for Safely Pushing a Trailer in a Desired Direction of Travel

(328) FIG. **11**A shows a modified trailer jack and transport system **1** in FIG. **1**A, arranged in its transport mode and being used to transport and jack a PWC trailer **5** located on a beach surface with a sandy ground.

(329) FIG. **11**B shows the trailer jack and transport system **1** of FIG. **11**A, arranged in its transport mode, removed from any trailer system or assembly, and having a push-type handle bar accessory **19**, with handle **193** and its rod **192** mounted on the jacking post assembly using a directional mount **191** allowing for the handle rod to be arranged in several different directions: (i) forward direction illustrated in FIG. **11**D; (ii) reverse direction illustrated in FIG. **11**E; and (iii) downward storage direction illustrated in FIGS. **11**F and **11**G.

(330) Specification of the Trailer Jack and Transport System of the Present Invention Having a Push-Type Ball Assembly for Safely Pushing a Trailer in a Desired Direction of Travel

(331) FIG. **11**H shows the trailer jack and transport system **1** shown mounted in FIG. **1**C, with a

push-type ball assembly **19′** atop the external jacking post member **122** for pushing, pulling or steering the trailer **5** to which it is attached.

(332) FIGS. **11**I and **11**J show the trailer jack and transport system **1** shown mounted in FIG. **11**H, with the push-type ball assembly **19′**.

(333) Specification of the Trailer Jack and Transport System of the Alternative Embodiment of the Present Invention with Simplified Rotational Mounting Mechanism Made Possible by Side-Mounted Jacking Post

(334) FIGS. **12**A and **12**C show the trailer jack and transport system of the alternative embodiment of the present invention **2** arranged in its transport mode and being used to transport and jack a trailer **5** during the hitching of the trailer to a hitching post mounted on a mobile vehicle such as a truck **9**.

(335) FIGS. **12**B and **12**D show the trailer jack and transport system of the alternative embodiment 2 shown in FIG. **12**A, arranged in its storage mode with the trailer **5** coupled to the trailer hitch **8** of the mobile vehicle **9** to demonstrate, by virtue of its design, the absence of interference between the spherical wheel assembly **20** and the frame of the trailer **5**. enabling compact storage.

(336) FIG. **12**E shows the trailer jack and transport system of the alternative embodiment 2 in FIG. **12**A, arranged in its transport mode, with the side-mounted trailer jack and transport system **2** offset from the trailer frame enabling the use of a simple rotational mounting mechanism, without telescopic or tilting mechanisms.

(337) Specification of the Trailer Jack and Transport System of the Present Invention Having Side-Mounted Design

(338) FIG. **13**A shows the trailer jack and transport system of the alternative embodiment 2 shown in FIG. **12**A, arranged in its transport mode, and removed from any trailer. The major difference of this system design **2** from the prior system design **1** is that the extendable jacking post assembly **22** is rigidly mounted to the side of the semispherical framework **203**, in contrast to axial mounting of the jacking post assembly **12** in system design **1** shown in FIG. **2**B.

(339) FIGS. **13**B and **13**C show the trailer jack and transport system **2** in FIG. **13**A, arranged in its transport mode, removed from any trailer.

(340) As shown in FIGS. **13**D and **13**E, the trailer jack and transport system **2** comprises: (i) a semispherical framework assembly **20** having a semispherical framework **203** supporting a rigid load-bearing transport ball **101** rotatably supported by bearing surfaces **102-1** through **102-5** mounted in ring structure **104**, and retained in the framework by a ball retainer ring structure **105**; (ii) a set of braking frames **106-1** and **106-2** mounted between the semispherical framework **203** and the load-bearing transport ball **101**, and having thumb slidable levers **106**A-**1** and **106**A-**2** extending through the semispherical framework **203**, and operable to be arranged in a braking-configuration or a non-braking configuration; and (iii) a hand-operated and rotatably jacking post assembly **22** rigidly mounted to the side of the semispherical framework **203** by welding or other fastening means, and having a rotatable mounting mechanism **223** that allows the outer post member **122** to rotate through at least 90 degrees of movement, from a storage configuration shown in FIGS. **12**B and **12**D, to a transport configuration shown in FIGS. **12**A, **12**C and **12**E, and a hand-crank **124** that extends the inner jacking post **221** relative to the outer jacking post member **122** in response to rotation of the hand-crank **124**.

(341) FIG. **13**E is an elevated side view of the trailer jack and transport system **2** shown in FIG. **13**A, comprising a semispherical framework integrated to an elongated jacking post member assembly having a rotational mounting mechanism, a set of five ball-bearing bearing surface pads mounted within a bearing pad mounting ring structure, a load-bearing transport ball, a set of braking frames and shoes, and a transport ball retention ring structure.

(342) Specification of a Trailer Equipped with a Jack and Transport System of the Present Invention Centrally Mounted within the Center of the A-Frame Portion of the Trailer

(343) FIGS. **14**A and **14**B show a trailer **55** equipped with a jack and transport system of the

present invention **3** centrally mounted within the center of the A-frame tongue portion of the trailer frame **6**, and arranged in its transport configuration, by lowering its load-bearing transport ball **101** within its semispherical framework **103** to contact the ground surface, when the coarse-adjusting pin **323**C is positioned high on the outer jacking post member **322** and the hand-crank mechanism **324** is rotated to lower the transport ball accordingly.
(344) As shown in FIGS. **14**C and **14**D, the A-frame portion of the trailer frame **6** is rotated into position with a trailer hitch **8** on a mobile vehicle **9**, and the trailer coupler **7** is lowered onto the same during the hitching process by rotating the hand-crank mechanism **324** to raise the spherical wheel assembly **10** and lower the trailer coupler **7** onto the hitch **8** o the motor vehicle **9**. The coarse adjustment pin **323**C is then disengaged and the outer jacking post member **322** is lifted until the spherical wheel assembly is arranged within the confines of the A-frame **6** and the A-frame trailer jack mount **323**A so as to provide ample ground clearance. The coarse adjustment pin **323**C is then inserted into a lower socket to securely position the trailer jack and transport system **3** in its storage position.
(345) FIG. **14**E shows the jack and transport system **3** in FIGS. **14**A and **14**B, with the semispherical framework and load-bearing transport ball stored within the A-frame portion of the trailer frame **6**, while the trailer is unhitched from any motor vehicle.
(346) FIGS. **14**F, **14**G and **14**H show the jack and transport system **3** in FIG. **14**E, with its semispherical framework **103** and load-bearing transport ball **101** lowered from the A-frame portion of the trailer frame **6** and engaging the ground surface, so that the trailer is supported for transport thereupon or can be jacked up for hitching purposes.
(347) Specification of a Motorized Jack and Transport System of the Present Invention Arranged in its Transport Configuration with a Human Operator Manually Controlling the Direction of Powered Motion of the Trailer, Via the Trailer Jack and Transport System, Towards the Hitching Post of a Nearby Motor Vehicle
(348) FIG. **15**A shows a trailer **5** equipped with a motorized (i.e. motor-powered) trailer jack and transport system **4** of an alternative embodiment of the present invention, shown arranged in its transport configuration and mode of operation, with a human operator manually controlling the direction of powered motion of the trailer, via the onboard controller **424** mounted on the top of the trailer jack and transport system, and guiding its travels towards the trailer hitch **8** on a nearby motor vehicle **9**.
(349) FIG. **15**B illustrates a trailer **5** for hitching to a motor vehicle and equipped with the powered trailer jack and transport system of the present invention **4** shown in FIGS. **15**, **16** and **17**, and illustrating the use of the onboard mounted controller **424** on the top of the jacking post member **422**, for moving the trailer **5** in the direction of desired travel for transport and hitching operations as applicable.
(350) FIG. **15**C shows the motor-powered jack and transport system spherical wheel assembly **40** in FIG. **18**A, showing in greater detail the three electric battery-powered motors **407-1**, **407-2** and **407-3** driving the load-bearing transport ball **101** supported within its semispherical framework **403**. The drive motors **407** may be spring biased, thereby maximizing and maintaining consistent engagement with the surface of the load-bearing transport ball **101**. Notably, in this embodiment, the braking system is not show as the electric motors can be operated so as to provide a safety braking function on the load-bearing transport ball **101** supported within the system **4**.
(351) Specification of a Method and System for Hitching a Trailer to a Motor Vehicle Using a Powered Trailer Jack and Transport System Having an Onboard Mounted Controller on the Top of the Jacking Post Member for Moving the Trailer in the Direction of Desired Travel for Transport and Jacking Operations
(352) FIG. **16** shows the motor-powered trailer jack and transport system **45** of FIG. **15**, comprising: (i) a controller **424** supporting a directional joystick, jacking controls, and controller electronics; (ii) a controller wiring harness **435**; and (iii) a jack and transport drive subsystem **44**

supporting a control and comminutions unit **425** provided with gears **425**C, jack motor **425**B, and battery module **425**D, and a control and communications electronics including a processor, firmware and memory **425**E, a battery charging unit **425**G and a position controller **425**H, and three drive motors **407-1** through **407-3** consisting each of a DC-powered motor **407**A and drive roller **407**C provided with a drive motor wiring interface **425**J.

(353) FIG. **17** describes the primary steps involved in the method of operating the powered trailer jack and transport system with onboard controls **45** mounted to a trailer according to the principles of the present invention.

(354) As shown in FIG. **17**, the system user starts at Block A and progresses to Block B and powers on the remote controller unit **424**.

(355) At Block C, the user controls powered transport of the trailer with the onboard joystick controller **424**B.

(356) At Block D, the user controls trailer jacking (i.e., raising and lowering relative to ground surface) with the jacking toggle switch **424**C on the remote controller **424**.

(357) At Block E, control signals are communicated to the control and communications unit (CCU) **425** via the controller wiring **425**K illustrated in FIG. **16**.

(358) At Block F, the controller determines whether or not the CCU has received an input signal from the joystick directional controller. If yes, the control process proceeds to Block G, and if not, the control process flow proceeds to Block L.

(359) At Block G, the CCU microprocessor **425**E receives input from joystick directional controller.

(360) At Block H, the CCU microprocessor **425**E processes directional information from joystick directional controller.

(361) At Block I, the CCU microprocessor **425**E sends directional information to position controller **425**H.

(362) At Block J, rotational speed information and power are transmitted to the KIWI drive motors **407-1**, **407-2** and **407-3**.

(363) At Block K, the drive motors **407-1**, **407-2** and **407-3** power and drive rotational motion of the transport ball through interaction of drive rotors **407**C-**1**, **407**C-**2** and **407**C-**3** and the surface of the spherical wheel **101** so as to enable transport of the system under the control of the user operator.

(364) At Block L, if the CCU has received an input signal from the jacking control toggle switch **424**C then at Block M these input signals are received by the CCU microprocessor from the jacking control toggle switch **424**C

(365) At Block N, the CCU microprocessor processes the directional information from the jacking control toggle switch **424**C

(366) At Block O, the CCU microprocessor sends rotational direction information to the jack motor **425**B

(367) At Block P, the driven jack motor **425**B raises or lowers the spherical wheel shaft (i.e., inner jacking post member) **121** through interaction of the gears **425**C with the telescoping jacking post members **422** and **121**, for use during hitching and other trailer related functions such as raising the front section of the trailer to allow water runoff of the trailer or its contents.

(368) If at Block L, the CCU **425** has not received an input signal from the jacking control toggle switch **424**C, then at Block Q, the powered trailer jack and transport system **4** remains stationary relative to the ground surface.

(369) At Block R, the CCU microcontroller **425**E determines whether or not power has been on without control signals received by the CCU, for more than the pre-configured time period. If power has not been supplied to the system without input control signal within the prespecified time period, then the control flow returns to Block F, where the controller determines whether or not the CCU has received an input signal from the joystick directional controller.

(370) At Block S, if power has been supplied to the system without input control signal (i.e., no user input has been detected when the system has been activated), then the powered trailer jack and transport system **4** is automatically deactivated (i.e., power is turned off) within the CCU module **425**.

(371) Specification of Motor-Powered Trailer Jack and Transport System of the Present Invention

(372) FIGS. **18**A, **18**B and **18**C show the motorized (i.e. motor-powered) trailer jack and transport system **4** in FIG. **15**A, but removed the trailer **5** and arranged in its transport configuration, revealing onboard controller **424** mounted atop of its extendable jacking post, **422** and its load-bearing transport ball **101** supported within its semispherical framework **403**, and driven into controlled motion by its three battery-powered drive motors mounted external to the semispherical framework **403** and controlled by the onboard controller **424**.

(373) FIG. **18**D reveals how the transport ball **101** is retained within the framework **403** by retaining ring structure **105** employed in the powered trailer jack and transport system **4** in FIG. **19**A.

(374) FIGS. **18**E, **18**F, **18**G and **18**H reveal the extendable electrical power cord/harness **425**J that extends from the battery-powered control and communications module **425**, to the electric motors **407-1**, **407-2** and **407-3** provided in the system **4**.

(375) FIGS. **18**I and **18**J show that the powered trailer jack and transport system in FIG. **19**A, comprises: (i) a load-bearing transport ball **101** rotatably and freely mounted within a semispherical framework **403** by way of a set of four ball-bearing bearing pads **102-1** through . . . **102-4** by way of bearing pad mounting/support ring structure **104** and the semispherical framework **403**; (ii) a powered jacking post **422** mounted and extending from the top central portion of the semispherical framework **403**, and bearing a rotatable mounting mechanism **123** around midsection of post member **422**, and supporting an onboard controller **424**; (iii) a set of three battery-powered motors **407-1**, **407-2**, **407-3** mounted on the semispherical framework **403**, operably connected to the onboard controls and the solar-charged control and communications unit **425**, and having rubber drive gears or bearing wheels **407**C-**1**, **407**C-**2** and **407**C-**3** configured for engaging the outer surface of the transport ball **101** as the motors and bearing gears/wheels rotate driving the transport ball into controlled motion in accordance with user-operator controls provided through the controller **424**; and (iv) a transport ball retention ring structure **105** that is threaded on matching threads formed on the end of the bearing pad mounting ring structure **104**.

(376) Specification of Powered Trailer Jack and Transport System Provided with Drive Motor Protective Housing Covers Adapted for Fitting Over the Rotating Motor Drive Gears

(377) FIGS. **18**K, **18**L, **18**M, **18**N and **18**O show the powered trailer jack and transport system **4** shown in FIG. **18**A, provided with drive motor protective housing covers **404** adapted for fitting over the rotating motor drive wheels **407**C for user safety protection. These covers may be fabricated from high impact plastic material, casted metal material, rubber materials and any combination of the above. While shown as individual housing covers **404**, alternative designs may call for a single housing that covers two or more of the motor drive gears to promote safety. Also, a single cover housing may be fabricated and mounted over the entire powered spherical wheel assembly **40**, to cover and shroud the technical machinery (e.g., drive motors **407**B and drive wheels **407**C) disposed beneath the housing covering, while providing an aesthetically pleasing, if not beautiful, industrial design for the trailer jack and transport system **4** of the present invention.

(378) Specification of a Method of and System for Remotely Controlling Transport and Jacking Functions of a Powered Trailer Jack and Transport System Employing an Outboard Controller Removable and Hand-Supportable within the Hand of an Operator, and Also Controllable by a Mobile Smartphone Running a Suitable Mobile Application (APP)

(379) FIG. **19**A shows a person **80** using a mobile phone **427**′ equipped with a mobile application of the present invention **427**AA illustrated in FIGS. **21**B through **21**E, to control and navigate a trailer **53** equipped with the motorized trailer jack and transport system of the present invention **4**′,

configured to serve and support various applications namely: (i) trailer-vehicle hitching operations illustrated in FIG. **19**B, wherein the trailer **53** is transported towards the trailer hitch of a truck or vehicle **9** and then jacked up so the trailer coupler **7** can be placed on the trailer hitch **8**; and (ii) trailer parking operations illustrated in FIG. **19**C, wherein the trailer **53** equipped with the jack and transport system **4**′ is transported from a starting position in a parking lot, across the parking lot and into an intended parking space, without the need to physically handle the trailer during the parking operation.

(380) As shown in FIG. **19**B, a motor vehicle **9** is provided with a trailer hitch located near a trailer **53** to be hitched to the motor vehicle. The trailer is equipped with a powered trailer jack and transport system of the present invention **4**′, as shown in FIGS. **19**A through **19**J, but suitably modified so that the trailer jack and transport system **4**′ employs an outboard controller **424**′ that is removable and hand-supportable within an operator's hand for wireless remote control, and which is also controllable by a wirelessly-interfaced mobile smartphone **427**′ running a suitable mobile application (APP) **427**AA and Bluetooth communications **500** to enable remote control operations with the trailer jack and trailer system **4**′ paired with the mobile phone system **427**. The remotely-controlled trailer-vehicle hitching process is shown using the wireless handheld controller **424**′ in FIG. **22**.

(381) As shown in FIG. **19**C, using the motorized trailer jack and transport system **4**′, the user uses her mobile phone **427**′ and mobile app **427**AA to operate the system **46** and navigate the motorized/powered transport of the trailer **53** from its stationary starting position in the parking lot, disconnected from its truck, across the parking lot surface and ultimately into its target parking space using the powered trailer jack and transport system **46** remotely and wirelessly controlled by the mobile app **427**AA running on mobile phone **427**′ in communication with the control and communications unit **425**′ aboard the trailer jack and transport system **4**′.

(382) As shown in FIG. **20**, the motor-powered trailer jack and transport system **46** in FIG. **19**A, comprises: (i) a removable remote controller **424**′ supporting a directional joystick **424**B, jacking controls **424**C, controller electronics including processor, firmware and memory **424**E′, battery storage module **424**D, recharging module **424**G, and wireless communications **424**F; (ii) a releasable electrical connector **425**L; (iii) a jack and transport drive subsystem **44** supporting a control and comminutions unit **425**′ provided with gears **425**C, jack motor **425**B, and battery module **425**D, and control and communications electronics including a processor, firmware and memory **425**E′, Bluetooth wireless communication module **424**F, Wireless Wide Area Networking module **425**W, a battery charging unit **425**G, a Solar Panel **435**S for trickle charging the battery and a position controller **425**H, and drive motors **407-1**, **407-2** and **407-3**; each provided with a motor **407**A and drive wheels **407**C and provided with a drive motor wiring interface **425**J; and (iii) a mobile computing system **427** (e.g. mobile smartphone as shown in FIG. **21**A, tablet computer such an Apple® iPad in FIG. **21**B, and/or a laptop computer in FIG. **21**C) with a touch-screen GUI display and running a mobile app **427**AA of the present invention, and equipped with the features and sensors shown in FIG. **21**D, including GPS tracking, Bluetooth communications and other communication features.

(383) As shown FIG. **21**D, each mobile client system **427** deployed on the system network of the present invention, comprises: a Processor(s) **427**L; a Memory Interface **427**M; Memory **427**N for storing Operating System Instructions **4270**, Electronic Messaging Instructions **427**T, Communication Instructions **427**P, GUI Instructions **427**Q, Sensor Processing Instructions **427**R, Phone Instructions **427**S, Web Browsing Instructions **427**U, Media Processing Instructions **427**V, GPS/Navigation Instructions **427**W, Camera Instructions **427**X, Other Software Instructions **427**Y, and GUI Adjustment Instructions **427**Z; Trailer Jack and Transport App Instructions; **427**AA; Peripherals Interface **427**K; Touch-Screen Controller; Other Input Controller(s) **427**B; Touch Screen **427**A; Other Input/Control Devices; I/O Subsystem **427**C; Other Sensor(s) **427**J; Motion Sensor **427**I; Light Sensor **427**H; Proximity Sensor **427**G; Camera Subsystem **427**F; Wireless

Communication Subsystem(s) **427**E; and Audio Subsystem **427**D. Details regarding the mobile computing devices of the present invention **427** can be found in U.S. Pat. No. 8,631,358 incorporated herein by reference.

(384) FIG. **21**D shows the system architecture of an exemplary mobile client computing system **427** that is deployed on the system network and supporting the many services offered by system network servers. As shown, the mobile smartphone device **427** can include a memory interface **427**M, one or more data processors, image processors and/or central processing units **427**L, and a peripherals interface **427**K. The memory interface **427**M, the one or more processors **427**L and/or the peripherals interface **427**K can be separate components or can be integrated in one or more integrated circuits. The various components in the mobile device can be coupled by one or more communication buses or signal lines. Sensors, devices, and subsystems can be coupled to the peripherals interface **427**K to facilitate multiple functionalities. For example, a motion sensor **427**I, a light sensor **427**H and a proximity sensor **427**G can be coupled to the peripherals interface **427**K to facilitate the orientation, lighting, and proximity functions. Other sensors **427**J can also be connected to the peripherals interface **427**K such as a positioning system (e.g., GPS receiver), a temperature sensor, a biometric sensor, a gyroscope, or other sensing device, to facilitate related functionalities. A camera subsystem **427**F and an optical sensor **427**J, e.g., a charged coupled device (CCD) or a complementary metal-oxide semiconductor (CMOS) optical sensor, can be utilized to facilitate camera functions, such as recording photographs and video clips. Communication functions can be facilitated through one or more wireless communication subsystems **427**E, which can include radio frequency receivers and transmitters and/or optical (e.g., infrared) receivers and transmitters. The specific design and implementation of the communication subsystem **427**E can depend on the communication network(s) over which the mobile device is intended to operate. For example, the mobile device **427** may include communication subsystems designed to operate over a GSM network, a GPRS network, an EDGE network, a Wi-Fi or WiMax network, and a Bluetooth™ network. In particular, the wireless communication subsystems **427**E may include hosting protocols such that the device **427** may be configured as a base station for other wireless devices. An audio subsystem **427**D can be coupled to a speaker and a microphone to facilitate voice-enabled functions, such as voice recognition, voice replication, digital recording, and telephony functions. The I/O subsystem **427**C can include a touch screen controller and/or other input controller(s). The touch-screen controller can be coupled to a touch screen **246**. The touch screen **427**A and touch screen controller can, for example, detect contact and movement or break thereof using any of a plurality of touch sensitivity technologies, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with the touch screen **427**A. The other input controller(s) **427**B can be coupled to other input/control devices **427**B, such as one or more buttons, rocker switches, thumb wheel, infrared port, USB port, and/or a pointer device such as a stylus. The one or more buttons (not shown) can include an up/down button for volume control of the speaker and/or the microphone. Such buttons and controls can be implemented as hardware objects, or touch-screen graphical interface objects, touched and controlled by the system user. Additional features of mobile smartphone device **427** can be found in U.S. Pat. No. 8,631,358 incorporated herein by reference in its entirety.

(385) Specification of Method of Using Hand-Supportable Remote Controller to Remotely Control the Jack and Transport Movement Operations of the System Via a Wireless Communication Protocol

(386) FIG. **22** shows the use of the hand-supportable remote controller **424**′ removed from the powered trailer jack and transport system **4**′ mounted to a trailer **53** and being used to remotely control the jack and transport movement operations of the system, when transporting and/or hitching the trailer **5** to a motor vehicle **9** or the like.

(387) FIG. **23** describes the method of remotely-controlling the powered trailer jack and transport

system **4′** in FIG. **21** mounted to a trailer **5**, using a removable wireless controller **424′** as illustrated in FIG. **22**.
(388) As shown in FIG. **23**, Block A indicates the user starts, and progresses to Block B and powers on the handheld wireless controller **424′**.
(389) At Block C, the wireless remote controller **424′** powers on the control and communications unit **425′**
(390) At Block D, the user unlocks the wireless hand-held controller **424′** from the trailer jack and transport system **4**.
(391) At Block E, the user controls the powered transport system **4′** with the joystick of the wireless handheld controller **424′**.
(392) At Block F, the user controls the powered jacking (raising and lowering) with the jacking toggle switch **424**C of the wireless handheld controller **424′**.
(393) At Block G, the control signals are communicated from the handheld controller **424′** to the control and communications unit **425′** via Bluetooth wireless communications **500**.
(394) At Block H, the CCU microcontroller **425**E′ determines whether or not it has received an input signal from the joystick directional controller **424**B.
(395) At Block I, the wireless CCU microprocessor **425**E′ receives input from joystick directional controller.
(396) At Block J, the wireless CCU microprocessor **425**E′ processes directional information from joystick directional controller **424**B.
(397) At Block K, the wireless CCU microprocessor **425**E′ sends rotational speed information and power to drive the kiwi electric motors **407** via the position controller **425**H.
(398) At Block L, the drive motors **407** power rotational motion through interaction of drive rotors **407**C-**1**, **407**C-**2**, and **407**C-**3** with the surface of spherical wheel (i.e., transport ball) **101** to enable transport of the system under the control of the user operator.
(399) At Block M, the CCU microcontroller **425**E′ determines whether or not there is any input signal received from the jacking control toggle switch **424**C, and if so then proceeds to Block N, and if not, then proceeds to Block R.
(400) At Block N, the wireless CCU microprocessor **425**E′ receives input from jacking control toggle switch **424**C.
(401) At Block O, the wireless CCU microprocessor **425**E′ processes directional information from jacking control toggle switch **424**C.
(402) At Block P, the wireless CCU microprocessor **425**E′ sends rotational direction information to jack motor **425**B.
(403) At Block Q, the jack motor **425**B raises or lowers the inner jacking post member **121** through interaction of the gears with the telescoping spherical outer shaft **422** and then the control flow returns to back to Block H.
(404) At Block M, after the CCU microprocessor **425**E′ determines there is no input signal from the jacking control toggle switch **424**C, then at Block R the powered trailer jack transport system **4′** remains stationary.
(405) As Block S, the CCU microprocessor **425**E′ determines whether or not power has been on, without control signals being received by the wireless CCU, for more than a predefined time period, and if so, then the system control progresses to Block T, where the battery powered subsystems of the trailer jack and transport system **4′** are automatically turned off to conserve electrical power, and if not, then system control returns to Block H, as shown.
(406) Specification of Method of Using Mobile Smart Phone Running a Suitable Mobile Application to Remotely Control the Jack and Transport Movement Operations of the System Via a Wireless Communication Protocol
(407) FIG. **24** shows the use of the mobile smart phone **427′** running a suitable mobile application **427**AA being used to remotely control the jack and transport movement operations of the system **4′**

via a wireless communication protocol (e.g., Bluetooth Wireless) **500**, when transporting and/or hitching the trailer **5** to a motor vehicle or the like.
(408) FIG. **25** describes the primary steps involved in the method of remotely controlling the powered trailer jack and transport system **4**′ of FIG. **19**A mounted to a trailer **53**, using a mobile smart phone system **427**′ running a suitable mobile application **427**AA according to the principles of the present invention.
(409) As shown in FIG. **24**, Block A indicates the user starts, and progresses to Block B ad powers on the handheld controller **424**.
(410) At Block C, the controller powers on the control and communications unit (CCU) **425**.
(411) At Block D, the user powers on mobile phone and opens trailer jack and transport system app **427**AA.
(412) At Block F, the user controls the powered jack and transport system **4**′ with arrow buttons on the mobile phone app **427**AA “move” GUI screen.
(413) At Block G, the user controls the powered jacking (raising and lowering) operations of the system with the jacking toggle switch **424**C provided the wireless handheld controller **424**′.
(414) At Block H, the control signals are communicated from the handheld controller **424**′ to the control and communications unit (CCU) **425**′ via Bluetooth wireless communication or like protocols.
(415) At Block I, the CCU microcontroller **425**E′ determines whether or not it has received an input signal from the mobile phone “Move” GUI screen; if so, then the control flow proceeds to Block J, and if not, then the control flow proceeds to Block N.
(416) At Block J, the wireless CCU microprocessor receives input from the mobile phone “Move” GUI screen.
(417) At Block K, the wireless CCU microprocessor process directional information from the mobile phone “Move” GUI screen.
(418) At Block L, the wireless CCU microprocessor sends rotational speed information and power to drive the kiwi electric motors **407-1**, **407-2** and **407-3**.
(419) At Block M, the drive motors power rotational motion through interaction of drive rotors **407**C with the surface of the spherical wheel to enable transport of the system under the control of the user operator.
(420) At Block N, the CCU microcontroller **425**E′ determines whether or not there is any input signal received from the mobile phone app “Lift” GUI screen, and if so then proceeds to Block O, and if not, then proceeds to Block S.
(421) At Block O, the wireless CCU microprocessor **425**E′ receives input from the mobile phone “Lift” GUI screen.
(422) At Block P, the wireless CCU microprocessor **425**E′ processes directional information from the mobile phone “Lift” GUI screen.
(423) At Block Q, the wireless CCU microprocessor **425**E′ sends rotational direction information to jack motor **425**B.
(424) At Block R, the jack motor raises or lowers the inner jacking post member **121** through interaction of the gears **425**C with the inner jacking post, and then the control flow returns to back to Block I.
(425) At Block S, after the CCU microprocessor determines there is no input signal from the mobile phone “Move” or “Lift” GUI screens of the mobile app **427**AA, then at Block S the powered trailer jack transport system remains stationary.
(426) As Block T, the CCU microprocessor **425**E′ determines whether or not power has been on, without control signals being received by the wireless CCU, for more than a predefined time period, and if so, then the system control progresses to Block U, where the battery powered subsystems of the trailer jack and transport system **4**′ are automatically turned off to conserve electrical power, and if not, then system control returns to Block H, as shown in FIG. **25**.

(427) Specification of an Automated Method of and System for Hitching a Trailer to a Motor Vehicle Using an Automated Trailer Jack and Transport System Employing an Intelligent GPS-Tracked and IR-Ranging Controller on the Trailer and a GPS-Tracked and IR-Ranging Module about the Trailer Hitch on the Motor Vehicle, During Wireless and Automated Transport, Docking and Hitching Operations

(428) FIG. **26**A shows a motor vehicle **9** provided with a trailer hitch **8** located near a trailer **53** to be hitched to the motor vehicle **9** via its coupler **7**. As shown, the trailer **53** is equipped with a powered trailer jack and transport system **4**″ of the present invention as shown in FIGS. **19**A through **19**J, but suitably modified so that this alternative trailer jack and transport system employs an intelligent GPS-tracked, IR-ranging, WWAN-connected wireless control and communications unit **425**″ and a GPS-tracked and IR-Ranging module **428** which mounts about the trailer hitch **8** on the motor vehicle **9**, that communicate in a wireless manner and support automated transport, docking and hitching operations, without user involvement after the user inputs an automated docking request to the wireless outboard controller **424**′, or to a suitable mobile application running on a smartphone **427**′ operated by the user.

(429) FIG. **26**B shows the motor vehicle shown in FIG. **26**, showing the GPS-tracking and IR ranging module **428** mounted about the trailer hitch **8**, receiving GPS signals and IR ranging signals, and communicating with the trailer jack and transport system **4**″ during the automated docking and hitching process of the present invention.

(430) FIG. **26**C illustrates a method of mounting the GPS-tracking and IR ranging hitch finding module (HFM) **428** about the trailer hitch (namely the ball thereof) **8** mounted on the rear end of the motor vehicle **9** in the system of FIG. **26**, intended to hitch to a trailer **53** equipped with the trailer jack and transport system **4**″.

(431) FIG. **26**D shows the GPS-tracking and IR ranging hitch finding module (HFM) **428** showing its centrally positioned mounting aperture, through which the shank of a trailer hitch ball is allowed to pass and then secured by a suitable nut.

(432) FIGS. **26**E and **26**F show the GPS-tracking and IR ranging hitch finding module (HFM) **428** with its left and right spaced apart IR transceiver diodes **425**I-**1** and **425**I-**2** spaced apart and allowing left and right channel IR signals to be received/transmitted through the IR light transmission window **428**K formed in the compact device housing.

(433) FIG. **26**G shows the trailer jack and transport system **4**′ and the hitch finding module **428** in wireless communication with each other during its automated hitch finding, tracking and docking operations. As shown, two channels of line-of-sight IR signal communications are established for unobstructed wireless communication during system operation. This allows the controller **425**″ aboard the trailer jack and transport system **4**″ to acquire (i) left and right channel distance information (L1 and L2) from the IR signal channels, and (ii) GPS coordinate information from the GPS-hitch module **428** via Bluetooth wireless communication link **500**. Using this acquired information from the hitch module **428**, the control and communications unit **425**″ aboard the trailer jack and transport system **4**″ can automatically generate drive control signals that are used to drive motors and accurately navigate the trailer towards the trailer hitch, using a hitch-finding algorithm running within the processor **425**E″ aboard the trailer jack and transport system.

(434) As shown in FIG. **27**, the automated motor-powered trailer jack and transport system **47** of FIG. **26**, interfaced with a cloud-computing environment and wireless data communication infrastructure **430**, and comprise: (i) a controller module **424**′ mounted upon the end of the jacking post member **422** and supporting a directional joystick, jacking controls and automated finding, docking and hitching mode button, and controller electronics, battery storage module, and wireless communications; (ii) an electrical connector **425**L; and (iii) a jack and transport drive subsystem **44** supporting a control and comminutions unit provided with gears, jack motor, and battery module, and control and communications electronics including a processor, firmware and memory **425**E″, an IR finding transceiver **425**I, a battery charging unit **425**G, a position controller **425**H, WWAN

communication **425**W, local Bluetooth wireless communication **424**F, GPS receiver **425**M and solar charging panel **425**S

(435) FIG. **28** describes a method of automated docking and hitching of a trailer **53** to a trailer hitch **8** on a motor vehicle **9**, using an automated electrically-powered, GPS- and IR-ranging-equipped trailer jack and transport system **4**″ of FIGS. **26** and **27** and operated with the removable wireless handheld controller **424**′ according to the principles of the present invention, as illustrated in FIG. **27**.

(436) As shown in FIG. **28**, Block A indicates the user starts, and progresses to Block B ad powers on the wireless handheld controller **424**′.

(437) At Block C, the wireless handheld controller **424**′ powers on the control and communications unit (CCU) **425**″ provided with a programmed processor and memory controlling most operations within the system.

(438) At Block D, the user unlocks the wireless handheld controller **424**′ from the trailer jack and transport system **4**′.

(439) At Block E, the user powers on the wireless hitch LIDAR finding and GPS positioning unit **428**.

(440) At Block F, the wireless hitch LIDAR finding and GPS positioning module **428** receives GPS positioning signals from the GNSS satellites **428**G, and local RTK station **428**I to provide position correction signals to enable more precise positioning at the module **428** and controller **425**″

(441) At Block G, the user presses and holds the “automatic find and hitch” button on the wireless handheld controller **424**′.

(442) At Block H, the control signals are communicated from the handheld controller **424**′ to the control and communications unit **425**″ via Bluetooth wireless communication or like protocols.

(443) At Block I, the CCU microcontroller **425**E″ determines whether or not a “hitch finding” signal is being received by the wireless CCU microprocessor, from the wireless hitch finding module **428** mounted on the motor vehicle participating in the local hitching process; if so, then the control flow proceeds to Block J, and if not, then the control flow proceeds to Block T, as shown in FIG. **28**.

(444) At Block J, the wireless CCU microprocessor receives relational positioning input from the wireless hitch finder GPS positioning module **428**.

(445) At Block K, the CCU microprocessor receive GPS signals from the GNSS satellite system **428**G and RTK station **428**I.

(446) At Block L, the wireless CCU microprocessor processes relational positioning input, GPS and RTK signals, so as to determine direction of travel required to properly position the transport system relative to the hitch finder module **428**.

(447) At Block M, the wireless CCU microprocessor send rotational speed information and power to drive motors **407-1**, **407-2** and **407-3**.

(448) At Block N, the drive motors power motion through interaction of drive rotors **407**C-**1**, **407**C-**2**, and **407**C-**3** with spherical wheel surface to enable transport of the system under the control of the user operator.

(449) At Block O, the CCU microprocessor determines whether or not the trailer jack transport system has reached its final destination as determined by GPS positioning signals, and if no, then the control flow returns to Block I, and if so, then the control process proceeds to Block P.

(450) At Block P, the CCU microprocessor stops delivery of electrical power to the drive motors **407-1**, **407-2** and **407-3** so as to bring the trailer jack and transport system **4**″ to a stop.

(451) At Block Q, the CCU microprocessor sends signal to jack motor **425**B to lower trailer **53** by retracting the load-bearing spherical wheel assembly **40**.

(452) At Block R, the jack motor **425**B retracts the spherical wheel shaft by rotating the gears which are in mechanical communication with the telescoping jack shafts **121**, **422**.

(453) At Block S, the CCU microprocessor determines whether or not the docking switch has been

activated so as to signal successful hitching operations, and if so, the process proceeds to Block T, and if not, then returns the control flow to Block Q, as shown in FIG. **28**.
(454) At Block T in FIG. **28**, the CCU microprocessor **425**E″ ends (i.e., terminates) the automated transport process and sends wireless CCU alerts to user to signal end of the process with audible indication, and then powers down the battery-powered subsystem **44** to conserve electrical battery power.
(455) FIG. **29** describes a method of automated docking and hitching of a trailer **53** to a trailer hitch **8** on a motor vehicle **9**, using an automated, electrically-powered, GPS- and IR-ranging-equipped trailer jack and transport system **4**″ of FIGS. **26** and **27** and operated with a mobile phone **427**′ running an appropriate mobile app **427**AA according to the principles of the present invention, as illustrated in FIG. **27**.
(456) As shown in FIG. **29**, Block A of the method indicates that the user starts and progresses to Block B and powers on the handheld controller **424**′.
(457) At Block C, the wireless handheld controller **424**′ powers on the control and communications unit (CCU) **425**″.
(458) At Block D, the user powers on mobile phone **427**′, opens trailer jack and transport system app **427**AA, and pairs wireless Bluetooth communication **500** with the wireless CCU **425**″.
(459) At Block E, the user powers on the wireless IR and GPS positioning hitch finding module **428**.
(460) At Block F, the wireless hitch IR finding and GPS positioning module **428** receives GPS positioning signals from the GNSS satellites (and local RTK station for more precise positioning), while IR signals are communicated between the CCU of the trailer jack and transport and the GPS hitch finding module **428**.
(461) At Block G, the user presses and holds the "automatic find and hitch" button on the mobile phone app **427**AA.
(462) At Block H, the control signals are communicated from mobile phone **427**′ to the control and communications unit (CCU) via Bluetooth wireless communication or like protocols.
(463) At Block I, the CCU **425**″ determines whether or not a "hitch finding" signal is being received by the wireless CCU microprocessor **425**E″, from the wireless hitch finding module **428** mounted on the motor vehicle **9** participating in the local hitching process; if so, then the control flow proceeds to Block J, and if not, then the control flow proceeds to Block T, as shown in FIG. **29**.
(464) At Block J, the wireless CCU microprocessor receives relational positioning input from the wireless hitch finder GPS positioning module **428**.
(465) At Block K, the CCU microprocessor receive GPS signals from the GNSS satellite system **428**G and RTK station **428**I.
(466) At Block L, the wireless CCU microprocessor processes relational positioning input, GPS and RTK signals, so as to determine direction of travel required to properly position the transport system relative to the hitch finder module **428**.
(467) At Block M, the wireless CCU microprocessor **425**E′ sends rotational speed information and power to drive motors **407-1**, **407-2** and **407-3**.
(468) At Block N, the drive motors power motion through interaction of drive rotors **407**C-**1**, **407**C-**2**, and **407**C-**3** with the surface of spherical wheel (i.e., transport ball) **101** to enable transport of the trailer jack and transport system **4**″ under the control of the user operator.
(469) At Block O, the CCU microprocessor determines whether or not the trailer jack transport system **4**′ has reached its final destination as determined by GPS positioning signals and local IR ranging signals **428**F, and if not, then the control flow returns to Block I, and if so, then the control process proceeds to Block P.
(470) At Block P, the CCU microprocessor stops delivery of electrical power to the drive motors **407-1**, **407-2** and **407-3** so as to bring the trailer jack and transport system **4**″ to a stop.

(471) At Block Q, the CCU microprocessor sends signal to jack **425**B to lower trailer **53** by retracting the jack post member **121** within **422**, and thus raise the load-bearing spherical (transport ball) wheel assembly **40**.
(472) At Block R, the jack motor **425**B retracts the spherical wheel shaft by rotating the gears which are in mechanical communication with the telescoping spherical wheel shaft (i.e., inner jacking post member) **121**.
(473) At Block S, the CCU microprocessor **425**E″ determines whether or not the docking switch **428**D on the module **428** has been activated so as to signal successful hitching operations, and if so, then proceeds to Block T, and if not, then returns the control flow to Block Q, as shown in FIG. **29**.
(474) At Block T in FIG. **29**, the CCU microprocessor **425**E″ ends (i.e., terminates) the automated transport process, and sends wireless CCU alerts to user to signal end of the process with audible indication, and a message on the mobile phone **427**′ and then powers down the battery-powered subsystem **44** to conserve electrical battery power.
(475) Specification of a Method of and System for Jacking and Transporting a Trailer Employing an Intelligent GPS-Tracked Trailer Jack and Transport System Remotely Controllable Using a Mobile Computing System Running a Mobile Application of the Present Invention
(476) FIG. **30**A shows a person standing in a marina parking lot near a parked vehicle and boat trailer **53** equipped with the intelligent GPS-tracking motorized trailer jack and transport system **4**′″, that is remotely-controllable using a mobile computing system, such as a smartphone **427**′ running a special mobile application **427**AA, and operational in a programmable mode of operation including an automated trailer parking method illustrated in FIG. **38**.
(477) As shown in FIG. **30**A, the trailer **53** is equipped with a GPS-tracking motorized trailer jack and transport system **4**′″ having a GPS-tracking trailer module (i.e. navigator beacon) **440** mounted on the rear of the trailer and in wireless (Bluetooth) communication **500** with the GPS-tracking motorized trailer jack and transport system **4**′″ mounted on the front of the trailer **53** creating a navigation axis **444** to enable automatic transport from a starting position in a parking lot, across the parking lot and into an intended parking space.
(478) Also, the GPS-coordinates of the trailer's boundaries, destination parking space and keep out zones are captured prior to automated parking operations so as to enable automated GPS-guided transport of the trailer **53** from a starting position in a parking lot, across the parking lot, and into an intended destination parking space using GPS-tracking and navigation methods without the need to physically handle the trailer during the parking operation.
(479) FIG. **30**B illustrates the automated trailer jack and transport system **4**′″ when configured and working in cooperation with a GPS-tracking and LIDAR-mapping navigator beacon module **440** mounted to the rear of the trailer **53**. As configured, these systems **4**′″ and **440** establish and maintain a virtual trailer navigation axis **444**, passing through the central axis of the trailer. About this virtual trailer navigation axis **444**, the spatial boundaries of the trailer are predefined and managed within local databases **425**E′″ and cloud-based servers **430**, for use when parking the trailer into a destination parking space while avoiding collisions in cooperation with the GPS-tracking and LIDAR-mapping navigator beacon module **440** mounted at the rear of trailer **53** to detect and prevent collisions with obstacles that may be presented along the trailer parking pathway computed by the system network.
(480) FIG. **30**C shows in more detail the GPS and LIDAR-mapping navigator beacon **440** used to create the virtual axis **444** when in communication and working in cooperation with the trailer jack and transport system **4**′″ as part of the system **48** as described in FIG. **30**A.
(481) FIG. **31** shows the automated motor-powered trailer jack and transport system **4**′″ of FIGS. **30**A through **30**C, interfaced with a cloud-computing environment and wireless data communication infrastructure **430** and further comprising: (i) a wireless controller module **424**′ mounted upon the end of the jacking post member and supporting a directional joystick **424**B, jacking controls **424**C and automated finding, docking and hitching mode button, and controller

electronics including a processor, firmware and memory **424**E′, battery storage module **424**D, battery charging circuit **424**G and wireless communications **424**F; (ii) a releasable electrical connector **425**L; (iii) a jack and transport drive subsystem **44** supporting a control and communications unit (CCU) **425**′″ provided with gears **425**C, jack motor **425**B, and battery module **425**D, battery recharging module **425**G, solar panel **425**S, and control and communications electronics including a processor, firmware, and memory (supporting all functions of the system) **425**E′″, a dual-channel IR hitch finding module **425**I, and a position controller **425**H, GPS signal receiver **425**M with antenna for receiving and processing GPS positioning signals, WWAN module **425**W, Bluetooth module **425**F, and drive motors **407-1**, **407-2**, **407-3** and a drive motor wiring interface **425**J; (iv) a GPS navigator beacon **440** for mounting on the rear of the trailer provided with a GPS receiver **425**M and antenna, a Bluetooth communication module **424**F, a processor, firmware and memory **440**C, and LIDAR ToF range finder **440**D with 180 degree FOV; and (v) cloud computing and wireless data communication infrastructure **430** having communication, application and database servers.

(482) FIG. **32** illustrates a method of automatic parking of a registered trailer (i.e. its GPS-coordinate-based virtual axis and boundary have been captured as described in FIGS. **34**A, **34**B and **35**) that is equipped with the automated trailer jack and transport system **4**′″ and GPS navigator beacon **440**, showing the trailer's navigation vector at different stages along a projected pathway of travel towards the destination parking space.

(483) FIG. **33** describes the primary steps carried out during the practice of the method of automatic parking of a registered trailer **53** equipped with the automated trailer jack and transport system of the present invention **4**′″ controlled within the cloud-based computing and communications infrastructure **430** of system **48**.

(484) As indicated at Block A, the user starts by progressing to Block B and user powers on wireless handheld controller of the registered automated trailer jack and transport system **4**′″

(485) At Block C, wireless handheld controller powers on wireless control and communications unit (CCU).

(486) At Block D, the user powers on mobile phone **427**′, opens trailer jack transport system app **427**AA, and pairs Bluetooth comm with the wireless CCU **425**″.

(487) At Block E, the user powers on the GPS navigator beacon **440**, and at Block F, the GPS navigator beacon receives GPS signals and communicates position information to CCU.

(488) At Block G, the user selects the “park” button in the mobile phone app **427**AA.

(489) At Block H, control signals are communicated from the mobile phone **427**′ to the wireless control and communications unit (CCU) **425**′″ via Bluetooth wireless communications **500**.

(490) At Block I, the CCU **425**′″ determines whether or not the position of the automated trailer jack and transport system and navigator beacon is known to the cloud server and mobile app, and if not, then the control returns to the start Block A and if so, the proceeds to Block J.

(491) At Block J, the CCU **425**′″ determines whether or not the path between the trailer and target parking space is visually clear of obstacles, and if not, returns to Block I, and if so, then proceeds to Block K.

(492) At Block K, the user executes the “park” command in the automated trailer jack and transport system mobile phone app.

(493) At Block L, the mobile phone **427**′ communicates park command to the wireless CCU via Bluetooth communications.

(494) At Block M, the wireless control and communications unit (CCU) microprocessor **425**E′″ processes park command, GPS signals, relative position of navigator beacon **440**, beacon LIDAR signals, trailer boundary, parking space boundary and keep-out (i.e., obstacle) zone boundaries.

(495) At Block N, the wireless CCU **425**′″ determines direction of travel required to transport the trailer system **4**′″ to the selected parking space

(496) At Block O, the CCU **425**′ determines whether or not the path as determined by the CCU and

the navigator beacon LIDAR is clear of obstacles, and if not, the returns to Block J, and if so, then proceeds to Block P.

(497) At Block P, the wireless CCU microprocessor **425**E′″ sends rotational speed information and power to drive motors **407-01**, **407-2**, **407-3**.

(498) At Block Q, the drive motors **407-01**, **407-2**, **407-3** power motion through interaction of drive rotors with the surface of spherical wheel (i.e., transport ball **101**) to enable transport of the system **4**′″ under the control of the GPS-assisted control system.

(499) At Block R, the wireless CCU **425**′″ processes GPS position with the GPS position of the navigator beacon **440** to determine and maintain direction of intended travel.

(500) At Block S, the CCU **425**′″ determines whether or not the intended travel path is clear of intersections between the trailer boundary and keep out boundaries as determined by the CCU, and if so, the proceeds to Block T, and if not advances to Block V.

(501) At Block T, the CCU **425**′″ determines whether or not the travel path is clear of clear of obstacles as determined by the LIDAR range finder in module **440**, and if yes, then proceeds to Block U, and if not, then proceeds to Block V.

(502) At Block U, the CCU **425**′″ determines whether or not the GPS coordinates of the trailer boundary is positioned within the set of GPS coordinates of the parking space boundary as determined by the CCU **425**′″ and the cloud-computing system **430**, and if not then returns to Block P, and if yes, then proceeds to Block V.

(503) At Block V, the CCU **425**′″ terminates the automated transport process and generates wireless CCU alerts to user to indicate the end of process with an audible indicator and mobile phone app message.

(504) Specification of the Method of Registering of GPS Coordinate Maps of a Trailer Equipped with the Automated Trailer Jack and Transport System, Destination Parking Spaces, and Keep Out (i.e., Obstacle) Zones, for Storage in a Network Database System on a Cloud-Based System Network

(505) FIG. **34**A illustrates the process of registering GPS coordinate maps of (i) the trailer **53** equipped with the automated trailer jack and transport system **4**′″ of the present invention and navigator beacon **440** and (ii) a destination parking space **91** associated with automated parking of so-equipped trailer **53**.

(506) FIG. **34**B illustrates the process of registering GPS coordinate maps of keep out (i.e., obstacle) zones (e.g., motorcycles, picnic table) associated with automated parking of a trailer **53** equipped with the automated trailer jack and transport system **4**′″ of the present invention.

(507) FIG. **35** describes a method of registering of GPS coordinate maps of several system network components, namely: (i) a trailer **53** equipped with the automated trailer jack and transport system **4**′″ illustrated in FIG. **32**; (ii) destination parking spaces **91** as illustrated in FIG. **34**A; and (iii) keep out (i.e., obstacle) zones **443** as illustrated in FIG. **34**B. Once captured, these GPS coordinate maps are stored in a database system maintained on the cloud-based system network.

(508) As shown in FIG. **35**, at Block A, the user starts the process and proceeds to Block B, and powers on the wireless control and communications unit (CCU) **425**′″ of the automated trailer jack and transport system **4**′″.

(509) At Block C, the user powers on mobile phone **427**′, opens trailer jack transport system app **427**AA and pairs Bluetooth comm **500** with the CCU **425**′″.

(510) At Block D, the mobile phone **427**′ and the CCU **425**′″ establishes a connection to the cloud-based server via wireless wide area network.

(511) At Block E, the wireless CCU **425**′″ and the mobile phone **427**′ each receive positioning data from the GNSS satellite system **428**G and RTK station **428**I

(512) At Block F, the user powers on GPS navigator beacon **440** which receives positioning data and communicates the same to the CCU via its established Bluetooth wireless communication link **500**.

(513) At Block G, the user enables “register trailer GPS boundary” function of the mobile app **427**AA.
(514) At Block H, the system determines whether or not the position of the automated trailer jack and transport system **4**′″ and navigator beacon **440** is known to the cloud server **430** and mobile app **427**AA, and if not, returns to Block B, and if so, then advances to Block I.
(515) At Block I, the user is prompted by images within the mobile app **427**AA to place the phone in a geospatial location relative to the trailer **53**.
(516) At Block J, the user selects “store point.”
(517) At Block K, the mobile app communicates stored position to the cloud via the mobile phone WWAN **425**W.
(518) At Block L, cloud-based server **430** stores position information as a point of the boundary.
(519) At Block M, the cloud server **430** determines whether or not the minimum number of points have been captured to produce the property boundary.
(520) At Block N, the user selects “generate trailer boundary” command and function supported by the system.
(521) At Block O, the cloud server **430** processes the command, generates and stores the GPS coordinate boundary data **441** with respect to automated trailer jack and transport system **4**′″ and navigator beacon **440**.
(522) At Block P, the cloud server **430** sends boundary data **441** to automated trailer jack and transport system **4**′″ for local storage.
(523) At Block Q, the user enables “register parking space GPS boundary” function of the mobile app **427**AA.
(524) At Block R, the user is prompted by the mobile app to place the mobile phone **427**′ in a geospatial location relative to the target space among parking spaces **91**.
(525) At Block S, the user selects “store point.”
(526) At Block T, the mobile app **427**AA communicates stored position to the cloud server **430** via the mobile phone WWAN **425**W.
(527) At Block U, the cloud-based server **430** stores position information as a point (e.g., GPS coordinate data) along the parking space boundary.
(528) At Block V, the cloud server determines whether or not the minimum number of points have been captured to produce the parking boundary.
(529) At Block W, the user selects “generate parking space boundary” and names the space for later recall.
(530) At Block X, cloud server processes the command, generates and stores the boundary **442**.
(531) At Block Y, the cloud server **430** sends GPS-specified boundary data **442** to automated jack and transport system **4**′″ for local storage.
(532) At block A, the user enables “keep out zone GPS boundary” function of the mobile app **427**AA.
(533) At Block AA, the user is prompted by the mobile app **427**AA to place the mobile phone **427**′ in a geospatial location relative to the keep out zone.
(534) At Block BB, the user selects “store point.”
(535) At Block CC, the mobile app **427**AA communicates stored position to the cloud server **430** via the mobile phone WWAN **425**W.
(536) At Block DD, the cloud server **430** determines whether or not the minimum number of points have been captured to produce the keep-out (i.e., obstacle) boundary or space.
(537) At Block EE, the user selects “generate keep out boundary” and names the space for later recall.
(538) At Block FF, the cloud server **430** processes the command, generates and stores the GPS coordinate boundary data **443** set to specify the keep-out (i.e., obstacle) space; and then sends the same to the CCU **425**′″ for local storage aboard the trailer jack and transport system **4**′″

(539) Specification of Method of Training an Automated Trailer Jack and Transport System During a Learning Mode Supported by the System for Subsequent Use in Automated Trailer Parking Operations

(540) Referring now to FIGS. **36** through **39**, a method will now be described for training the cloud-based trailer jack and transport system **4**‴ in FIGS. **30**A through **31** (System **48**), while configured in an automated learning mode so as to automatically capture and store the GPS coordinates of a trailer **53** and parking locations **442** where the trailer **5** has been parked by a human driver during training operations, as illustrated in FIG. **36**. The method involves using cloud-based servers **430** and/or local memory of an automated trailer jack and transport system **4**‴. Subsequently, when operating the trailer jack and transport system **4**‴ in its automated parking mode illustrated in FIG. **38** and taught in FIG. **39**, the captured GPS coordinates stored in the cloud-based server system **430** are then used by motorized trailer jack and transport system **4**‴ to automatically guide and transport the trailer **53** safely to its intended parking location without the use of a human driver, as illustrated in FIG. **38**. The details of this alternative method and mode of system operation will be described in great technical detail below.

(541) FIG. **36** illustrates the method of training the automated trailer jack and transport system **53** in FIG. **30**A during the learning mode supported by the system, for subsequent use in automated trailer parking operations illustrated in FIG. **38**. As illustrated in FIG. **36**, training of the system starts when the trailer **53** located at a trailer home position at time T**1**, while the trailer is being towed by a motor vehicle **9** along a pathway to a trailer finish position at time Tx at some later time period. As illustrated, the cloud-based system **48** comprises the RTK base station **428***i*, GNSS satellite system **428**G, mobile computing system **427**, trailer jack and transport drive subsystem **44**, and other system components configured as shown in FIG. **31**. When operating together, the system **48** is capable of capturing the GPS position of the trailer equipped with system **4**‴, the tow vehicle **9**, and parking spaces **91** down to centimeter resolution.

(542) FIG. **37** describes the method of training the system **48** to capture and store the GPS coordinates of a home position (i.e., trailer start position) and points along an associated entrance path of the trailer **53** equipped with the automated trailer jack and transport system **48**, all the way towards the trailer finish position at some time later.

(543) As shown in FIG. **37**, the user starts at Block A, and proceeds to Block B.

(544) At Block B, the user hitches a trailer **53** equipped with automated trailer jack and transport system **4**‴ to tow vehicle **9** and rotates the jack and transport system **4**‴ into its storage mode.

(545) At Block C, the user positions the trailer equipped with automated trailer jack and transport system **4**‴ into a desired parking location with tow vehicle and powers on the wireless control and communications unit (CCU) **425**‴.

(546) At Block D, the user selects “set parking space” in the automated trailer jack and transport system mobile app **427**AA.

(547) At Block E, the mobile phone **427**′ communicates command via Bluetooth wireless communication **500** to the control and communications unit **425**‴ and further to navigator beacon **440**.

(548) At Block F, the wireless CCU **425**‴ processes GPS and RTK positioning signals of the CCU itself and the navigator beacon **440** and stores the same in non-volatile memory with a logical address as well as in the cloud servers **430**.

(549) At Block G, the mobile app **427**AA prompts user to select “generate parking path”.

(550) At Block H, the user tows trailer **53** forward with tow vehicle **9**.

(551) At Block I, the wireless control and communications unit (CCU) **425**‴ continually processes GPS and RTK positioning signals of the CCU and navigator beacon and stores positioning information in non-volatile memory **425**E‴ in the CCU and the cloud servers **430**, with logical addresses.

(552) At Block J, the driver of tow vehicle **9** determines whether or not the trailer **53** completely

clear of its home parking position and every obstacle in its egress path.
(553) At Block K, the user brings tow vehicle **9** and trailer **53** to a stop.
(554) At Block L, user selects “done” in the mobile app.
(555) At Block M, the mobile app and cloud server store the positional information of the parking space and associated parking path and prompts user to name the sequence.
(556) At Block N, the user names the sequence for later retrieval.
(557) At Block O, the store function is complete; and battery-powered assemblies of the automated trailer jack and transport system **4**′ are powered down.
(558) By virtue of the method of the present invention, it is now possible for the system to automatically learn, capture and store/remember the GPS coordinates of a trailer's parking path, including a parking boundary **442** and the opposite end of the path at T (x), and other parking locations where the trailer **53** may be parked by a human driver. These GPS coordinates specifying the geometrical boundaries of each spatial region where the trailer **53** can or may be stored along its journey, can be stored in cloud-based servers **430**, and/or local memory of an automated trailer jack and transport system **4**′″. Subsequently, when operating in an automated parking mode, these GPS-specified trailer parking boundary coordinates stored on cloud servers **430** or locally can be quickly accessed and used by the cloud system **430** and/or transport system **4**′″ to automatically guide and transport the trailer **53**, via transport system **4**′″, safely to its intended registered parking location **442** without the use of a human driver, thereby advancing the state of the art in cloud-based trailer parking automation.
(559) Specification of Method of Automatic Parking of Trailer Equipped with the Automated Cloud-Based Trailer Jack and Transport System of the Present Invention and a Registered Parking Path
(560) FIG. **38** illustrates an automated method of and system **48** for parking of trailer **53** equipped with the automated cloud-based trailer jack and transport system **4**′″ of system **48** with a registered (stored) parking path as described in FIGS. **37** and **38**.
(561) FIG. **39** describes the method of automatic parking of trailer **53** of the automated trailer jack and transport system **48** illustrated in FIG. **38**.
(562) As shown in FIG. **39**, at Block A, the user starts and proceeds to Block B.
(563) At Block B, the user rotates automated trailer jack and transport system **4**′″ from its storage mode/configuration, into its transport mode so there is contact between the ground surface and its load-bearing transport ball **101**, and is capable of rolling freely under motor control.
(564) At Block C, the user decouples trailer and automated trailer jack and transport system **4**′″ from tow vehicle **9** and powers on control and communications unit (CCU) **425**″.
(565) At Block D, the user presses “park” button in the trailer jack and transport system mobile phone app **427**AA.
(566) At Block E, the mobile phone app **427**AA prompts user to confirm the parking path is clear of obstacles before permitting further operation.
(567) At Block F, the mobile phone app **427**AA offers available stored parking spaces based on GPS positioning information of the automated trailer jack and transport system **4**′″ and stored location information.
(568) At Block G, the mobile phone **427**′ communicates park command via Bluetooth wireless communication **500**.
(569) At Block H, the wireless control and communications unit (CCU) **425**′″ processes park command and determines direction of travel required to properly transport the trailer **53** to its parking space **442**.
(570) At Block I, the wireless CCU microprocessor **425**E′″ sends rotational speed information and power to drive motors.
(571) At Block J, the drive motors **407-1**, **407-2** and **407-3** power motion through interaction of drive rotors with the surface of the spherical wheel **101** to enable transport of the system **4**′″ under

the control of the cloud-based automated trailer parking system **48**.
(572) At Block K, system cloud servers **430** automatically monitor and determines whether or not the trailer has reached its parked position in the selected parking space.
(573) At Block L, to terminate the automated transport process, wireless CCU **425**‴ alerts user about the end of process with an audible indicator and generating a mobile phone app message, and driving the battery-powered sub-systems **44** to their power saving mode.
(574) By virtue of the method of the present invention, it is now possible for a person to use a cloud-based trailer jack and transport system **4**‴ to automatically learn, capture and store the GPS coordinates of a trailer **53** and parking locations **442** where the trailer has been parked by a human driver using cloud-based servers **430** and/or local memory of an automated trailer jack and transport system **4**‴, as illustrated in FIG. **36**. Subsequently, when operating the trailer jack and transport system **4**‴ in its automated parking mode, the stored GPS coordinates can be used to automatically guide and transport the trailer **53** safely to its intended parking location without the use of a human driver or a mobile vehicle **9**, as illustrated in FIG. **38**.
(575) Specification of a Personal Watercraft (PWC) Trailer Equipped with the Trailer Jack and Transport System of the Present Invention
(576) FIGS. **40**A and **40**B show a personal watercraft (PWC) **50** and trailer **5** equipped with the trailer jack and transport system **1** shown and described in FIGS. **2**B through **2**Q.
(577) FIG. **40**C shows a personal watercraft (PWC) and trailer **5** equipped with the trailer jack and transport system **1** shown and described in FIGS. **2**B through **2**Q, arranged in its transport configuration, and located near a motor vehicle **9** preparing to be hitched to the trailer **5** via its hitch ball.
(578) Specification of a Snowmobile Trailer Equipped with the Trailer Jack and Transport System of the Present Invention
(579) FIGS. **41**A and **41**B show the snowmobile trailer **5** carrying a snowmobile **51** and equipped with the trailer jack and transport system **1**, shown arranged in its transport mode, and not hitched to any motor vehicle **9**.
(580) By virtue of the large surface area of the transport ball **101** contacting the snow surface, surface pressure can be reduced in such use applications and the reduce the likelihood of the trailer from getting stuck in the snow.
(581) Specification of a Multi-Axle Boat Trailer Equipped with the Trailer Jack and Transport System of the Present Invention
(582) FIG. **42**A shows a multi-axle boat trailer **53** equipped with the trailer jack and transport system **1** arranged in its transport mode and preparing for hitching up with a motor vehicle **9**.
(583) FIG. **42**B shows the multi-axle boat trailer **53** equipped with the trailer jack and transport system **1** shown arranged in its transport mode, with the trailer coupler **7** positioned over and lowered onto the trailer hitch **8** of the motor vehicle **9**, ready to be fully hitched.
(584) FIG. **42**C shows the multi-axle boat trailer **53** equipped with the trailer jack and transport system **1** shown arranged in its storage mode, with the trailer **53** hitched to the motor vehicle **9**.
(585) Specification of a Camper Trailer Equipped with the Trailer Jack and Transport System of the Present Invention
(586) FIGS. **43**A AND **43**B show a camper trailer **54** equipped with the trailer jack and transport system **1**, arranged in its transport mode, having been positioned after being decoupled from a tow vehicle.
(587) By virtue of the large surface area of the transport ball **101** contacting the dirt campground surface, surface pressure can be reduced in such use applications and the reduce the likelihood of the trailer from getting stuck in the dirt.
(588) Specification of a Flatbed Utility Trailer Equipped with the Trailer Jack and Transport System of the Present Invention Mounted within its A-Frame Structure
(589) FIG. **44**A shows the flatbed trailer **55** equipped with the trailer jack and transport system **3**

arranged in its transport mode, and not hitched to any motor vehicle.
(590) FIG. **44**B shows the flatbed trailer **55** equipped with the trailer jack and transport system **3**, arranged in its storage mode, shown not hitched to any motor vehicle for illustrative purposes.
(591) Specification of an Industrial Tool with Integrated Trailer Equipped with the Trailer Jack and Transport System of the Present Invention Mounted within its A-Frame
(592) FIGS. **45**A and **45**B show an industrial tool (e.g., a towable generator) **56** equipped with the trailer jack and transport system **3** arranged in its transport and jack mode, shown on a construction site with a heavy gravel ground surface.
(593) By virtue of the large surface area of the transport ball **101** contacting the gravel surface, surface pressure can be reduced in such applications, reducing the likelihood of the trailer jack and transport system of getting stuck.
(594) Specification of First Transportable System Equipped with Load-Bearing Transport Ball Constructions of the Present Invention Mounted within the Corners of the Platform
(595) FIG. **46**A shows a first transportable system **60** having trapezoidal platform of rigid construction **601**, with three corners and built-in grab handles **603-1** and **603-2**. Each platform corner has a load-bearing transportation ball construction **10-1**, **10-2**, **10-3**, side-mounted from its side surface, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.).
(596) FIG. **46**B shows the first transportable system **60** of FIG. **45**A, carrying and transporting a heavy cylindrical beer keg or barrel **62** across a lawn surface. A hinged handle **602** is attached to the platform, for easy pulling of the platform across ground, sandy and lawn surfaces. Apertures **603-1** and **603-2** are formed in the side edges of the platform for fastening straps or ropes that may be used to fasten any object being transported by the transportable system.
(597) Specification of Second Transportable System Equipped with Load-Bearing Transport Ball Constructions of the Present Invention Mounted within the Corners of the Platform
(598) FIG. **47**A shows a second transportable system **60**′ having trapezoidal platform **601**′ of rigid construction, with three corners and built-in grab handles **603**. Each platform corner has a load-bearing transportation ball construction **10-1**, **10-2**, **10-3**, top-mounted through its top surface, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.).
(599) FIG. **47**B shows the second transportable system, **60**′ of FIG. **47**A, carrying and transporting a heavy cylindrical beer keg or barrel across a lawn surface. A hinged handle **602** is attached to the platform, for easy pulling of the platform across ground, sandy and lawn surfaces. Apertures **603-1** and **603-2** are formed in the side edges of the platform for fastening straps or ropes that may be used to fasten any object being transported by the transportable system.
(600) Specification of Third Transportable System Equipped with Load-Bearing Transport Ball Constructions of the Present Invention Mounted within the Corners of the Platform
(601) FIG. **48**A shows a third transportable system **61** having trapezoidal platform of rigid construction **611**, with four corners and integrated grab handles **603**. Each platform corner has a load-bearing transportation ball construction **10-1**, through **10-4**, edge-mounted to its side surface, defining a plane of load-bearing support for carrying a load (e.g. such as barrel, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.).
(602) FIG. **48**B shows the third trapezoidal transportable platform of FIG. **48**A, carrying and transporting a heavy object, such as barbeque gas grill **63**, across a lawn surface. A hinged handle

602 is attached to the platform, for easy pulling of the platform across ground, sandy and lawn surfaces. Apertures **603-1** and **603-2** are formed in the side edges of the platform for fastening straps or ropes that may be used to fasten any object being transported by the transportable system.
(603) Specification of Fourth Transportable System Equipped with Load-Bearing Transport Ball Constructions of the Present Invention Mounted within the Corners of the Platform
(604) FIG. **49**A shows a fourth transportable system **61**′ having trapezoidal platform of rigid construction **611**′, with four corners and integrated grab handles **603**. Each platform corner has a load-bearing transportation ball construction **10-1** through **10-4**, top-mounted through its top surface as shown, defining a plane of load-bearing support for carrying a load (e.g. such as an ice chest, container, vessel, tools, or other objects) to be transported across ground surfaces that may be smooth and hard (e.g. concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.).
(605) FIG. **49**B shows the fourth transportable system **61**′ of FIG. **49**A, carrying and transporting a heavy object, such as a full ice chest, across a lawn surface. A hinged handle **602** is attached to the platform, for easy pulling of the platform across ground, sandy and lawn surfaces. Apertures **603-1** and **603-2** are formed in the side edges of the platform for fastening straps or ropes that may be used to fasten any object being transported by the transportable system.
(606) Specification of the Battery-Powered Remotely-Controlled (RC) Transportable System Equipped with any Arrangement of Multiple Load-Bearing Transport Ball Constructions of the Present Invention Mounted within a Framework Supporting Trailer Hitching Apparatus
(607) Referring to FIGS. **50**A and **50**C, there is shown another alternative embodiment of the present invention in the form of a battery-powered remotely-controlled (RC) transportable system **70** equipped with an arrangement of multiple load-bearing transport ball constructions of the present invention **40-1**, **40-2**, and **40-3**, mounted within a framework or chassis **701** and integrated within a computer-based control system, supported by battery power storage module and control electronics **725** for supporting trailer hitching apparatus **8** and its various jacking, transport and navigation functions described hereinabove, in connection with a trailer **53**.
(608) As shown, the battery-powered remotely-controlled (RC) transport system **49** comprises: f a dual-axle boat trailer **53** operably coupled to a remotely controlled powered (i.e., motorized) transport system **70** being controlled by a human operator remotely controlling the direction of powered motion of the transport system and hitched trailer via a Bluetooth wireless connection **500** from a mobile phone **427**′ equipped with a suitable mobile app **427**AA.
(609) As shown, the remote-controlled (RC) powered transport system **70** utilizes a configuration of multiple load-bearing spherical wheel assemblies of the present invention **40-1**, **40-2** and **40-3**, mounted on a strong chassis **701** for mounting and supporting the spherical wheel assemblies, and preferably supporting a common battery storage cell and control and navigation electronics **725** illustrated in the system diagrams of FIGS. **20** and/or **27**, both of which are incorporated herein by reference to this illustrative embodiment. Its jacking post **8** is centrally mounted within the central portion of the chassis **701**, with a trailer ball at its top, and may include automated/powered hydraulics or linear motors, to enable the automated raising and lowering of the hitching post **8** as required to couple up with its target trailer **53**. Preferably, this function will be remotely controlled using the mobile app **427**AA along with trailer transport and navigation function of the present invention, disclosed herein.
(610) In FIG. **50**B, the system **70** is shown arranged and hitched to a trailer **53**, or other recreational or work vehicle, is being remotely controlled by a user using a mobile phone **427**′ running associated mobile app **427**AA configured for supporting navigation, (optionally jacking) and transport functions in accordance with the principles of the present invention, on a ground surface **90** that may be hard (e.g., concrete, asphalt, wooden, etc.) as well as rough and soft (e.g. lawns, dirt paths, and sandy ground surfaces such as beaches, etc.).
(611) FIG. **50**C shows the battery-powered remotely-controlled transport system **70** comprising: a

rugged weight/load-bearing frame/chassis **701** supported and driven by three battery-motor powered spherical wheel assemblies **40-1**, **40-2**, and **40-3**; a trailer hitch and ball **8**; and a battery-powered control and communications unit **725** electrically connected to and driving the electric-drive motors **407** employed in the powered spherical wheel assemblies **40**, adapted for pulling and transporting (i.e. trucking) a trailer **53** equipped with a coupler, without the use of a traditional motor vehicle.

(612) As shown, the system **70** supports all of the automated navigation and transport functions of all other power system embodiments described herein using a single powered spherical wheel assembly of the present invention, but just performs such function using a cluster of spherical wheel assemblies to support great trailer loads and great pulling power required by the application at hand. Powered spherical wheel clusters, comprising two, three, four or more powered spherical wheel assemblies of the present invention, may be configured and deployed about any kind of framework structure such as **701**, to suit and support the trailer trucking and transport application at hand. The chassis framework **70**, made from welded metal materials, or composite materials having high strength for the application at hand, may be realized as a standalone unit as shown in FIG. **50**C, or the chassis framework **70** may also be mounted to the trailer frame **53** and support both transport and storage modes of configuration and operation, as described in the single battery-powered spherical wheel assembly embodiments of the present invention, specified in great technical detail hereinabove.

(613) It is understood that the battery-powered remotely-controlled transport system **70** may be integrated within the A-type or other frame portion of a trailer framework, and adapted for retraction within the trailer frame or body during the storage mode as taught in FIG. **14**A through **14**D, and protracted from the trailer body during the transport mode of operation when configured and operation in such a manner.

(614) Also instead of using a mobile phone and mobile app to remotely control the battery-powered remotely-controlled transport system **70**, it is understood that the system can be adapted to support the use of an integrated remote control unit such as RC unit **424** shown in FIGS. **18**J, **19**A and **19**B, or RC unit **424**′ shown in FIG. **22**, which may be used to also remotely control the trailer jacking, transport and navigation functions as described in detail herein. Such variations and modifications will readily occur to those skilled in the art having the benefit of the present invention disclosure.

(615) When using multiple spherical wheel assemblies **40** to design, construct and operate a manually powered (i.e. push powered) transport system, a mechanically-powered transport system (e.g. using a gear-drive mechanism driven battery-powered motor or drill) or a battery-powered motor driven transport system, the RC function can be provided and offer great benefits and advantages to the user in terms of convenience and safety. Such multi-spherical wheel supported transport systems **70** and variations thereof, can also provide interconnected and in communication with the cloud-based servers and services illustrated in FIGS. **26**A, **27**, **30**A, and **36**, to provide GPS-guided navigation capabilities to system **70**, as well as LIDAR-based environmental awareness.

(616) In general, the system networks described in FIGS. **26**A, **27**, **30**A, and **36**, and which are used to practice the various embodiments of the present invention disclosure herein, comprise various system components, including: a cellular phone and SMS messaging systems; and one or more industrial-strength data centers, preferably mirrored with each other and running Border Gateway Protocol (BGP) between its router gateways, in a manner well known in the data center art. In FIGS. **26**A, **27**, **30**A, and **36**, and other illustrative embodiments, each data center typically comprises: a cluster of communication servers for supporting http and other TCP/IP based communication protocols on the Internet; cluster of application servers; a cluster of email processing servers; cluster of SMS servers; and a cluster of RDBMS servers configured within an distributed file storage and retrieval ecosystem/system, and interfaced around the TCP/IP infrastructure of the Internet well known in the art. For details regarding such system engineering

considerations and requirements, reference should be made to US Patent Application Publication No. 20160197993, incorporated herein by reference in its entirety as if set forth fully herein.

(617) As shown, the system network architecture also comprises: a plurality of Web-enabled mobile client machines **427** (e.g. mobile computers such as iPad, and other Internet-enabled computing devices with graphics display capabilities, etc.) running native mobile applications **427**AA and mobile web browser applications supported modules supporting client-side and server-side processes on the system network of the present invention; and numerous media servers (e.g. Google, Facebook, NOAA, etc.) operably connected to the infrastructure of the Internet. The network of mobile computing systems will run enterprise-level mobile application software, operably connected to the TCP/IP infrastructure of the Internet. Each mobile computing system is provided with GPS-tracking and having wireless internet connectivity with the TCP/IP infrastructure of the Internet, using various communication technologies (e.g. GSM, Bluetooth and other wireless networking protocols well known in the wireless communications arts).

(618) In general, regardless of the method of implementation employed, the system network of the present invention will be in almost all instances, realized upon an industrial-strength, carrier-class Internet-based network of object-oriented system design. Also, the system network will be deployed over a global data packet-switched communication network comprising numerous computing systems and networking components, as shown. As such, the information networks of the present invention is often referred to herein as the “system” or “system network”.

(619) Preferably, although not necessary, the system network of the present invention would be designed according to object-oriented systems engineering (OOSE) methods using UML-based modeling tools such as ROSE by Rational Software, Inc. using an industry-standard Rational Unified Process (RUP) or Enterprise Unified Process (EUP), both well known in the art. Implementation programming languages can include C, Objective C, C, Java, PHP, Python, Google's GO, and other computer programming languages known in the art. The Internet-based system network can be implemented using any object-oriented integrated development environment (IDE) such as for example: the Java Platform, Enterprise Edition, or Java EE (formerly J2EE); Websphere IDE by IBM; Weblogic IDE by BEA; a non-Java IDE such as Microsoft's .NET IDE; or other suitably configured development and deployment environment well known in the art. Preferably, the system network is deployed as a three-tier server architecture with a double-firewall, and appropriate network switching and routing technologies well known in the art. In some deployments, private/public/hybrid cloud service providers, such Amazon Web Services (AWS), may be used to deploy Kubernetes, an open-source software container/cluster management/orchestration system, for automating deployment, scaling, and management of containerized software applications, such as the mobile enterprise-level application described above. Such practices are well known in the computer programming, networking and digital communication arts.

Modifications of the Illustrative Embodiments of the Present Invention

(620) The present invention has been described in great detail with reference to the above illustrative embodiments. It is understood, however, that numerous modifications will readily occur to those with ordinary skill in the art having had the benefit of reading the present disclosure.

(621) For example, in alternative embodiments of the present invention described hereinabove, the system can be used for a wide variety of transport applications while not illustrated, will readily occur to those skilled in the art having the benefit of the present invention disclosure. Such alternative applications and/or system configurations will depend on particular end-user application requirements, and target markets for products and services.

(622) These and all other such modifications and variations are deemed to be within the scope and spirit of the present invention as defined by the accompanying Claims to Invention.

Claims

1. A trailer transport system for mounting to a trailer that can be coupled to a motor-driven vehicle, said trailer transport system comprising: a semispherical framework having an interior volume contained by the boundaries of the semispherical framework; a set of bearing surfaces provided within the interior volume of said semispherical framework; a load-bearing transport ball installed within the interior volume of said semispherical framework, and supported by the bearing surfaces provided therein, and allowing said load-bearing transport ball to rotate freely within said semispherical framework, as defined by the bearing surfaces; and a support mechanism for supporting said semispherical framework in a transport configuration during transport operations, and also in a storage configuration during storage operations; wherein said load-bearing transport ball is able to roll freely in all directions within said semispherical framework when said load-bearing transport ball contacts the ground surface and enables rollable transport of the trailer; wherein the bearing surfaces comprise a set of ball-bearing pads mounted within the interior volume of said semispherical framework; and wherein each said ball-bearing pad comprises a plurality of spring-biased ball bearings that are mounted on a mounting plate having a surface curvature that approximates a portion of a spherical object.
2. The trailer transport system of claim 1, a jack assembly mounting mechanism integrated with an extendable jacking post member mounted to said semispherical framework; and a user-operated mechanism integrated within the extendable jacking post member allowing the jacking post member to elongate, and setting said load-bearing transport ball on a ground surface during said transport configuration.
3. The trailer transport system of claim 2, wherein the extendable jacking post member comprises a telescopically-extending jacking post member having a free end fixedly mounted to said semispherical framework.
4. The trailer transport system of claim 1, wherein said support mechanism comprises a jack assembly mounting mechanism integrated with an extendable jacking post member mounted to said semispherical framework; and a user-operated mechanism integrated within the extendable jacking post member allowing the jacking post member to elongate, and setting said load-bearing transport ball on a ground surface during said transport mode.
5. The trailer transport system of claim 4, wherein said extendable jacking post member comprises a telescopically-extending jacking post member having a free end fixedly mounted to said semispherical framework.
6. The trailer transport system of claim 4, wherein said user-operated mechanism comprises: a hand-cranked jack handle integrated within said telescopically-extending jacking post member allowing the post member to elongate in response to certain rotation of the hand-cranked jack handle, and setting said load-bearing transport ball on a ground surface.
7. The trailer transport system of claim 4, wherein said user-operated mechanism comprises a hand-operated electronic device for sending signals to an electrically-driven motor configured to automatically elongate said extendible jacking post member along its axis and enable jacking operating of the trailer jack and transport system.
8. The trailer transport system of claim 4, wherein said semispherical framework has a substantially closed semi-spherical dome-like structure to which said extendible jacking post member is fixedly attached, and in which said load-bearing transport ball is rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.
9. The trailer transport system of claim 4, wherein said semispherical framework has a substantially open semi-spherical dome-like structure to which said extendible jacking post member is fixedly attached, and in which the load-bearing transport ball is rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.

10. The trailer transport system of claim 4, wherein a retainer mechanism is mounted to said semispherical framework for securely retaining said load-bearing transport ball rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.
11. The trailer transport system of claim 1, wherein said load-bearing transport ball comprises a hollow interior volume occupied by an inert gas.
12. The trailer transport system of claim 1, wherein said load-bearing transport ball comprises a solid interior volume occupied by a solid mass material.
13. The trailer transport system of claim 1, which further comprises a transport ball braking system having mechanically-operated safety-levers to arrange for non-braking and braking modes of operation; wherein when configured in the non-braking mode of operation, a set of braking shoes are disengaged from said load-bearing transport ball allowing the transport ball to roll freely within said semispherical framework, and wherein when configured in the braking mode of operation, the set of braking shoes are engaged with said load-bearing transport ball, restricting said load-bearing transport ball from rolling freely within said semispherical framework.
14. A trailer transport system for mounting to a trailer that can be coupled to a motor-driven vehicle, said trailer transport system comprising: a semispherical framework having an interior volume contained by the boundaries of the semispherical framework; a set of bearing surfaces provided within the interior volume of the semispherical framework; a load-bearing transport ball installed within the interior volume of the semispherical framework, and supported by the bearing surfaces provided therein, and allowing said load-bearing transport ball to rotate freely within said semispherical framework, as defined by the bearing surfaces; and a support mechanism for supporting said semispherical framework in a transport configuration during transport operations, and also in a storage configuration during storage operations; wherein the load-bearing transport ball is free to roll freely in all directions within said semispherical framework when said load-bearing transport ball contacts the ground surface and enables rollable transport of the trailer; and wherein the bearing surfaces comprise a plurality of spring-biased ball bearings that are mounted between a set of mounting plates having a surface curvature that approximates a portion of a spherical object.
15. The trailer transport system of claim 14, which further comprises an extendable jacking post member having a free end fixedly mounted to said semispherical framework.
16. The trailer transport system of claim 14, which further comprises a transport ball braking system having mechanically-operated safety-levers to arrange for non-braking and braking modes of operation; wherein when configured in the non-braking mode of operation, a set of braking shoes are disengaged from said load-bearing transport ball allowing the transport ball to roll freely within said semispherical framework, and wherein when configured in the braking mode of operation, the set of braking shoes are engaged with said load-bearing transport ball, restricting said load-bearing transport ball to roll from rolling freely within said semispherical framework.
17. The trailer transport system of claim 14, wherein said support mechanism comprises a jack assembly mounting mechanism integrated with an extendable jacking post member mounted to said semispherical framework; and a user-operated mechanism integrated within the extendable jacking post member allowing the jacking post member to elongate, and setting said load-bearing transport ball on a ground surface during said transport mode.
18. The trailer transport system of claim 17, wherein the said extendable jacking post member comprises a telescopically-extending jacking post member having a free end fixedly mounted to said semispherical framework.
19. The trailer transport system of claim 17, wherein said user-operated mechanism comprises: a hand-cranked jack handle integrated within said extendable jacking post member allowing the post member to elongate in response to certain rotation of the hand-cranked jack handle, and setting said load-bearing transport ball on a ground surface.

20. The trailer transport system of claim 17, wherein said user-operated mechanism comprises a hand-operated electronic device for sending signals to an electrically-driven motor configured to automatically elongate said extendible jacking post member along its axis and enable the jacking operation of the trailer jack and transport system.
21. The trailer transport system of claim 17, wherein said semispherical framework has a substantially closed semi-spherical dome-like structure to which said extendible jacking post member is fixedly attached, and in which the load-bearing transport ball is rotatably mounted and supported on the bearing surfaces provided within the interior volume of the semispherical framework.
22. The trailer transport system of claim 17, wherein said semispherical framework has a substantially open semi-spherical dome-like structure to which said extendible jacking post member is fixedly attached, and in which said load-bearing transport ball is rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.
23. The trailer transport system of claim 17, wherein said jack assembly mounting mechanism comprises a telescoping rotational mounting mechanism that elongates in a direction orthogonal to the longitudinal extent of an extendible jacking post member, when the extendible jacking post member is rotated through 90 degrees of rotation so as to allow said semispherical framework to have dimensions larger than said extendible jacking post member and be accommodated when arranged in a storage configuration.
24. The trailer transport system of claim 14, wherein a retainer mechanism is mounted to said semispherical framework for securely retaining the load-bearing transport ball rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.
25. The trailer transport system of claim 14, wherein said load-bearing transport ball comprises a hollow interior volume occupied by an inert gas.
26. The trailer transport system of claim 14, wherein said load-bearing transport ball comprises a solid interior volume occupied by a solid mass material.
27. A trailer transport system for mounting to a trailer that can be coupled to a motor-driven vehicle, said trailer transport system comprising: a semispherical framework having an interior volume contained by the boundaries of the semispherical framework; a set of bearing surfaces provided within the interior volume of said semispherical framework; a load-bearing transport ball installed within the interior volume of said semispherical framework, and supported by the bearing surfaces provided therein, and allowing said load-bearing transport ball to rotate freely within said semispherical framework, as defined by the bearing surfaces; and a support mechanism for supporting said semispherical framework in a transport configuration during transport operations, and also in a storage configuration during storage operations; wherein said load-bearing transport ball is free to roll freely in all directions within said semispherical framework when said load-bearing transport ball contacts the ground surface and enables rollable transport of the trailer; and wherein a brushing mechanism is mounted about the bearing pads said bearing surfaces for brushing loose debris from the surfaces of said load-bearing transport ball while rotating and supported on the bearing surfaces provided within the interior volume of said semispherical framework.
28. A trailer transport system for mounting to a trailer that can be coupled to a motor-driven vehicle, said trailer transport system comprising: a semispherical framework having an interior volume contained by the boundaries of said semispherical framework; a set of bearing surfaces provided within the interior volume of said semispherical framework; a load-bearing transport ball installed within the interior volume of said semispherical framework, and supported by the bearing surfaces provided therein, and allowing said load-bearing transport ball to rotate freely within said semispherical framework, as defined by the bearing surfaces; and a support mechanism for

supporting said semispherical framework in a transport configuration during transport operations, and also in a storage configuration during storage operations; wherein said load-bearing transport ball is free to roll freely in all directions within said semispherical framework when said load-bearing transport ball contacts the ground surface and enables rollable transport of the trailer; and wherein said load-bearing transport ball comprises a rigid spherical structure capable of withstanding a predetermined weight-bearing load, and coated with a surface texture providing a threading surface for channeling water and preventing slippage and sliding while rolling on the ground surface.

29. A trailer transport system for mounting to a trailer that can be coupled to a motor-driven vehicle, said trailer transport system comprising: a semispherical framework having an interior volume contained by the boundaries of said semispherical framework; a set of bearing surfaces provided within the interior volume of said semispherical framework; a load-bearing transport ball installed within the interior volume of said semispherical framework, and supported by the bearing surfaces provided therein, and allowing said load-bearing transport ball to rotate freely within said semispherical framework, as defined by the bearing surfaces; and a support mechanism for supporting said semispherical framework in a transport configuration during transport operations, and also in a storage configuration during storage operations; wherein the load-bearing transport ball is enabled to roll freely in all directions within said semispherical framework when said load-bearing transport ball contacts the ground surface and enables rollable transport of said trailer; and wherein said jack assembly mounting mechanism comprises an angulated joint that allows an elongated jacking post member to rotate away from said trailer when rotated towards a storage configuration, wherein the load-bearing transport ball, supported within said semispherical framework, is allowed to store be stored alongside said trailer.

30. A trailer transport system for mounting to a trailer that can be coupled to a motor-driven vehicle, said trailer transport system comprising: a semispherical framework having an interior volume contained by the boundaries of said semispherical framework; a set of bearing surfaces provided within the interior volume of said semispherical framework; a load-bearing transport ball installed within the interior volume of said semispherical framework, and supported by the bearing surfaces provided therein, and allowing said load-bearing transport ball to rotate freely within said semispherical framework, as defined by the bearing surfaces; and a support mechanism for supporting said semispherical framework in a transport configuration during transport operations, and also in a storage configuration during storage operations; wherein said load-bearing transport ball is free to roll freely in all directions within said semispherical framework when said load-bearing transport ball contacts the ground surface and enables rollable transport of said trailer; wherein a set of electrically-powered motors are mounted relative to said semispherical framework to engage the exterior surface of said load-bearing transport ball, to drive and cause said load-bearing transport ball to rotate in a direction or pathway on the ground surface controlled by a user, to enable the automated transport of said trailer.

31. The trailer jack and transport system of claim 30, wherein a control device supported on an extendible jacking post member is provided to operate and control said set of electrically-powered motors.

32. The trailer transport system of claim 31, wherein said control device is a removable wireless communication device provided for use in operating and controlling said set of electrically-powered motors remotely.

33. The trailer transport system of claim 30, wherein said support mechanism comprises a jack assembly mounting mechanism integrated with an extendable jacking post member mounted to said semispherical framework; and a user-operated mechanism integrated within said extendable jacking post member allowing the extendable jacking post member to elongate, and setting said load-bearing transport ball on a ground surface during said transport mode.

34. The trailer transport system of claim 33, wherein said extendable jacking post member

comprises a telescopically-extending jacking post member having a free end fixedly mounted to said semispherical framework.

35. The trailer transport system of claim 33, wherein said user-operated mechanism comprises: a hand-cranked jack handle integrated within the telescopically-extending jacking post member allowing the post member to elongate in response to certain rotation of the hand-cranked jack handle, and setting said load-bearing transport ball on a ground surface.

36. The trailer transport system of claim 33, wherein said user-operated mechanism comprises a hand-operated electronic device for sending signals to an electrically-driven motor configured to automatically elongate the extendible jacking post member along its axis and enable jacking operating of said trailer jack and transport system.

37. The trailer transport system of claim 33, wherein said semispherical framework has a substantially closed semi-spherical dome-like structure to which said extendible jacking post member is fixedly attached, and in which said load-bearing transport ball is rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.

38. The trailer transport system of claim 33, wherein said semispherical framework has a substantially open semi-spherical dome-like structure to which said extendible jacking post member is fixedly attached, and in which the load-bearing transport ball is rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.

39. The trailer transport system of claim 30, wherein a retainer mechanism is mounted to said semispherical framework for securely retaining the load-bearing transport ball rotatably mounted and supported on the bearing surfaces provided within the interior volume of said semispherical framework.

40. The trailer transport system of claim 30, wherein said load-bearing transport ball comprises a hollow interior volume occupied by an inert gas.

41. The trailer transport system of claim 30, wherein said load-bearing transport ball comprises a solid interior volume occupied by a solid mass material.

42. The trailer transport system of claim 30, wherein a brushing mechanism is mounted about the bearing surfaces for brushing loose debris from the surfaces of said load-bearing transport ball while rotating and supported on the bearing surfaces provided within the interior volume of said semispherical framework.

43. The trailer transport system of claim 30, wherein said load-bearing transport ball comprises a rigid spherical structure capable of withstanding a predetermined weight-bearing load, and coated with a surface texture providing a threading surface for channeling water and preventing slippage and sliding while rolling on the ground surface.